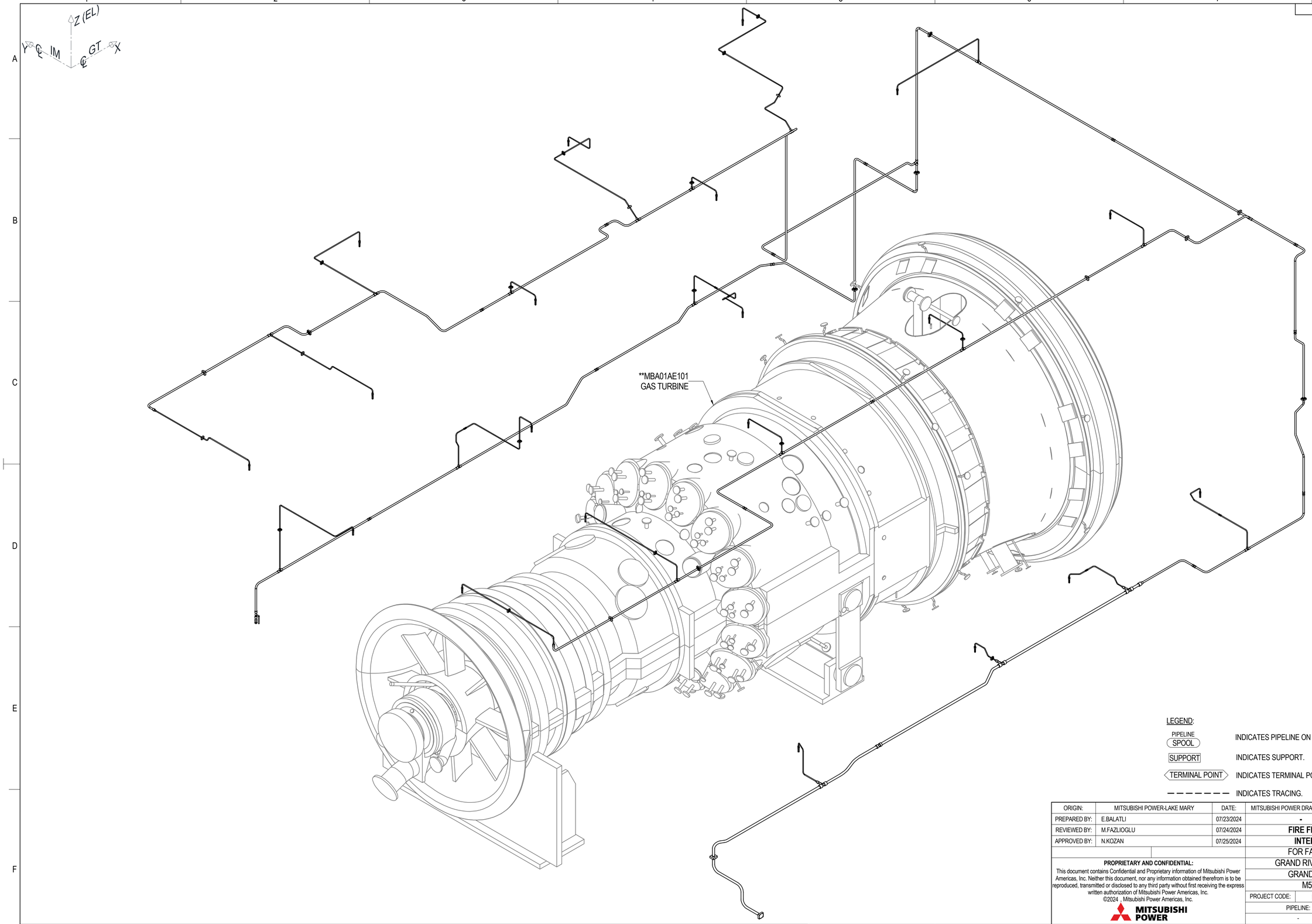
REVISION RECORD



LEGEND:

PIPELINE / SPOOL	INDICATES PIPELINE ON THE TOP AND SPOOL ON THE BOTTOM.
SUPPORT	INDICATES SUPPORT.
TERMINAL POINT	INDICATES TERMINAL POINT.
– – – – – – –	INDICATES TRACING.

ORIGIN:	MITSUBISHI POWER-LAKE MARY	DATE:	MITSUBISHI POWER DRAWING NUMBER:	REV. NO.:	SYSTEM CODE:
PREPARED BY:	E.BALATLI	07/23/2024	-	0	64A0A
REVIEWED BY:	M.FAZLIOGLU	07/24/2024	FIRE FIGHTING SYSTEM PIPING		
APPROVED BY:	N.KOZAN	07/25/2024	INTERNAL TO GT PACKAGE		
			FOR FABRICATION & ERECTION		
			GRAND RIVER ENERGY CENTER UNIT 4		
			GRAND RIVER DAM AUTHORITY		
			M501JAC GAS TURBINE		
PROJECT CODE:	K-GREC1	P&ID:	-		
PIPELINE:	-	PAGE NUMBER:	1 OF 4		

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MITSUBISHI POWER

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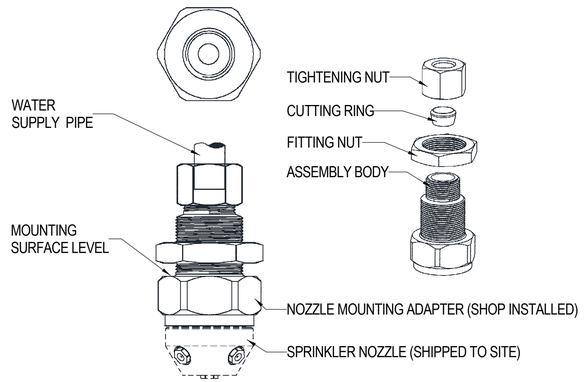
REVISION RECORD

SPRINKLER ASSEMBLY BODY SET FOR HI-FOG® 1000 SERIES SPRINKLER AND SPRAY HEADS



LEGEND:

PIPELINE SPOOL	INDICATES PIPELINE ON THE TOP AND SPOOL ON THE BOTTOM.
SUPPORT	INDICATES SUPPORT.
TERMINAL POINT	INDICATES TERMINAL POINT.
– – – – – – –	INDICATES TRACING.

ORIGIN:	MITSUBISHI POWER-LAKE MARY	DATE:	MITSUBISHI POWER DRAWING NUMBER:	REV. NO.:	SYSTEM CODE:
PREPARED BY:	E.BALATLI	07/23/2024	-	0	64A0A
REVIEWED BY:	M.FAZLIOGLU	07/24/2024	FIRE FIGHTING SYSTEM PIPING		
APPROVED BY:	N.KOZAN	07/25/2024	INTERNAL TO GT PACKAGE		
			FOR FABRICATION & ERECTION		
			GRAND RIVER ENERGY CENTER UNIT 4		
			GRAND RIVER DAM AUTHORITY		
			M501JAC GAS TURBINE		
PROJECT CODE:	K-GREC1	P&ID:	-		
PIPELINE:	-	PAGE NUMBER:	2 OF 4		

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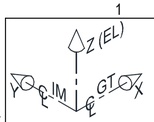
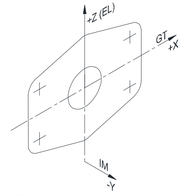
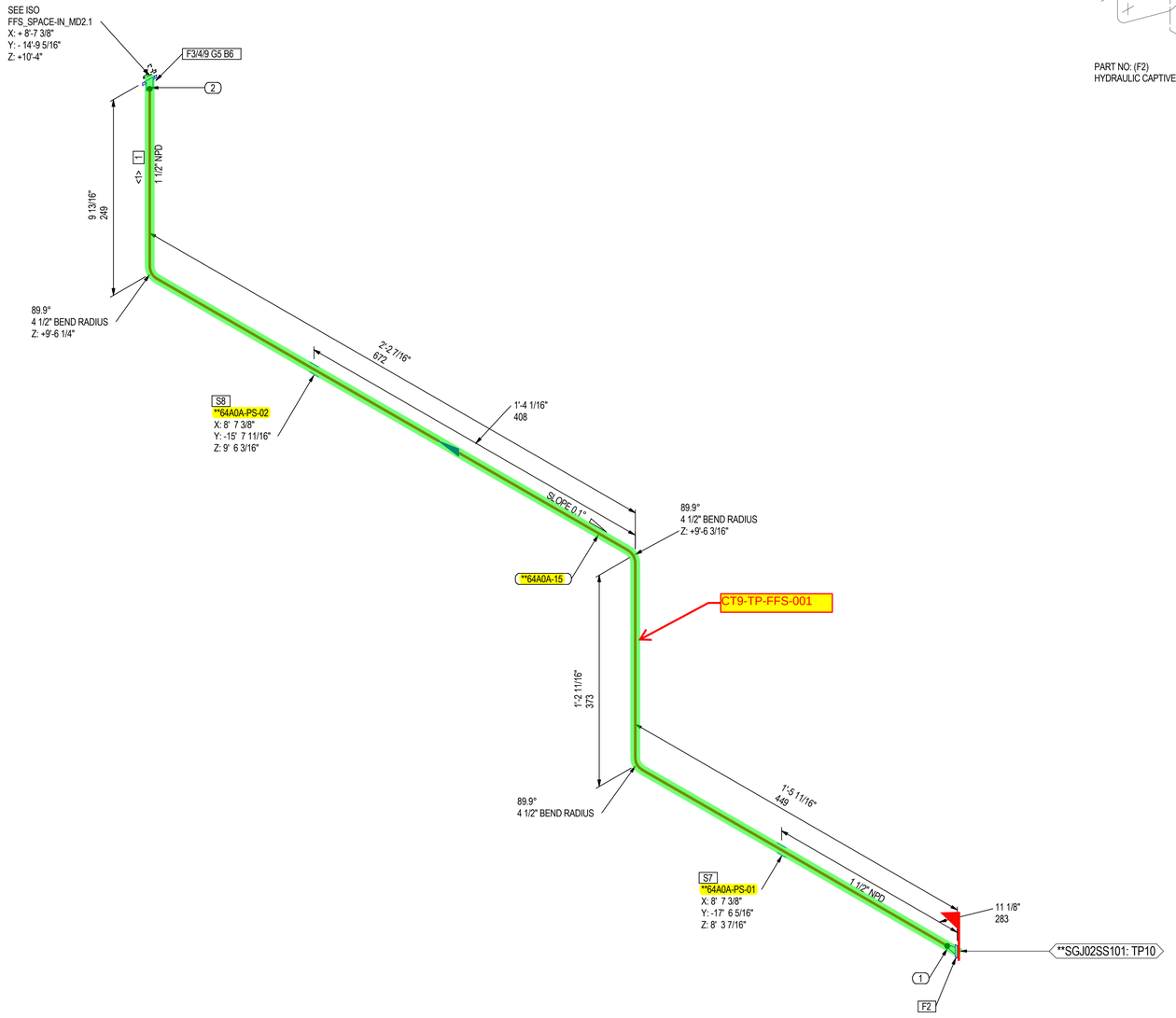




PART NO: (F2)
HYDRAULIC CAPTIVE FLANGE OREINTATION



SHOP MATERIALS - FABRICATION ONLY

PT NO	DESCRIPTION	NPD (IN)	CMDTY CODE	QTY
1	TUBE, 0.12 in THK, PE, ASTM A213-TP316, SEAMLESS	1 1/2	MTAAZZBPZZZZZXAQ	5.2'
2	FLANGE HYDRAULIC, 3000 PSI, FFFE x SWE, SAE J518-1 CODE 61, ASTM A182-F316, CAPTIVE FLANGE	1 1/2	MFFBAWYGZZIARALS	1
3	FLANGE HYDRAULIC, 3000 PSI, GROOVED END x SWE, SAE J518-1 CODE 61, ASTM A182-F316, FLANGE HEAD	1 1/2	MFFAAWYFZZIARALS	1
	MISCELLANEOUS COMPONENTS			
4	FLANGE HYDRAULIC, 3000 PSI, SAE J518-1 CODE 61, ASTM A182-F316, SPLIT FLANGE CLAMP (ONE PAIR)	1 1/2	MWABAWYHZZIARALS	1

OTHER THAN SHOP MATERIALS - ERECTION ONLY

PT NO	DESCRIPTION	NPD (IN)	CMDTY CODE	QTY
5	GASKET, ORING, 0.125" THK, AS 568, VITON HS90 (DUROMETER90)	1 1/2	MKBZZACIAQXANUS	1
6	SCREW, DRILLED HEX SOCKET HEAD CAP (2 HOLES), ASME B18.3, ASTM A193-B7, 13-UNC - 2.25 in. Length	1/2	MBCZZACZZIALBEUZ	4
	PIPE SUPPORTS			
7	PIPE SUPPORT	1 1/2	**64A0A-PS-01	1
8	PIPE SUPPORT	1 1/2	**64A0A-PS-02	1
	MISCELLANEOUS COMPONENTS			
9	LOCK WIRE, 1mm DIA x 1m LENGTH, ASTM A580-304	1 1/2	MWRZZZZBAIASXAPU	1
	SPOOLS			
	**64A0A-15			

CUT PIPE LENGTH

PIECE NO	LENGTH (FT-INS)	NPD (IN)	REMARKS	END ONE	END TWO	ITEM CODE
<1>	5'2 3/16"	1 1/2	PLD BEND(3)	BEVEL	SQ.CUT	MTAAZZBPZZZZZXAQ

ACTUAL QUANTITY OF WASHERS SHALL BE CONSIDERED AS HALF OF THE QUANTITY SHOWN ON ISOMETRICS

NOTES:
1. THE X AND Y COORDINATES ARE WITH RESPECT TO THE INTERSECTION OF THE CENTERLINE OF THE GAS TURBINE (GT) AND THE INLET MANIFOLD (IM). THE Z (EL) COORDINATE IS WITH RESPECT TO THE TOP OF CONCRETE, WHICH IS 15'-9" [4800mm] BELOW THE CENTERLINE OF THE GT AND IS 0'-0".
2. ALL PIPE SUPPORT COORDINATES ARE WITH RESPECT TO THE CENTERLINE OF THE INSTALLED SUPPORTED PIPE.
3. THE MARK 'FFW' (FIELD FIT WELD) IN THE DRAWING INDICATES EXCESS LENGTH OF APPROX. 4 INCHES IDENTIFIED AS LOOSE IN THE DRAWING TO BE CUT AND ADJUSTED AT THE FIELD.
4. ALL WELDING TO BE DONE AND INSPECTED PER LATEST APPLICABLE EDITION OF ASME B31.1 AND SECTION IX AVAILABLE ON THE PROJECT AWARD DATE.
5. PIPING SHALL BE INSTALLED PER THE INDICATED SLOPE.
6. FOR ANY FURTHER INFORMATION REQUIRED DURING INSTALLATION, CONSULT MITSUBISHI POWER TFA (TECHNICAL FIELD ADVISOR) AT SITE.
7. XXX## ### INDICATES PIPING TERMINAL POINT.

WELD NO	WELD TYPE	WELD CAT	NPD	QTY
1	BW	S	1 1/2"	1
2	BW	S	1 1/2"	1

PIPING SPEC:	SDQ	PAINTING SPEC: PER PROJECT SPECIFICATION	REVISION RECORD				
FLUID CODE:	HW, FIRE FIGHTING WATER	INTERIOR SURFACE TREATMENT SPEC: LU-00129	REV. #	DESCRIPTION	BY	CHK	APP	DATE
MAX. PRESS.:	2030.00 psig / 14.00 MPag							
TEST PRESS.:	3045.00 psig / 20.99 MPag							
MAX. TEMP:	100 F / 38 C							
INSULATION SPEC:	NOT INSULATED							

WELD TYPE IDENTIFIER:
BW - BUTT WELD
SW - SOCKET WELD
SOF - SLIP-ON FLANGE
SOB - SET-ON BRANCH
LET - OLET, LATROLET OR HALF COUPLING
TW - TACK WELD
MW - MITER WELD

WELD CATEGORY IDENTIFICATION:
S - SHOP
FW - FIELD WELD
FFW - FIELD FIT WELD

SYMBOLOGY:
WELD NUMBER
PART NUMBER (PT NO)
< > CUT LIST PIECE NUMBER

SYMBOL IDENTIFIER:
F - FLANGE
G - GASKET
B - BOLT
N - NUT
W - WASHER
O - O-RING
L - LOCK WIRE

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MITSUBISHI POWER

ORIGIN:	MITSUBISHI POWER-LAKE MARY	DATE:	MITSUBISHI POWER DRAWING NUMBER: / REV. NO.: / SYSTEM CODE:
PREPARED BY:	E.BALATLI	07/23/2024	- / 0 / 64A0A
REVIEWED BY:	M.FAZLIOGLU	07/24/2024	FIRE FIGHTING SYSTEM PIPING
APPROVED BY:	N.KOZAN	07/25/2024	INTERNAL TO GT PACKAGE
			FOR FABRICATION & ERECTION
			GRAND RIVER ENERGY CENTER UNIT 4
			GRAND RIVER DAM AUTHORITY
			M501JAC GAS TURBINE
PROJECT CODE:	K-GREC1	P&ID:	
PIPELINE:	FFS_SPACE-IN_MD1.1	PAGE NUMBER:	1 OF 1

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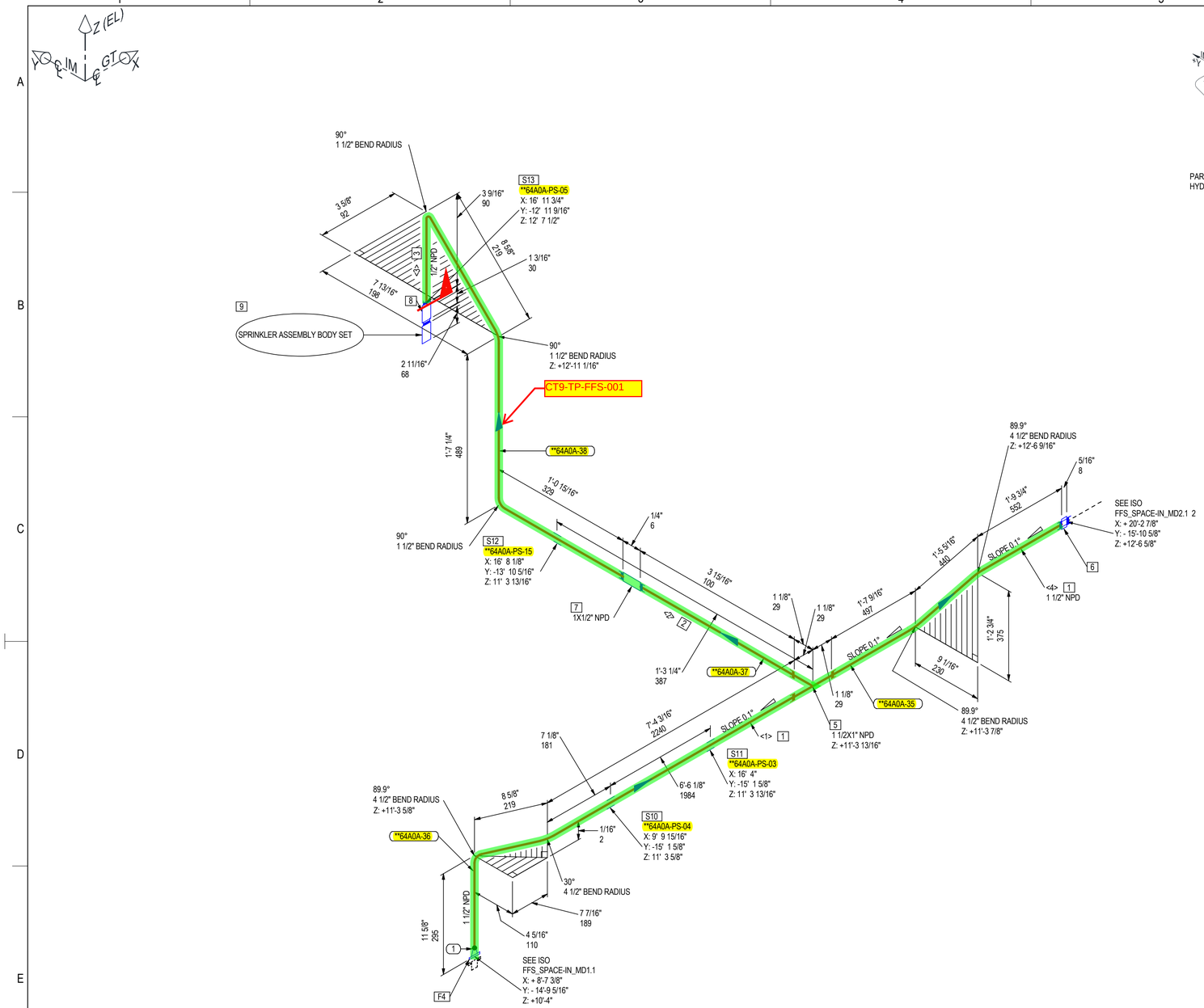

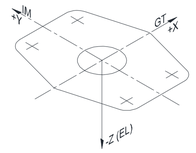

PART NO: (F4)
HYDRAULIC CAPTIVE FLANGE ORIENTATION

SHOP MATERIALS - FABRICATION ONLY

PT NO	DESCRIPTION	NPD (IN)	CMDTY CODE	QTY
1	TUBE, 0.12 in THK, PE, ASTM A213-TP316, SEAMLESS	1 1/2	MTAAZZBPZZZZZXAQ	13.4'
2	TUBE, 0.083 in THK, PE, ASTM A213-TP316, SEAMLESS	1	MTAAZZBPZZZZZXAQ	0.4'
3	TUBE, 0.049 in THK, PE, ASTM A213-TP316, SEAMLESS	1/2	MTAAZZBPZZZZZXAQ	3.6'
4	FLANGE HYDRAULIC, 3000 PSI, FFFE x SWE, SAE J518-1 CODE 61, ASTM A182-F316, CAPTIVE FLANGE	1 1/2	MFFBAWYGZZIARALS	1

OTHER THAN SHOP MATERIALS - ERECTION ONLY

PT NO	DESCRIPTION	NPD (IN)	CMDTY CODE	QTY
5	TUBE REDUCING UNION TEE, SWGLKE END, SWAGELOK, ASTM A182-F316	1 1/2X1	MWBCZZCGZZAKWALS	1
6	TUBE UNION, SWGLKE END, SWAGELOK PART NO. SS-2400-6, ASTM A182-F316	1 1/2	MWADZZCGZZAKWALS	1
7	REDUCING TUBE UNION, SWGLKE END, SWAGELOK PART NO. SS-1610-6-8, ASTM A182-F316	1X1/2	MWARZZCGZZAKWALS	1
8	TUBE REDUCER (FRACTIONAL TO METRIC), SWGLKE END x PE, SWAGELOK PART NO.SS-810-R-12M, ASTM A182-F316	1/2	MWAOZZYDZZAKWALS	1
	INSTRUMENTS			
9	SPRINKLER ASSEMBLY BODY SET FOR HI-FOG® 1000 SERIES SPRINKLER AND SPRAY HEADS, A1000S1CT00D12S10 SPRINKLER ASSEMBLY BODY SET	1/2		1
	PIPE SUPPORTS			
10	PIPE SUPPORT	1 1/2	**64A0A-PS-04	1
11	PIPE SUPPORT	1 1/2	**64A0A-PS-03	1
12	PIPE SUPPORT	1/2	**64A0A-PS-15	1
13	NOZZLE MOUNTING ADAPTER SUPPORT	1/2	**64A0A-PS-05	1
	SPOOLS			
	**64A0A-35, **64A0A-36, **64A0A-37, **64A0A-38			

CUT PIPE LENGTH

PIECE NO	LENGTH (FT-INS)	NPD (IN)	REMARKS	END ONE	END TWO	ITEM CODE
<1>	8'9 7/8"	1 1/2	PLD BEND(2)	BEVEL	SQ.CUT	MTAAZZBPZZZZZXAQ
<2>	3 15/16"	1		SQ.CUT	SQ.CUT	MTAAZZBPZZZZZXAQ
<3>	3'6 7/16"	1/2	PLD BEND(3)	SQ.CUT	SQ.CUT	MTAAZZBPZZZZZXAQ
<4>	4'6 13/16"	1 1/2	PLD BEND(2)	SQ.CUT	SQ.CUT	MTAAZZBPZZZZZXAQ

ACTUAL QUANTITY OF WASHERS SHALL BE CONSIDERED AS HALF OF THE QUANTITY SHOWN ON ISOMETRICS

NOTES:
1. THE X AND Y COORDINATES ARE WITH RESPECT TO THE INTERSECTION OF THE CENTERLINE OF THE GAS TURBINE (GT) AND THE INLET MANIFOLD (IM). THE Z (EL) COORDINATE IS WITH RESPECT TO THE TOP OF CONCRETE, WHICH IS 15'-9" [4800mm] BELOW THE CENTERLINE OF THE GT AND IS 0'-0".
2. ALL PIPE SUPPORT COORDINATES ARE WITH RESPECT TO THE CENTERLINE OF THE INSTALLED SUPPORTED PIPE.
3. THE MARK 'FFW' (FIELD FIT WELD) IN THE DRAWING INDICATES EXCESS LENGTH OF APPROX. 4 INCHES IDENTIFIED AS LOOSE IN THE DRAWING TO BE CUT AND ADJUSTED AT THE FIELD.
4. ALL WELDING TO BE DONE AND INSPECTED PER LATEST APPLICABLE EDITION OF ASME B31.1 AND SECTION IX AVAILABLE ON THE PROJECT AWARD DATE.
5. PIPING SHALL BE INSTALLED PER THE INDICATED SLOPE.
6. FOR ANY FURTHER INFORMATION REQUIRED DURING INSTALLATION, CONSULT MITSUBISHI POWER TFA (TECHNICAL FIELD ADVISOR) AT SITE.
7. 〈XXX## ###〉 INDICATES PIPING TERMINAL POINT.

ADDITIONAL NOTE:
TUBE REDUCER (FRACTIONAL TO METRIC) SHALL BE IDENTIFIED BY COLOR OR TAGGING FOR QUICK NOTATION IN THE FIELD.

WELD NO	WELD TYPE	WELD CAT	NPD	QTY
1	BW	S	1 1/2"	1

PIPING SPEC:	SDQ	PAINTING SPEC: PER PROJECT SPECIFICATION	REVISION RECORD					
FLUID CODE:	HW, FIRE FIGHTING WATER	INTERIOR SURFACE TREATMENT SPEC: LU-00129	REV. #	DESCRIPTION	BY	CHK	APP	DATE
MAX. PRESS.:	2030.00 psig / 14.00 MPag							
TEST PRESS.:	3045.00 psig / 20.99 MPag							
MAX. TEMP:	100 F / 38 C							
INSULATION SPEC:	NOT INSULATED							

WELD TYPE IDENTIFIER:
BW - BUTT WELD
SW - SOCKET WELD
SOF - SLIP-ON FLANGE
SOB - SET-ON BRANCH
LET - OLET, LATROLET OR HALF COUPLING
TW - TACK WELD
MW - MITER WELD

WELD CATEGORY IDENTIFICATION:
S - SHOP
FW - FIELD WELD
FFW - FIELD FIT WELD

SYMBOLOGY:
WELD NUMBER
PART NUMBER (PT NO)
< > CUT LIST PIECE NUMBER

SYMBOL IDENTIFIER:
F - FLANGE
G - GASKET
B - BOLT
N - NUT
W - WASHER
O - O-RING
L - LOCK WIRE

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MITSUBISHI POWER

ORIGIN:	MITSUBISHI POWER-LAKE MARY	DATE:	MITSUBISHI POWER DRAWING NUMBER:	REV. NO.:	SYSTEM CODE:
PREPARED BY:	E.BALATLI	07/23/2024	-	0	64A0A
REVIEWED BY:	M.FAZLIOGLU	07/24/2024	FIRE FIGHTING SYSTEM PIPING		
APPROVED BY:	N.KOZAN	07/25/2024	INTERNAL TO GT PACKAGE		
			FOR FABRICATION & ERECTION		
			GRAND RIVER ENERGY CENTER UNIT 4		
			GRAND RIVER DAM AUTHORITY		
			M501JAC GAS TURBINE		
PROJECT CODE:	K-GREC1	P&ID:			
PIPELINE:	FFS_SPACE-IN_MD2.1	PAGE NUMBER:	1 OF 5		

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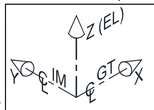
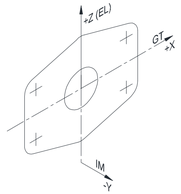
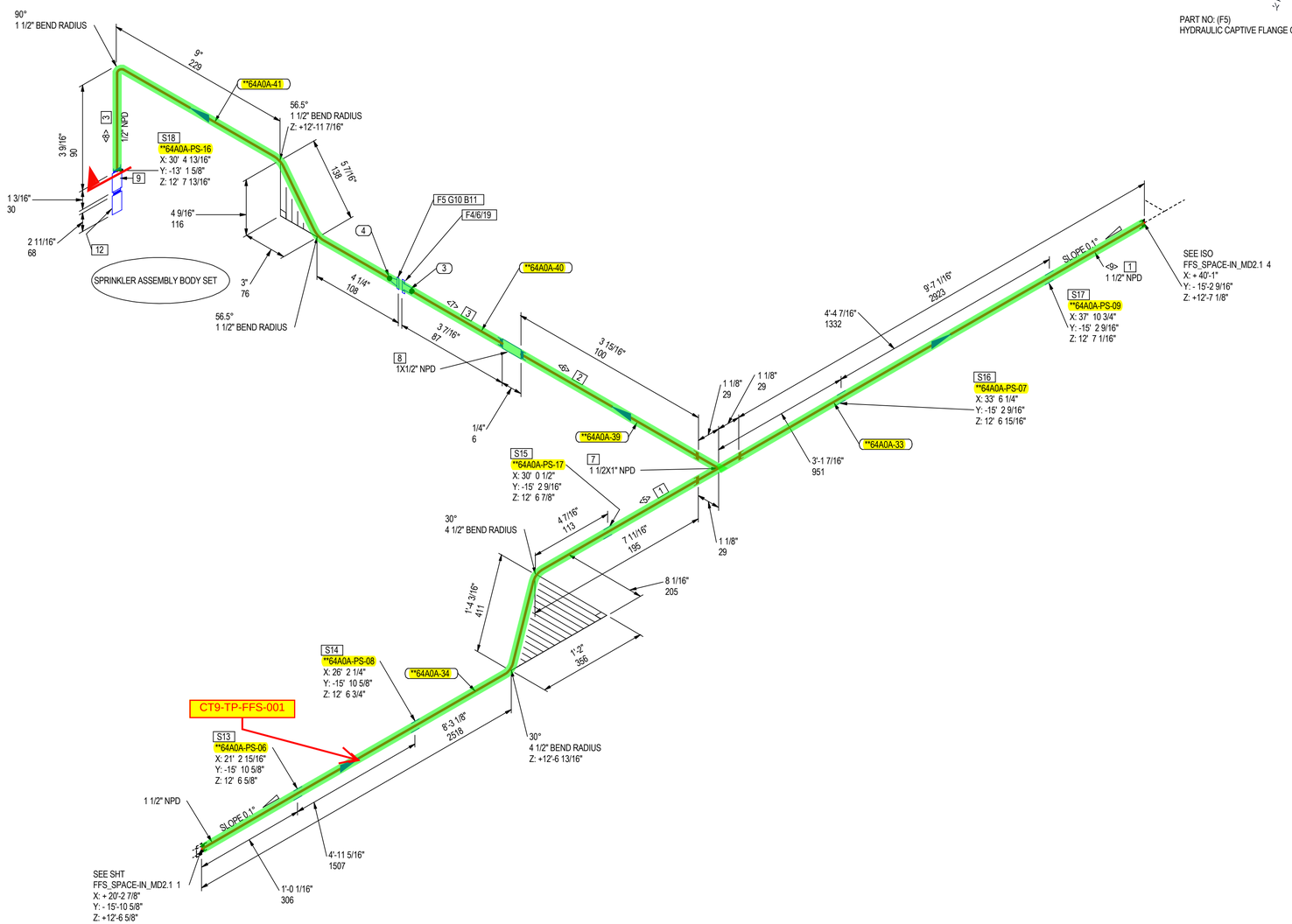




PART NO: (F5)
HYDRAULIC CAPTIVE FLANGE ORIENTATION



ADDITIONAL NOTE:
TUBE REDUCER (FRACTIONAL TO METRIC) SHALL BE IDENTIFIED BY COLOR OR TAGGING FOR QUICK NOTATION IN THE FIELD.

SHOP MATERIALS - FABRICATION ONLY

PT NO	DESCRIPTION	NPD (IN)	CMDTY CODE	QTY
1	TUBE, 0.12 in THK, PE, ASTM A213-TP316, SEAMLESS	1 1/2	MTAAZZBPZZZZZXAQ	19.9'
2	TUBE, 0.083 in THK, PE, ASTM A213-TP316, SEAMLESS	1	MTAAZZBPZZZZZXAQ	0.4'
3	TUBE, 0.049 in THK, PE, ASTM A213-TP316, SEAMLESS	1/2	MTAAZZBPZZZZZXAQ	2.1'
4	FLANGE HYDRAULIC, 3000 PSI, GROOVED END x SWE, SAE J518-1 CODE 61, ASTM A182-F316, FLANGE HEAD	1/2	MFFAAWYFZZIARALS	1
5	FLANGE HYDRAULIC, 3000 PSI, FFFE x SWE, SAE J518-1 CODE 61, ASTM A182-F316, CAPTIVE FLANGE	1/2	MFFBAWYGZZIARALS	1
	MISCELLANEOUS COMPONENTS			
6	FLANGE HYDRAULIC, 3000 PSI, SAE J518-1 CODE 61, ASTM A182-F316, SPLIT FLANGE CLAMP (ONE PAIR)	1/2	MWABAWYHZZIARALS	1

OTHER THAN SHOP MATERIALS - ERECTION ONLY

PT NO	DESCRIPTION	NPD (IN)	CMDTY CODE	QTY
7	TUBE REDUCING UNION TEE, SWGLKE END, SWAGELOK, ASTM A182-F316	1 1/2X1	MWBCZZCGZZAKWALS	1
8	REDUCING TUBE UNION, SWGLKE END, SWAGELOK PART NO. SS-1610-6-8, ASTM A182-F316	1X1/2	MWARZZCGZZAKWALS	1
9	TUBE REDUCER (FRACTIONAL TO METRIC), SWGLKE END x PE, SWAGELOK PART NO.SS-810-R-12M, ASTM A182-F316	1/2	MWAOZZYDZZAKWALS	1
10	GASKET, ORING, 0.125" THK, AS 568, VITON HS90 (DUROMETER90)	1/2	MKBZZACIAQXANUS	1
11	SCREW, DRILLED HEX SOCKET HEAD CAP (2 HOLES), ASME B18.3, ASTM A193-B7, 18-UNC - 1.5 in. Length	5/16	MBCZZBRZZIALBEUZ	4
	INSTRUMENTS			
12	SPRINKLER ASSEMBLY BODY SET FOR HI-FOG® 1000 SERIES SPRINKLER AND SPRAY HEADS, A1000S1CT00D12S10 SPRINKLER ASSEMBLY BODY SET	1/2		1
	PIPE SUPPORTS			
13	PIPE SUPPORT	1 1/2	**64A0A-PS-06	1
14	PIPE SUPPORT	1 1/2	**64A0A-PS-08	1
15	PIPE SUPPORT	1 1/2	**64A0A-PS-17	1
16	PIPE SUPPORT	1 1/2	**64A0A-PS-07	1
17	PIPE SUPPORT	1 1/2	**64A0A-PS-09	1
18	NOZZLE MOUNTING ADAPTER SUPPORT	1/2	**64A0A-PS-16	1

SEE SHT 3

ACTUAL QUANTITY OF WASHERS SHALL BE CONSIDERED AS HALF OF THE QUANTITY SHOWN ON ISOMETRICS

NOTES:
1. THE X AND Y COORDINATES ARE WITH RESPECT TO THE INTERSECTION OF THE CENTERLINE OF THE GAS TURBINE (GT) AND THE INLET MANIFOLD (IM). THE Z (EL) COORDINATE IS WITH RESPECT TO THE TOP OF CONCRETE, WHICH IS 15'-9" [4800mm] BELOW THE CENTERLINE OF THE GT AND IS 0'-0".
2. ALL PIPE SUPPORT COORDINATES ARE WITH RESPECT TO THE CENTERLINE OF THE INSTALLED SUPPORTED PIPE.
3. THE MARK 'FFW' (FIELD FIT WELD) IN THE DRAWING INDICATES EXCESS LENGTH OF APPROX. 4 INCHES IDENTIFIED AS LOOSE IN THE DRAWING TO BE CUT AND ADJUSTED AT THE FIELD.
4. ALL WELDING TO BE DONE AND INSPECTED PER LATEST APPLICABLE EDITION OF ASME B31.1 AND SECTION IX AVAILABLE ON THE PROJECT AWARD DATE.
5. PIPING SHALL BE INSTALLED PER THE INDICATED SLOPE.
6. FOR ANY FURTHER INFORMATION REQUIRED DURING INSTALLATION, CONSULT MITSUBISHI POWER TFA (TECHNICAL FIELD ADVISOR) AT SITE.
7. XXX## ### INDICATES PIPING TERMINAL POINT.

WELD NO	WELD TYPE	WELD CAT	NPD	QTY
3	BW	S	1/2"	1
4	BW	S	1/2"	1

PIPING SPEC:	SDQ	PAINTING SPEC: PER PROJECT SPECIFICATION	REVISION RECORD				
FLUID CODE:	HW, FIRE FIGHTING WATER	INTERIOR SURFACE TREATMENT SPEC: LU-00129	REV. #	DESCRIPTION	BY	CHK	APP	DATE
MAX. PRESS.:	2030.00 psig / 14.00 MPag							
TEST PRESS.:	3045.00 psig / 20.99 MPag							
MAX. TEMP:	100 F / 38 C							
INSULATION SPEC:	NOT INSULATED							

WELD TYPE IDENTIFIER:
BW - BUTT WELD
SW - SOCKET WELD
SOF - SLIP-ON FLANGE
SOB - SET-ON BRANCH
LET - OLET, LATROLET OR HALF COUPLING
TW - TACK WELD
MW - MITER WELD

WELD CATEGORY IDENTIFICATION:
S - SHOP
FW - FIELD WELD
FFW - FIELD FIT WELD

SYMBOLOGY:
WELD NUMBER
PART NUMBER (PT NO)
< > CUT LIST PIECE NUMBER

SYMBOL IDENTIFIER:
F - FLANGE
G - GASKET
B - BOLT
N - NUT
W - WASHER
O - O-RING
L - LOCK WIRE

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MITSUBISHI POWER

ORIGIN:	MITSUBISHI POWER-LAKE MARY	DATE:	MITSUBISHI POWER DRAWING NUMBER:	REV. NO.:	SYSTEM CODE:
PREPARED BY:	E.BALATLI	07/23/2024	-	0	64A0A
REVIEWED BY:	M.FAZLIOGLU	07/24/2024	FIRE FIGHTING SYSTEM PIPING		
APPROVED BY:	N.KOZAN	07/25/2024	INTERNAL TO GT PACKAGE		
			FOR FABRICATION & ERECTION		
			GRAND RIVER ENERGY CENTER UNIT 4		
			GRAND RIVER DAM AUTHORITY		
			M501JAC GAS TURBINE		
			PROJECT CODE: K-GREC1	P&ID:	
			PIPELINE: FFS_SPACE-IN_MD2.1	PAGE NUMBER: 2 OF 5	

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PLANT ENGINEERING

DEPARTMENT:
SECTION:

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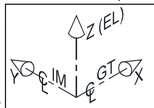

SEE SHT 2

MISCELLANEOUS COMPONENTS

19	LOCK WIRE, 1mm DIA x 1m LENGTH, ASTM A580-304	1/2	MWRZZZZBAIASXAPU	1

SPOOLS

**64A0A-33, **64A0A-34, **64A0A-39, **64A0A-40, **64A0A-41

CUT PIPE LENGTH

PIECE NO	LENGTH (FT-INS)	NPD (IN)	REMARKS	END ONE	END TWO	ITEM CODE
<5>	10'2 7/8"	1 1/2	PLD BEND(2)	SQ.CUT	SQ.CUT	MTAAZZBPZZZZZXAQ
<6>	3 15/16"	1		SQ.CUT	SQ.CUT	MTAAZZBPZZZZZXAQ
<7>	3 3/16"	1/2		SQ.CUT	SQ.CUT	MTAAZZBPZZZZZXAQ
<8>	1'9 1/16"	1/2	PLD BEND(3)	BEVEL	SQ.CUT	MTAAZZBPZZZZZXAQ
<9>	9'7 1/16"	1 1/2		SQ.CUT	SQ.CUT	MTAAZZBPZZZZZXAQ

ACTUAL QUANTITY OF WASHERS SHALL BE CONSIDERED AS HALF OF THE QUANTITY SHOWN ON ISOMETRICS

NOTES:
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2. ALL PIPE SUPPORT COORDINATES ARE WITH RESPECT TO THE CENTERLINE OF THE INSTALLED SUPPORTED PIPE.
3. THE MARK 'FFW' (FIELD FIT WELD) IN THE DRAWING INDICATES EXCESS LENGTH OF APPROX. 4 INCHES IDENTIFIED AS LOOSE IN THE DRAWING TO BE CUT AND ADJUSTED AT THE FIELD.
4. ALL WELDING TO BE DONE AND INSPECTED PER LATEST APPLICABLE EDITION OF ASME B31.1 AND SECTION IX AVAILABLE ON THE PROJECT AWARD DATE.
5. PIPING SHALL BE INSTALLED PER THE INDICATED SLOPE.
6. FOR ANY FURTHER INFORMATION REQUIRED DURING INSTALLATION, CONSULT MITSUBISHI POWER TFA (TECHNICAL FIELD ADVISOR) AT SITE.
7. <XXX## ###> INDICATES PIPING TERMINAL POINT.

ORIGIN:	MITSUBISHI POWER-LAKE MARY	DATE:	MITSUBISHI POWER DRAWING NUMBER:	REV. NO.:	SYSTEM CODE:
PREPARED BY:	E.BALATLI	07/23/2024	-	0	64A0A
REVIEWED BY:	M.FAZLIOGLU	07/24/2024	FIRE FIGHTING SYSTEM PIPING		
APPROVED BY:	N.KOZAN	07/25/2024	INTERNAL TO GT PACKAGE		

FOR FABRICATION & ERECTION
GRAND RIVER ENERGY CENTER UNIT 4
GRAND RIVER DAM AUTHORITY
M501JAC GAS TURBINE

PROJECT CODE:	K-GREC1	P&ID:	
PIPELINE:		PAGE NUMBER:	
FFS_SPACE-IN_MD2.1		3 OF 5	

PIPING SPEC:	SDQ
FLUID CODE:	HW, FIRE FIGHTING WATER
MAX. PRESS.:	2030.00 psig / 14.00 MPag
TEST PRESS.:	3045.00 psig / 20.99 MPag
MAX. TEMP:	100 F / 38 C
INSULATION SPEC:	NOT INSULATED

PAINTING SPEC: PER PROJECT SPECIFICATION
INTERIOR SURFACE TREATMENT SPEC: LU-00129

REV. #	DESCRIPTION	BY	CHK	APP	DATE

WELD TYPE IDENTIFIER:
BW - BUTT WELD
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SOF - SLIP-ON FLANGE
SOB - SET-ON BRANCH
LET - OLET, LATROLET OR HALF COUPLING
TW - TACK WELD
MW - MITER WELD

WELD CATEGORY IDENTIFICATION:
S - SHOP
FW - FIELD WELD
FFW - FIELD FIT WELD

SYMBOLOGY:
WELD NUMBER
PART NUMBER (PT NO)
< > CUT LIST PIECE NUMBER

SYMBOL IDENTIFIER:
F - FLANGE
G - GASKET
B - BOLT
N - NUT
W - WASHER
O - O-RING
L - LOCK WIRE

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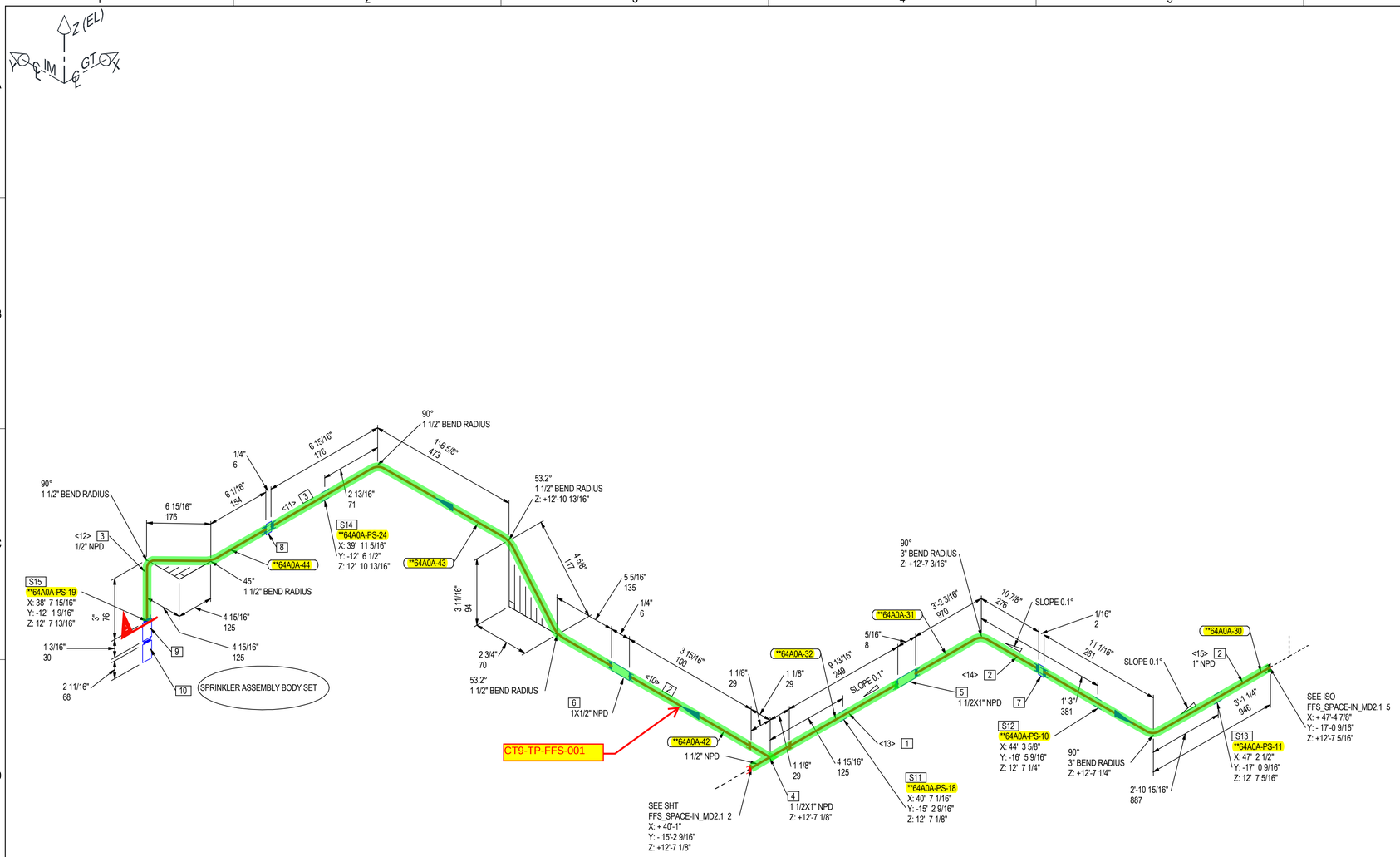

ADDITIONAL NOTE:
TUBE REDUCER (FRACTIONAL TO METRIC) SHALL BE IDENTIFIED BY COLOR OR TAGGING FOR QUICK NOTATION IN THE FIELD.

SHOP MATERIALS - FABRICATION ONLY

PT NO	DESCRIPTION	NPD (IN)	CMDTY CODE	QTY
1	TUBE, 0.12 in THK, PE, ASTM A213-TP316, SEAMLESS	1 1/2	MTAAZZBPZZZZZXAQ	0.9'
2	TUBE, 0.083 in THK, PE, ASTM A213-TP316, SEAMLESS	1	MTAAZZBPZZZZZXAQ	8.3'
3	TUBE, 0.049 in THK, PE, ASTM A213-TP316, SEAMLESS	1/2	MTAAZZBPZZZZZXAQ	4.2'

OTHER THAN SHOP MATERIALS - ERECTION ONLY

PT NO	DESCRIPTION	NPD (IN)	CMDTY CODE	QTY
4	TUBE REDUCING UNION TEE, SWGLKE END, SWAGELOK, ASTM A182-F316	1 1/2X1	MWBCZZCGZZAKWALS	1
5	REDUCING TUBE UNION, SWGLKE END, SWAGELOK PART NO. SS-2400-6-16, ASTM A182-F316	1 1/2X1	MWARZZCGZZAKWALS	1
6	REDUCING TUBE UNION, SWGLKE END, SWAGELOK PART NO. SS-1610-6-8, ASTM A182-F316	1X1/2	MWARZZCGZZAKWALS	1
7	TUBE UNION, SWGLKE END, SWAGELOK PART NO. , ASTM A182-F316	1	MWADZZCGZZAKWALS	1
8	TUBE UNION, SWGLKE END, SWAGELOK PART NO. SS-810-6, ASTM A182-F316	1/2	MWADZZCGZZAKWALS	1
9	TUBE REDUCER (FRACTIONAL TO METRIC), SWGLKE END x PE, SWAGELOK PART NO.SS-810-R-12M, ASTM A182-F316	1/2	MWAOZZYDZZAKWALS	1
	INSTRUMENTS			
10	SPRINKLER ASSEMBLY BODY SET FOR HI-FOG® 1000 SERIES SPRINKLER AND SPRAY HEADS, A1000S1CT00D12S10 SPRINKLER ASSEMBLY BODY SET	1/2		1
	PIPE SUPPORTS			
11	PIPE SUPPORT	1 1/2	**64A0A-PS-18	1
12	PIPE SUPPORT	1	**64A0A-PS-10	1
13	PIPE SUPPORT	1	**64A0A-PS-11	1
14	PIPE SUPPORT	1/2	**64A0A-PS-24	1
15	NOZZLE MOUNTING ADAPTER SUPPORT	1/2	**64A0A-PS-19	1

SPOOLS

**64A0A-30, **64A0A-31, **64A0A-32, **64A0A-42, **64A0A-43, **64A0A-44

CUT PIPE LENGTH

PIECE NO	LENGTH (FT-INS)	NPD (IN)	REMARKS	END ONE	END TWO	ITEM CODE
<10>	3 15/16"	1		SQ.CUT	SQ.CUT	MTAAZZBPZZZZZXAQ
<11>	2'10 11/16"	1/2	PLD BEND(3)	SQ.CUT	SQ.CUT	MTAAZZBPZZZZZXAQ
<12>	1'3 5/16"	1/2	PLD BEND(2)	SQ.CUT	SQ.CUT	MTAAZZBPZZZZZXAQ
<13>	9 13/16"	1 1/2		SQ.CUT	SQ.CUT	MTAAZZBPZZZZZXAQ
<14>	3'11 3/4"	1	PLD BEND(1)	SQ.CUT	SQ.CUT	MTAAZZBPZZZZZXAQ
<15>	3'11"	1	PLD BEND(1)	SQ.CUT	SQ.CUT	MTAAZZBPZZZZZXAQ

ACTUAL QUANTITY OF WASHERS SHALL BE CONSIDERED AS HALF OF THE QUANTITY SHOWN ON ISOMETRICS

NOTES:
1. THE X AND Y COORDINATES ARE WITH RESPECT TO THE INTERSECTION OF THE CENTERLINE OF THE GAS TURBINE (GT) AND THE INLET MANIFOLD (IM). THE Z (EL) COORDINATE IS WITH RESPECT TO THE TOP OF CONCRETE, WHICH IS 15'-9" [4800mm] BELOW THE CENTERLINE OF THE GT AND IS 0'-0".
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4. ALL WELDING TO BE DONE AND INSPECTED PER LATEST APPLICABLE EDITION OF ASME B31.1 AND SECTION IX AVAILABLE ON THE PROJECT AWARD DATE.
5. PIPING SHALL BE INSTALLED PER THE INDICATED SLOPE.
6. FOR ANY FURTHER INFORMATION REQUIRED DURING INSTALLATION, CONSULT MITSUBISHI POWER TFA (TECHNICAL FIELD ADVISOR) AT SITE.
7. XXX## ### INDICATES PIPING TERMINAL POINT.

PIPING SPEC:	SDQ	PAINTING SPEC: PER PROJECT SPECIFICATION			
FLUID CODE:	HW, FIRE FIGHTING WATER	INTERIOR SURFACE TREATMENT SPEC: LU-00129			
MAX. PRESS.:	2030.00 psig / 14.00 MPag				
TEST PRESS.:	3045.00 psig / 20.99 MPag				
MAX. TEMP:	100 F / 38 C				
INSULATION SPEC:	NOT INSULATED				

REVISION RECORD

REV. #	DESCRIPTION	BY	CHK	APP	DATE

WELD TYPE IDENTIFIER:
BW - BUTT WELD
SW - SOCKET WELD
SOF - SLIP-ON FLANGE
SOB - SET-ON BRANCH
LET - OLET, LATROLET OR HALF COUPLING
TW - TACK WELD
MW - MITER WELD

WELD CATEGORY IDENTIFICATION:
S - SHOP
FW - FIELD WELD
FFW - FIELD FIT WELD

SYMBOLOGY:
WELD NUMBER
PART NUMBER (PT NO)
< > CUT LIST PIECE NUMBER

SYMBOL IDENTIFIER:
F - FLANGE
G - GASKET
B - BOLT
N - NUT
W - WASHER
O - O-RING
L - LOCK WIRE

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MITSUBISHI POWER

ORIGIN:	MITSUBISHI POWER-LAKE MARY	DATE:	MITSUBISHI POWER DRAWING NUMBER:	REV. NO.:	SYSTEM CODE:
PREPARED BY:	E.BALATLI	07/23/2024	-	0	64A0A
REVIEWED BY:	M.FAZLIOGLU	07/24/2024	FIRE FIGHTING SYSTEM PIPING		
APPROVED BY:	N.KOZAN	07/25/2024	INTERNAL TO GT PACKAGE		
			FOR FABRICATION & ERECTION		
			GRAND RIVER ENERGY CENTER UNIT 4		
			GRAND RIVER DAM AUTHORITY		
			M501JAC GAS TURBINE		
PROJECT CODE:	K-GREC1	P&ID:			
PIPELINE:	FFS_SPACE-IN_MD2.1	PAGE NUMBER:	4 OF 5		

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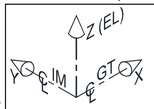
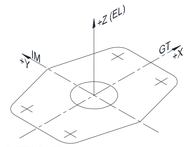
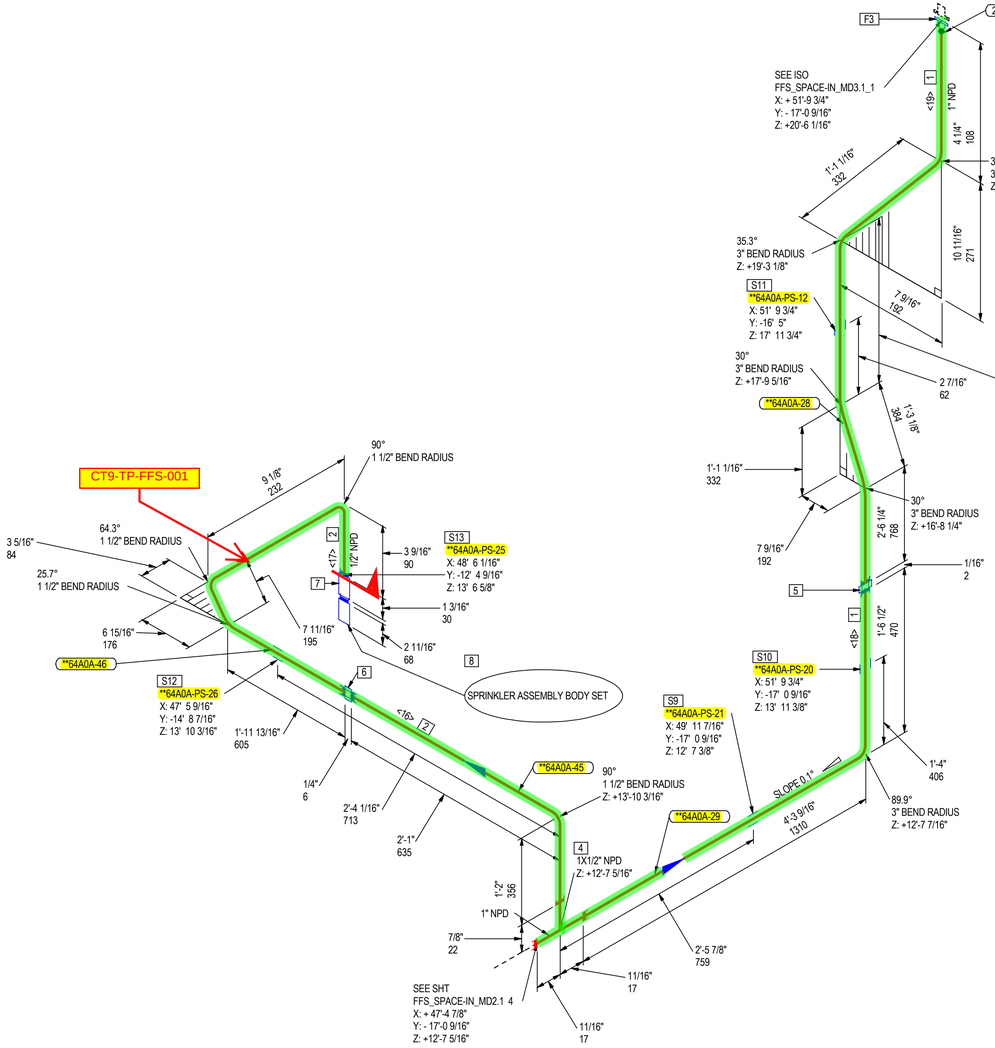




PART NO: (F3)
HYDRAULIC CAPTIVE FLANGE ORIENTATION



SHOP MATERIALS - FABRICATION ONLY

PT NO	DESCRIPTION	NPD (IN)	CMDTY CODE	QTY
1	TUBE, 0.083 in THK, PE, ASTM A213-TP316, SEAMLESS	1	MTAAZZBPZZZZZXAQ	12.4'
2	TUBE, 0.049 in THK, PE, ASTM A213-TP316, SEAMLESS	1/2	MTAAZZBPZZZZZXAQ	6.9'
3	FLANGE HYDRAULIC, 3000 PSI, FFFE x SWE, SAE J518-1 CODE 61, ASTM A182-F316, CAPTIVE FLANGE	1	MFFBAWYGZZIARALS	1

OTHER THAN SHOP MATERIALS - ERECTION ONLY

PT NO	DESCRIPTION	NPD (IN)	CMDTY CODE	QTY
4	TUBE REDUCING UNION TEE, SWGLKE END, SWAGELOK, ASTM A182-F316	1X1/2	MWBCZZCGZZAKWALS	1
5	TUBE UNION, SWGLKE END, SWAGELOK PART NO. , ASTM A182-F316	1	MWADZZCGZZAKWALS	1
6	TUBE UNION, SWGLKE END, SWAGELOK PART NO. SS-810-6, ASTM A182-F316	1/2	MWADZZCGZZAKWALS	1
7	TUBE REDUCER (FRACTIONAL TO METRIC), SWGLKE END x PE, SWAGELOK PART NO.SS-810-R-12M, ASTM A182-F316	1/2	MWAOZZYDZZAKWALS	1
	INSTRUMENTS			
8	SPRINKLER ASSEMBLY BODY SET FOR HI-FOG® 1000 SERIES SPRINKLER AND SPRAY HEADS, A1000S1CT00D12S10 SPRINKLER ASSEMBLY BODY SET	1/2		1
	PIPE SUPPORTS			
9	PIPE SUPPORT	1	**64A0A-PS-21	1
10	PIPE SUPPORT	1	**64A0A-PS-20	1
11	PIPE SUPPORT	1	**64A0A-PS-12	1
12	PIPE SUPPORT	1/2	**64A0A-PS-26	1
13	NOZZLE MOUNTING ADAPTER SUPPORT	1/2	**64A0A-PS-25	1

SPOOLS

**64A0A-28, **64A0A-29, **64A0A-45, **64A0A-46

CUT PIPE LENGTH

PIECE NO	LENGTH (FT-INS)	NPD (IN)	REMARKS	END ONE	END TWO	ITEM CODE
<16>	3'2 3/8"	1/2	PLD BEND(1)	SQ.CUT	SQ.CUT	MTAAZZBPZZZZZXAQ
<17>	3'7 7/16"	1/2	PLD BEND(3)	SQ.CUT	SQ.CUT	MTAAZZBPZZZZZXAQ
<18>	5'8 3/4"	1	PLD BEND(1)	SQ.CUT	SQ.CUT	MTAAZZBPZZZZZXAQ
<19>	6'8"	1	PLD BEND(4)	SQ.CUT	BEVEL	MTAAZZBPZZZZZXAQ

ACTUAL QUANTITY OF WASHERS SHALL BE CONSIDERED AS HALF OF THE QUANTITY SHOWN ON ISOMETRICS

NOTES:
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7. XXX## ### INDICATES PIPING TERMINAL POINT.

ADDITIONAL NOTE:
TUBE REDUCER (FRACTIONAL TO METRIC) SHALL BE IDENTIFIED BY COLOR OR TAGGING FOR QUICK NOTATION IN THE FIELD.

WELD NO	WELD TYPE	WELD CAT	NPD	QTY
2	BW	S	1"	1

PIPING SPEC:	SDQ	PAINTING SPEC: PER PROJECT SPECIFICATION	REVISION RECORD					
FLUID CODE:	HW, FIRE FIGHTING WATER	INTERIOR SURFACE TREATMENT SPEC: LU-00129	REV. #	DESCRIPTION	BY	CHK	APP	DATE
MAX. PRESS.:	2030.00 psig / 14.00 MPag							
TEST PRESS.:	3045.00 psig / 20.99 MPag							
MAX. TEMP:	100 F / 38 C							
INSULATION SPEC:	NOT INSULATED							

WELD TYPE IDENTIFIER:
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S - SHOP
FW - FIELD WELD
FFW - FIELD FIT WELD

SYMBOLOGY:
WELD NUMBER
PART NUMBER (PT NO)
< > CUT LIST PIECE NUMBER

SYMBOL IDENTIFIER:
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G - GASKET
B - BOLT
N - NUT
W - WASHER
O - O-RING
L - LOCK WIRE

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MITSUBISHI POWER

ORIGIN:	MITSUBISHI POWER-LAKE MARY	DATE:	MITSUBISHI POWER DRAWING NUMBER:	REV. NO.:	SYSTEM CODE:
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REVIEWED BY:	M.FAZLIOGLU	07/24/2024	FIRE FIGHTING SYSTEM PIPING		
APPROVED BY:	N.KOZAN	07/25/2024	INTERNAL TO GT PACKAGE		
			FOR FABRICATION & ERECTION		
			GRAND RIVER ENERGY CENTER UNIT 4		
			GRAND RIVER DAM AUTHORITY		
			M501JAC GAS TURBINE		
PROJECT CODE:	K-GREC1	P&ID:			
PIPELINE:	FFS_SPACE-IN_MD2.1	PAGE NUMBER:	5 OF 5		

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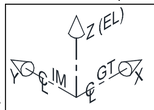

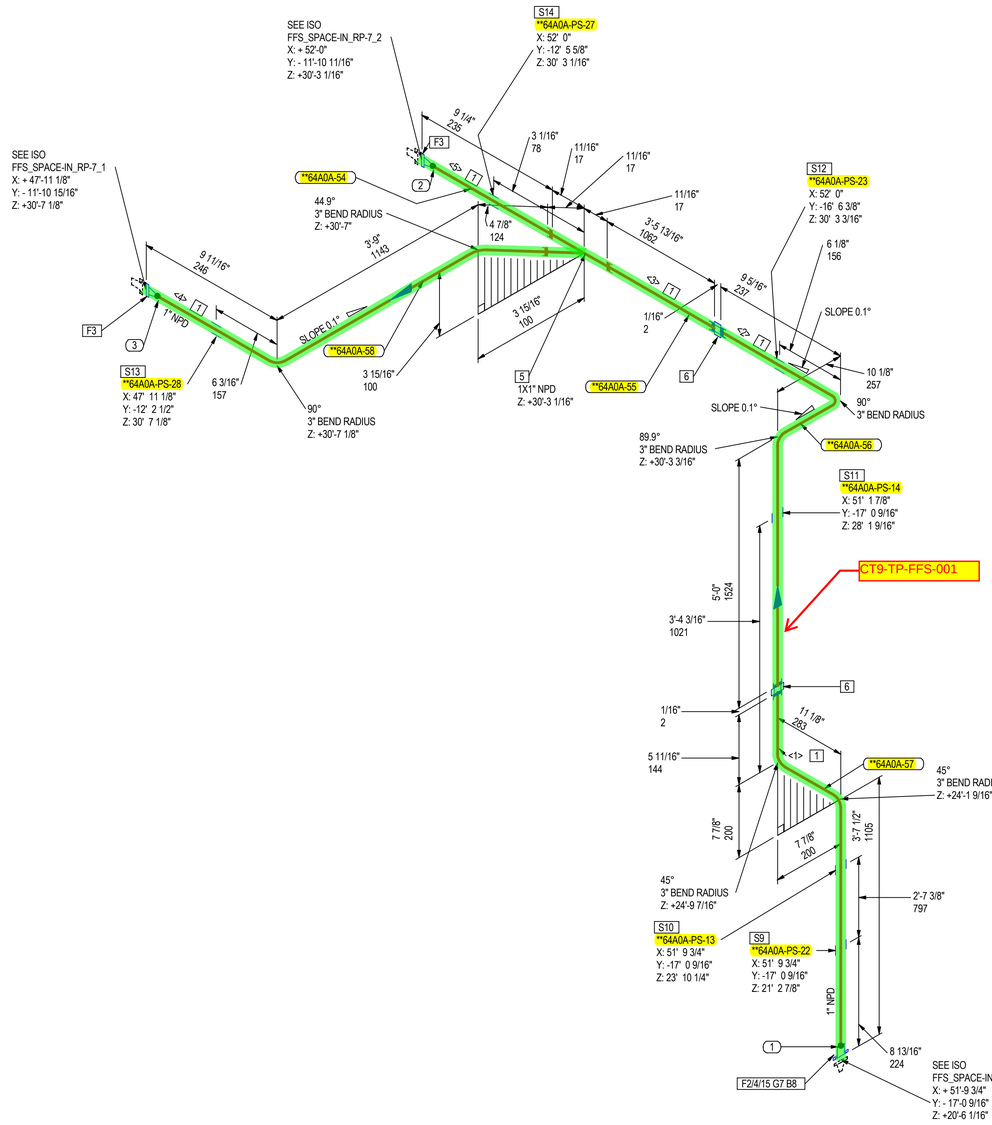

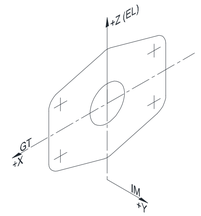

PART NO: (F3)
HYDRAULIC CAPTIVE FLNAGE ORIENTATION

SHOP MATERIALS - FABRICATION ONLY

PT NO	DESCRIPTION	NPD (IN)	CMDTY CODE	QTY
1	TUBE, 0.083 in THK, PE, ASTM A213-TP316, SEAMLESS	1	MTAAZZBPZZZZZXAQ	20.5'
2	FLANGE HYDRAULIC, 3000 PSI, GROOVED END x SWE, SAE J518-1 CODE 61, ASTM A182-F316, FLANGE HEAD	1	MFFAAWYFZZIARALS	1
3	FLANGE HYDRAULIC, 3000 PSI, FFFE x SWE, SAE J518-1 CODE 61, ASTM A182-F316, CAPTIVE FLANGE	1	MFFBAWYGZZIARALS	2
	MISCELLANEOUS COMPONENTS			
4	FLANGE HYDRAULIC, 3000 PSI, SAE J518-1 CODE 61, ASTM A182-F316, SPLIT FLANGE CLAMP (ONE PAIR)	1	MWABAWYHZZIARALS	1

OTHER THAN SHOP MATERIALS - ERECTION ONLY

PT NO	DESCRIPTION	NPD (IN)	CMDTY CODE	QTY
5	TUBE UNION TEE, SWGLKE END, SWAGELOK PART NO. , ASTM A182-F316	1X1	MWAEZZCGZZAKWALS	1
6	TUBE UNION, SWGLKE END, SWAGELOK PART NO. , ASTM A182-F316	1	MWADZZCGZZAKWALS	2
7	GASKET, ORING, 0.125" THK, AS 568, VITON HS90 (DUROMETER90)	1	MKBZZACIAQXANUS	1
8	SCREW, DRILLED HEX SOCKET HEAD CAP (2 HOLES), ASME B18.3, ASTM A193-B7, 16-UNC - 1.75 in. Length	3/8	MBCZZABZZIALBEUZ	4
	PIPE SUPPORTS			
9	PIPE SUPPORT	1	**64A0A-PS-22	1
10	PIPE SUPPORT	1	**64A0A-PS-13	1
11	PIPE SUPPORT	1	**64A0A-PS-14	1
12	PIPE SUPPORT	1	**64A0A-PS-23	1
13	PIPE SUPPORT	1	**64A0A-PS-28	1
14	PIPE SUPPORT	1	**64A0A-PS-27	1
	MISCELLANEOUS COMPONENTS			
15	LOCK WIRE, 1mm DIA x 1m LENGTH, ASTM A580-304	1	MWRZZZZBAIASXAPU	1
	SPOOLS			

**64A0A-54, **64A0A-55, **64A0A-56, **64A0A-57, **64A0A-58

CUT PIPE LENGTH

PIECE NO	LENGTH (FT-INS)	NPD (IN)	REMARKS	END ONE	END TWO	ITEM CODE
<1>	4'11 13/16"	1	PLD BEND(2)	SQ.CUT	SQ.CUT	MTAAZZBPZZZZZXAQ
<2>	6'4 13/16"	1	PLD BEND(2)	SQ.CUT	SQ.CUT	MTAAZZBPZZZZZXAQ
<3>	3'5 13/16"	1		SQ.CUT	SQ.CUT	MTAAZZBPZZZZZXAQ
<4>	4'9 13/16"	1	PLD BEND(2)	SQ.CUT	BEVEL	MTAAZZBPZZZZZXAQ
<5>	8 15/16"	1		SQ.CUT	BEVEL	MTAAZZBPZZZZZXAQ

ACTUAL QUANTITY OF WASHERS SHALL BE CONSIDERED AS HALF OF THE QUANTITY SHOWN ON ISOMETRICS

NOTES:
1. THE X AND Y COORDINATES ARE WITH RESPECT TO THE INTERSECTION OF THE CENTERLINE OF THE GAS TURBINE (GT) AND THE INLET MANIFOLD (IM). THE Z (EL) COORDINATE IS WITH RESPECT TO THE TOP OF CONCRETE, WHICH IS 15'-9" [4800mm] BELOW THE CENTERLINE OF THE GT AND IS 0'-0".
2. ALL PIPE SUPPORT COORDINATES ARE WITH RESPECT TO THE CENTERLINE OF THE INSTALLED SUPPORTED PIPE.
3. THE MARK 'FFW' (FIELD FIT WELD) IN THE DRAWING INDICATES EXCESS LENGTH OF APPROX. 4 INCHES IDENTIFIED AS LOOSE IN THE DRAWING TO BE CUT AND ADJUSTED AT THE FIELD.
4. ALL WELDING TO BE DONE AND INSPECTED PER LATEST APPLICABLE EDITION OF ASME B31.1 AND SECTION IX AVAILABLE ON THE PROJECT AWARD DATE.
5. PIPING SHALL BE INSTALLED PER THE INDICATED SLOPE.
6. FOR ANY FURTHER INFORMATION REQUIRED DURING INSTALLATION, CONSULT MITSUBISHI POWER TFA (TECHNICAL FIELD ADVISOR) AT SITE.
7. ⟨XXX## ###⟩ INDICATES PIPING TERMINAL POINT.

WELD NO	WELD TYPE	WELD CAT	NPD	QTY
1	BW	S	1"	1
3	BW	S	1"	1
2	BW	S	1"	1

PIPING SPEC:	SDQ	PAINTING SPEC: PER PROJECT SPECIFICATION	REVISION RECORD					
FLUID CODE:	HW, FIRE FIGHTING WATER	INTERIOR SURFACE TREATMENT SPEC: LU-00129	REV. #	DESCRIPTION	BY	CHK	APP	DATE
MAX. PRESS.:	2030.00 psig / 14.00 MPag							
TEST PRESS.:	3045.00 psig / 20.99 MPag							
MAX. TEMP:	100 F / 38 C							
INSULATION SPEC:	NOT INSULATED							

WELD TYPE IDENTIFIER:
BW - BUTT WELD
SW - SOCKET WELD
SOF - SLIP-ON FLANGE
SOB - SET-ON BRANCH
LET - OLET, LATROLET OR HALF COUPLING
TW - TACK WELD
MW - MITER WELD

WELD CATEGORY IDENTIFICATION:
S - SHOP
FW - FIELD WELD
FFW - FIELD FIT WELD

SYMBOLOGY:
WELD NUMBER
PART NUMBER (PT NO)
< > CUT LIST PIECE NUMBER

SYMBOL IDENTIFIER:
F - FLANGE
G - GASKET
B - BOLT
N - NUT
W - WASHER
O - O-RING
L - LOCK WIRE

ORIGIN:	MITSUBISHI POWER-LAKE MARY	DATE:	MITSUBISHI POWER DRAWING NUMBER:	REV. NO.:	SYSTEM CODE:
PREPARED BY:	E.BALATLI	07/23/2024	-	0	64A0A
REVIEWED BY:	M.FAZLIOGLU	07/24/2024	FIRE FIGHTING SYSTEM PIPING		
APPROVED BY:	N.KOZAN	07/25/2024	INTERNAL TO GT PACKAGE		
			FOR FABRICATION & ERECTION		
			GRAND RIVER ENERGY CENTER UNIT 4		
			GRAND RIVER DAM AUTHORITY		
			M501JAC GAS TURBINE		
			PROJECT CODE: K-GREC1	P&ID:	
			PIPELINE:	PAGE NUMBER:	
			FFS_SPACE-IN_MD3.1_1	1 OF 1	

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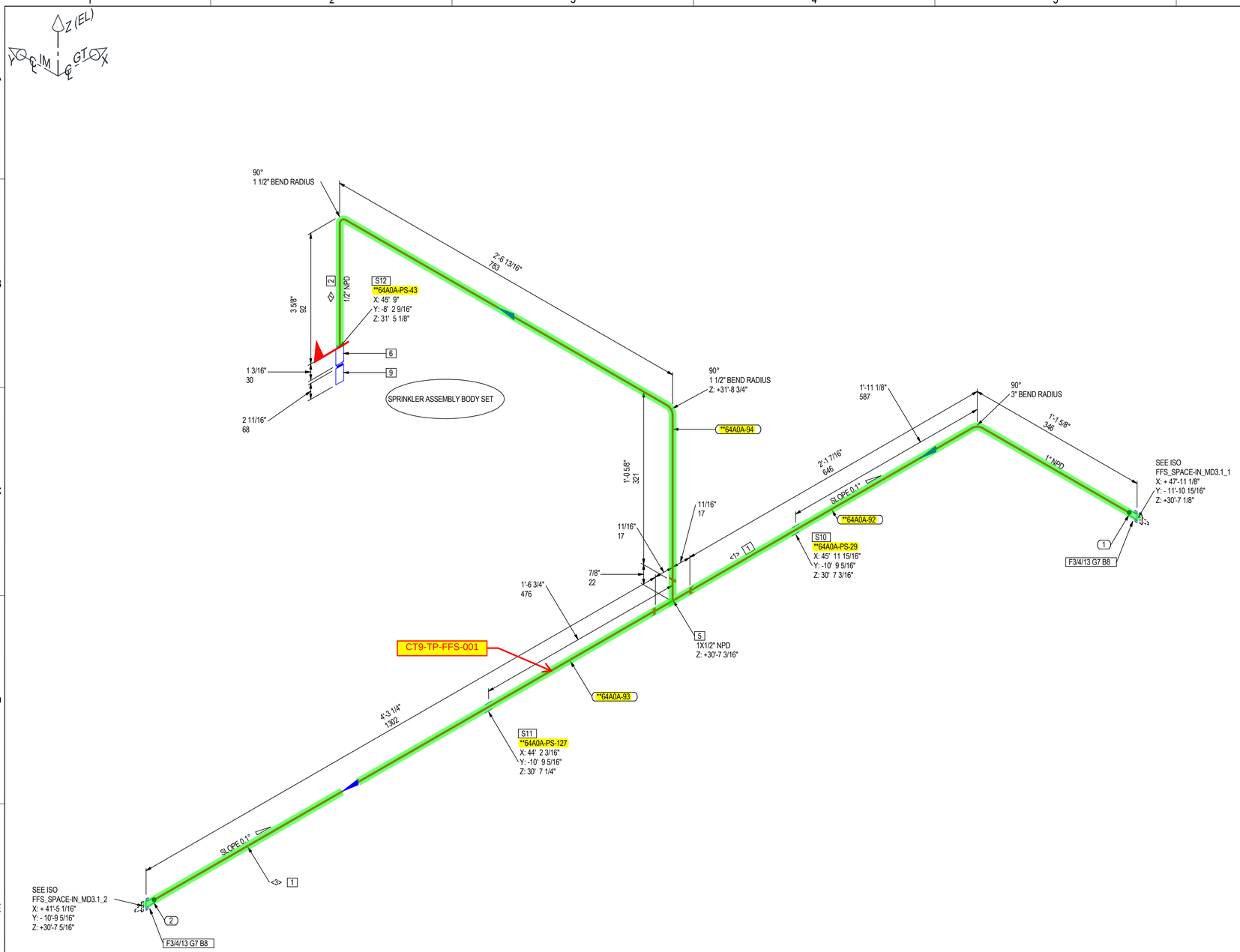

SHOP MATERIALS - FABRICATION ONLY

PT NO	DESCRIPTION	NPD (IN)	CMDTY CODE	QTY
1	TUBE, 0.083 in THK, PE, ASTM A213-TP316, SEAMLESS	1	MTAAZZBPZZZZZXAQ	7.4'
2	TUBE, 0.049 in THK, PE, ASTM A213-TP316, SEAMLESS	1/2	MTAAZZBPZZZZZXAQ	3.9'
3	FLANGE HYDRAULIC, 3000 PSI, GROOVED END x SWE, SAE J518-1 CODE 61, ASTM A182-F316, FLANGE HEAD	1	MFFAAWYFZZIARALS	2
	MISCELLANEOUS COMPONENTS			
4	FLANGE HYDRAULIC, 3000 PSI, SAE J518-1 CODE 61, ASTM A182-F316, SPLIT FLANGE CLAMP (ONE PAIR)	1	MWABAWYHZZIARALS	2

OTHER THAN SHOP MATERIALS - ERECTION ONLY

PT NO	DESCRIPTION	NPD (IN)	CMDTY CODE	QTY
5	TUBE REDUCING UNION TEE, SWGLKE END, SWAGELOK, ASTM A182-F316	1X1/2	MWBCZZCGZZAKWALS	1
6	TUBE REDUCER (FRACTIONAL TO METRIC), SWGLKE END x PE, SWAGELOK PART NO.SS-810-R-12M, ASTM A182-F316	1/2	MWAOZZYDZZAKWALS	1
7	GASKET, ORING, 0.125" THK, AS 568, VITON HS90 (DUROMETER90)	1	MKBZZACIAQXANUS	2
8	SCREW, DRILLED HEX SOCKET HEAD CAP (2 HOLES), ASME B18.3, ASTM A193-B7, 16-UNC - 1.75 in. Length	3/8	MBCZZABZZIALBEUZ	8
	INSTRUMENTS			
9	SPRINKLER ASSEMBLY BODY SET FOR HI-FOG® 1000 SERIES SPRINKLER AND SPRAY HEADS, A1000S1CT00D12S10 SPRINKLER ASSEMBLY BODY SET	1/2		1
	PIPE SUPPORTS			
10	PIPE SUPPORT	1	**64A0A-PS-29	1
11	PIPE SUPPORT	1	**64A0A-PS-127	1
12	NOZZLE MOUNTING ADAPTER SUPPORT	1/2	**64A0A-PS-43	1
	MISCELLANEOUS COMPONENTS			
13	LOCK WIRE, 1mm DIA x 1m LENGTH, ASTM A580-304	1	MWRZZZZBAIASXAPU	2

SPOOLS

**64A0A-92, **64A0A-93, **64A0A-94

CUT PIPE LENGTH

PIECE NO	LENGTH (FT-INS)	NPD (IN)	REMARKS	END ONE	END TWO	ITEM CODE
<1>	3'1 9/16"	1	PLD BEND(1)	SQ.CUT	SQ.CUT	MTAAZZBPZZZZZXAQ
<2>	3'9 3/4"	1/2	PLD BEND(2)	SQ.CUT	SQ.CUT	MTAAZZBPZZZZZXAQ
<3>	4'3"	1		SQ.CUT	SQ.CUT	MTAAZZBPZZZZZXAQ

ACTUAL QUANTITY OF WASHERS SHALL BE CONSIDERED AS HALF OF THE QUANTITY SHOWN ON ISOMETRICS

NOTES:
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4. ALL WELDING TO BE DONE AND INSPECTED PER LATEST APPLICABLE EDITION OF ASME B31.1 AND SECTION IX AVAILABLE ON THE PROJECT AWARD DATE.
5. PIPING SHALL BE INSTALLED PER THE INDICATED SLOPE.
6. FOR ANY FURTHER INFORMATION REQUIRED DURING INSTALLATION, CONSULT MITSUBISHI POWER TFA (TECHNICAL FIELD ADVISOR) AT SITE.
7. <XXX## ###> INDICATES PIPING TERMINAL POINT.

ADDITIONAL NOTE:
TUBE REDUCER (FRACTIONAL TO METRIC) SHALL BE IDENTIFIED BY COLOR OR TAGGING FOR QUICK NOTATION IN THE FIELD.

WELD NO	WELD TYPE	WELD CAT	NPD	QTY
1	BW	S	1"	1
2	BW	S	1"	1

PIPING SPEC:	SDQ	PAINTING SPEC: PER PROJECT SPECIFICATION	REVISION RECORD			
FLUID CODE:	HW, FIRE FIGHTING WATER	INTERIOR SURFACE TREATMENT SPEC: LU-00129	REV. #	DESCRIPTION	BY	CHK	APP	DATE
MAX. PRESS.:	2030.00 psig / 14.00 MPag							
TEST PRESS.:	3045.00 psig / 20.99 MPag							
MAX. TEMP:	100 F / 38 C							
INSULATION SPEC:	NOT INSULATED							

WELD TYPE IDENTIFIER:
BW - BUTT WELD
SW - SOCKET WELD
SOF - SLIP-ON FLANGE
SOB - SET-ON BRANCH
LET - OLET, LATROLET OR HALF COUPLING
TW - TACK WELD
MW - MITER WELD

WELD CATEGORY IDENTIFICATION:
S - SHOP
FW - FIELD WELD
FFW - FIELD FIT WELD

SYMBOLOGY:
WELD NUMBER
PART NUMBER (PT NO)
< > CUT LIST PIECE NUMBER

SYMBOL IDENTIFIER:
F - FLANGE
G - GASKET
B - BOLT
N - NUT
W - WASHER
O - O-RING
L - LOCK WIRE

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MITSUBISHI POWER

ORIGIN:	MITSUBISHI POWER-LAKE MARY	DATE:	MITSUBISHI POWER DRAWING NUMBER:	REV. NO.:	SYSTEM CODE:
PREPARED BY:	E.BALATLI	07/23/2024	-	0	64A0A
REVIEWED BY:	M.FAZLIOGLU	07/24/2024	FIRE FIGHTING SYSTEM PIPING		
APPROVED BY:	N.KOZAN	07/25/2024	INTERNAL TO GT PACKAGE		
			FOR FABRICATION & ERECTION		
			GRAND RIVER ENERGY CENTER UNIT 4		
			GRAND RIVER DAM AUTHORITY		
			M501JAC GAS TURBINE		
PROJECT CODE:	K-GREC1	P&ID:			
PIPELINE:	FFS_SPACE-IN_RP-7_1	PAGE NUMBER:	1 OF 1		

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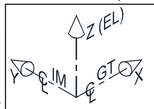
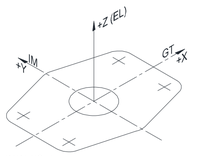
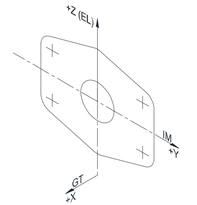
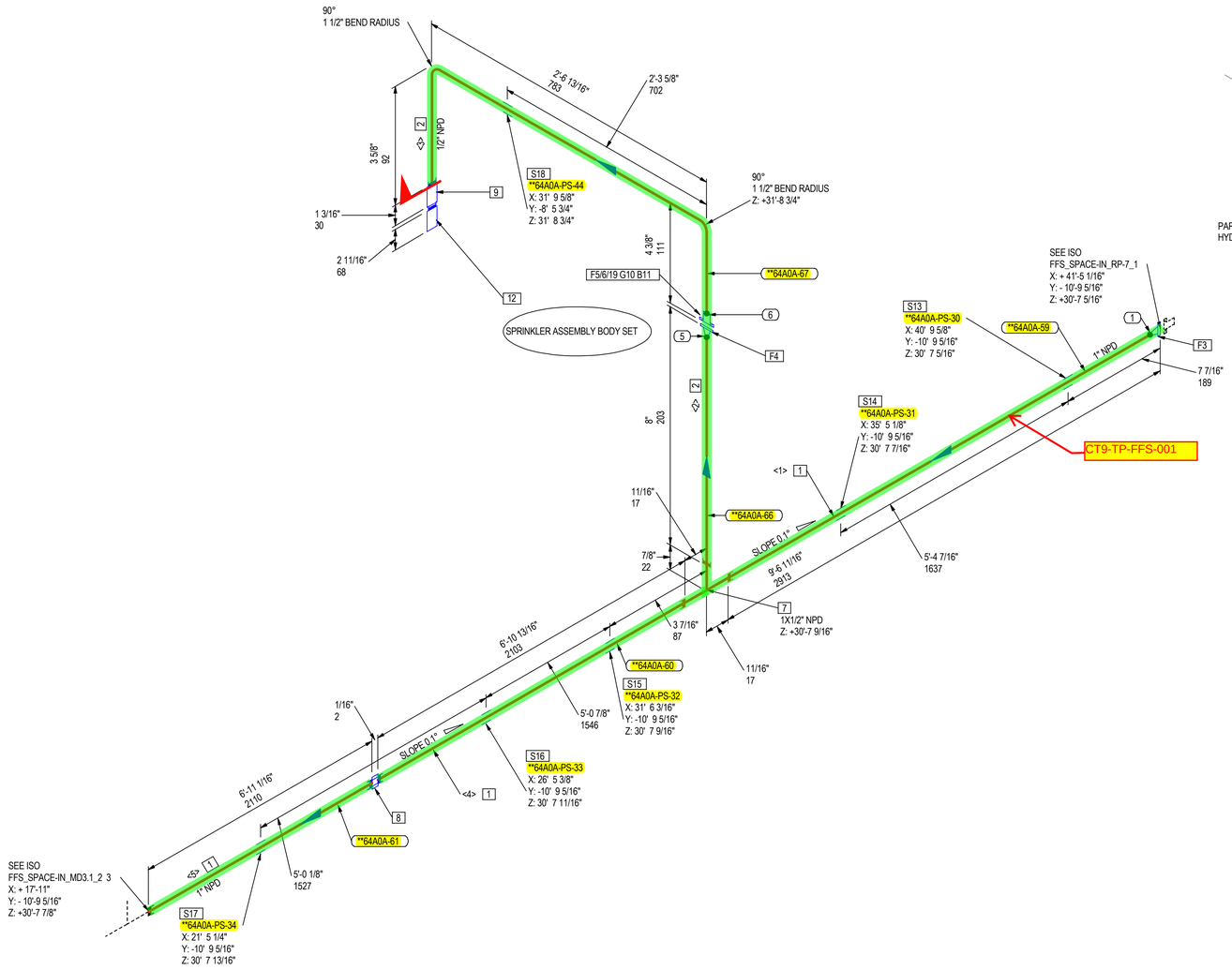




PART NO: (F4)
HYDRAULIC CAPTIVE FLANGE ORIENTATION



PART NO: (F3)
HYDRAULIC CAPTIVE FLANGE ORIENTATION



ADDITIONAL NOTE:
TUBE REDUCER (FRACTIONAL TO METRIC) SHALL BE IDENTIFIED BY COLOR OR TAGGING FOR QUICK NOTATION IN THE FIELD.

WELD NO	WELD TYPE	WELD CAT	NPD	QTY
1	BW	S	1"	1
5	BW	S	1/2"	1
6	BW	S	1/2"	1

SHOP MATERIALS - FABRICATION ONLY

PT NO	DESCRIPTION	NPD (IN)	CMDTY CODE	QTY
1	TUBE, 0.083 in THK, PE, ASTM A213-TP316, SEAMLESS	1	MTAAZZBPZZZZZXAQ	23.4'
2	TUBE, 0.049 in THK, PE, ASTM A213-TP316, SEAMLESS	1/2	MTAAZZBPZZZZZXAQ	3.8'
3	FLANGE HYDRAULIC, 3000 PSI, FFFE x SWE, SAE J518-1 CODE 61, ASTM A182-F316, CAPTIVE FLANGE	1	MFFBAWYGZZIARALS	1
4	FLANGE HYDRAULIC, 3000 PSI, FFFE x SWE, SAE J518-1 CODE 61, ASTM A182-F316, CAPTIVE FLANGE	1/2	MFFBAWYGZZIARALS	1
5	FLANGE HYDRAULIC, 3000 PSI, GROOVED END x SWE, SAE J518-1 CODE 61, ASTM A182-F316, FLANGE HEAD	1/2	MFFAAWYFZZIARALS	1
	MISCELLANEOUS COMPONENTS			
6	FLANGE HYDRAULIC, 3000 PSI, SAE J518-1 CODE 61, ASTM A182-F316, SPLIT FLANGE CLAMP (ONE PAIR)	1/2	MWABAWYHZZIARALS	1

OTHER THAN SHOP MATERIALS - ERECTION ONLY

PT NO	DESCRIPTION	NPD (IN)	CMDTY CODE	QTY
7	TUBE REDUCING UNION TEE, SWGLKE END, SWAGELOK, ASTM A182-F316	1X1/2	MWBCZZCGZZAKWALS	1
8	TUBE UNION, SWGLKE END, SWAGELOK PART NO. , ASTM A182-F316	1	MWADZZCGZZAKWALS	1
9	TUBE REDUCER (FRACTIONAL TO METRIC), SWGLKE END x PE, SWAGELOK PART NO.SS-810-R-12M, ASTM A182-F316	1/2	MWAOZZYDZZAKWALS	1
10	GASKET, ORING, 0.125" THK, AS 568, VITON HS90 (DUROMETER90)	1/2	MKBZZACIAQXANUS	1
11	SCREW, DRILLED HEX SOCKET HEAD CAP (2 HOLES), ASME B18.3, ASTM A193-B7, 18-UNC - 1.5 in. Length	5/16	MBCZZBRZZIALBEUZ	4
	INSTRUMENTS			
12	SPRINKLER ASSEMBLY BODY SET FOR HI-FOG® 1000 SERIES SPRINKLER AND SPRAY HEADS, A1000S1CT00D12S10 SPRINKLER ASSEMBLY BODY SET	1/2		1
	PIPE SUPPORTS			
13	PIPE SUPPORT	1	**64A0A-PS-30	1
14	PIPE SUPPORT	1	**64A0A-PS-31	1
15	PIPE SUPPORT	1	**64A0A-PS-32	1
16	PIPE SUPPORT	1	**64A0A-PS-33	1
17	PIPE SUPPORT	1	**64A0A-PS-34	1
18	PIPE SUPPORT	1/2	**64A0A-PS-44	1

SEE SHT 2

ACTUAL QUANTITY OF WASHERS SHALL BE CONSIDERED AS HALF OF THE QUANTITY SHOWN ON ISOMETRICS

NOTES:
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5. PIPING SHALL BE INSTALLED PER THE INDICATED SLOPE.
6. FOR ANY FURTHER INFORMATION REQUIRED DURING INSTALLATION, CONSULT MITSUBISHI POWER TFA (TECHNICAL FIELD ADVISOR) AT SITE.
7. XXX##-### INDICATES PIPING TERMINAL POINT.

PIPING SPEC:	SDQ	PAINTING SPEC:	PER PROJECT SPECIFICATION
FLUID CODE:	HW, FIRE FIGHTING WATER	INTERIOR SURFACE TREATMENT SPEC:	LU-00129
MAX. PRESS.:	2030.00 psig / 14.00 MPag		
TEST PRESS.:	3045.00 psig / 20.99 MPag		
MAX. TEMP:	100 F / 38 C		
INSULATION SPEC:	NOT INSULATED		

REV. #	DESCRIPTION	BY	CHK	APP	DATE

WELD TYPE IDENTIFIER:
BW - BUTT WELD
SW - SOCKET WELD
SOF - SLIP-ON FLANGE
SOB - SET-ON BRANCH
LET - OLET, LATROLET OR HALF COUPLING
TW - TACK WELD
MW - MITER WELD

WELD CATEGORY IDENTIFICATION:
S - SHOP
FW - FIELD WELD
FFW - FIELD FIT WELD

SYMBOLOGY:
WELD NUMBER
PART NUMBER (PT NO)
< > CUT LIST PIECE NUMBER

SYMBOL IDENTIFIER:
F - FLANGE
G - GASKET
B - BOLT
N - NUT
W - WASHER
O - O-RING
L - LOCK WIRE

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MITSUBISHI POWER

ORIGIN:	MITSUBISHI POWER-LAKE MARY	DATE:	MITSUBISHI POWER DRAWING NUMBER:	REV. NO.:	SYSTEM CODE:
PREPARED BY:	E.BALATLI	07/23/2024	-	0	64A0A
REVIEWED BY:	M.FAZLIOGLU	07/24/2024	FIRE FIGHTING SYSTEM PIPING		
APPROVED BY:	N.KOZAN	07/25/2024	INTERNAL TO GT PACKAGE		
			FOR FABRICATION & ERECTION		
			GRAND RIVER ENERGY CENTER UNIT 4		
			GRAND RIVER DAM AUTHORITY		
			M501JAC GAS TURBINE		
PROJECT CODE:	K-GREC1	P&ID:			
PIPELINE:		PAGE NUMBER:	FFS_SPACE-IN_MD3.1_2	1 OF 4	

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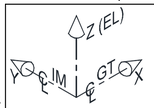

SEE SHT 1

MISCELLANEOUS COMPONENTS

19	LOCK WIRE, 1mm DIA x 1m LENGTH, ASTM A580-304	1/2	MWRZZZZBAIASXAPU	1

SPOOLS

**64A0A-59, **64A0A-60, **64A0A-61, **64A0A-66, **64A0A-67

CUT PIPE LENGTH

PIECE NO	LENGTH (FT-INS)	NPD (IN)	REMARKS	END ONE	END TWO	ITEM CODE
<1>	9'6 3/8"	1		BEVEL	SQ.CUT	MTAAZZBPZZZZZXAQ
<2>	7 3/4"	1/2		SQ.CUT	BEVEL	MTAAZZBPZZZZZXAQ
<3>	3'1 3/16"	1/2	PLD BEND(2)	SQ.CUT	SQ.CUT	MTAAZZBPZZZZZXAQ
<4>	6'10 13/16"	1		SQ.CUT	SQ.CUT	MTAAZZBPZZZZZXAQ
<5>	6'11 1/16"	1		SQ.CUT	SQ.CUT	MTAAZZBPZZZZZXAQ

ACTUAL QUANTITY OF WASHERS SHALL BE CONSIDERED AS HALF OF THE QUANTITY SHOWN ON ISOMETRICS

NOTES:
1. THE X AND Y COORDINATES ARE WITH RESPECT TO THE INTERSECTION OF THE CENTERLINE OF THE GAS TURBINE (GT) AND THE INLET MANIFOLD (IM). THE Z (EL) COORDINATE IS WITH RESPECT TO THE TOP OF CONCRETE, WHICH IS 15'-9" [4800mm] BELOW THE CENTERLINE OF THE GT AND IS 0'-0".
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6. FOR ANY FURTHER INFORMATION REQUIRED DURING INSTALLATION, CONSULT MITSUBISHI POWER TFA (TECHNICAL FIELD ADVISOR) AT SITE.
7. XXX## ### INDICATES PIPING TERMINAL POINT.

ORIGIN:	MITSUBISHI POWER-LAKE MARY	DATE:	MITSUBISHI POWER DRAWING NUMBER:	REV. NO.:	SYSTEM CODE:
PREPARED BY:	E.BALATLI	07/23/2024	-	0	64A0A
REVIEWED BY:	M.FAZLIOGLU	07/24/2024	FIRE FIGHTING SYSTEM PIPING		
APPROVED BY:	N.KOZAN	07/25/2024	INTERNAL TO GT PACKAGE		
			FOR FABRICATION & ERECTION		
			GRAND RIVER ENERGY CENTER UNIT 4		
			GRAND RIVER DAM AUTHORITY		
			M501JAC GAS TURBINE		
			PROJECT CODE: K-GREC1	P&ID:	
			PIPELINE: FFS_SPACE-IN_MD3.1_2	PAGE NUMBER: 2 OF 4	

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MITSUBISHI POWER

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SL2

DEPARTMENT: MITSUBISHI POWER ENGINEERING
SECTION: PLANT ENGINEERING

DRAWING DISTRIBUTION LIST

PIPING SPEC:	SDQ
FLUID CODE:	HW, FIRE FIGHTING WATER
MAX. PRESS.:	2030.00 psig / 14.00 MPag
TEST PRESS.:	3045.00 psig / 20.99 MPag
MAX. TEMP:	100 F / 38 C
INSULATION SPEC:	NOT INSULATED

PAINTING SPEC: PER PROJECT SPECIFICATION
INTERIOR SURFACE TREATMENT SPEC: LU-00129

REVISION RECORD

REV. #	DESCRIPTION	BY	CHK	APP	DATE

WELD TYPE IDENTIFIER:
BW - BUTT WELD
SW - SOCKET WELD
SOF - SLIP-ON FLANGE
SOB - SET-ON BRANCH
LET - OLET, LATROLET OR HALF COUPLING
TW - TACK WELD
MW - MITER WELD

WELD CATEGORY IDENTIFICATION:
S - SHOP
FW - FIELD WELD
FFW - FIELD FIT WELD

SYMBOLOGY:
WELD NUMBER
PART NUMBER (PT NO)
< > CUT LIST PIECE NUMBER

SYMBOL IDENTIFIER:
F - FLANGE
G - GASKET
B - BOLT
N - NUT
W - WASHER
O - O-RING
L - LOCK WIRE

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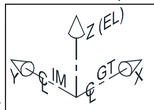
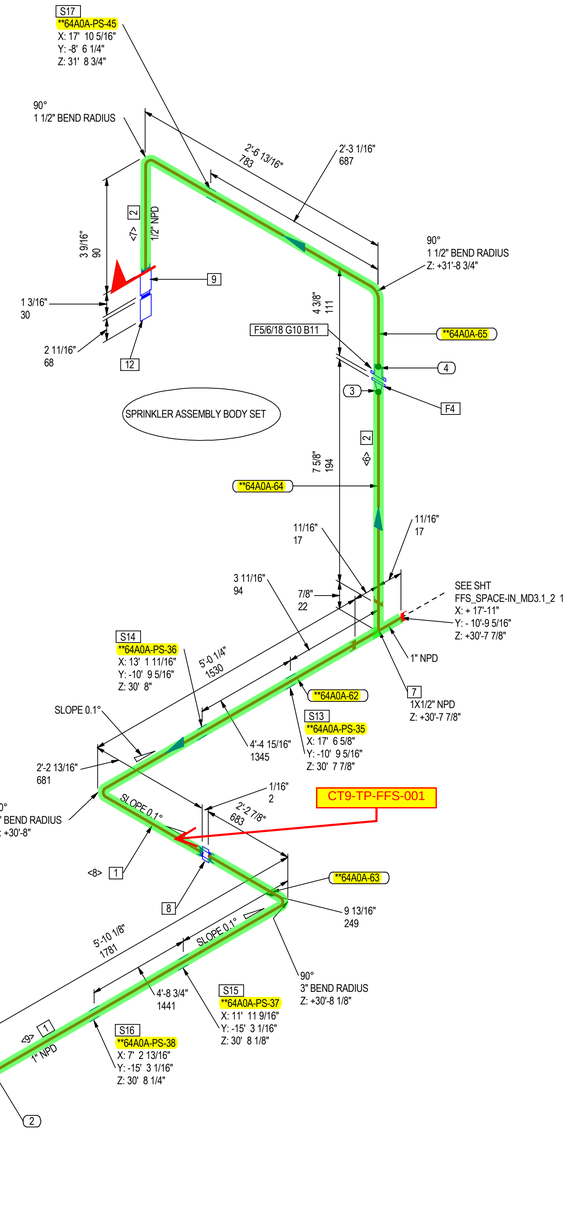
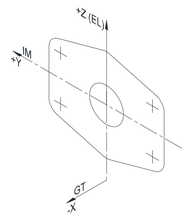
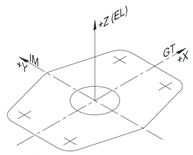






PART NO: (F3)
HYDRAULIC CAPTIVE FLANGE ORIENTATION



PART NO: (F4)
HYDRAULIC CAPTIVE FLANGE ORIENTATION

SHOP MATERIALS - FABRICATION ONLY

PT NO	DESCRIPTION	NPD (IN)	CMDTY CODE	QTY
1	TUBE, 0.083 in THK, PE, ASTM A213-TP316, SEAMLESS	1	MTAAZZBPZZZZZXAQ	15.1'
2	TUBE, 0.049 in THK, PE, ASTM A213-TP316, SEAMLESS	1/2	MTAAZZBPZZZZZXAQ	3.8'
3	FLANGE HYDRAULIC, 3000 PSI, FFFE x SWE, SAE J518-1 CODE 61, ASTM A182-F316, CAPTIVE FLANGE	1	MFFBAWYGZZIARALS	1
4	FLANGE HYDRAULIC, 3000 PSI, FFFE x SWE, SAE J518-1 CODE 61, ASTM A182-F316, CAPTIVE FLANGE	1/2	MFFBAWYGZZIARALS	1
5	FLANGE HYDRAULIC, 3000 PSI, GROOVED END x SWE, SAE J518-1 CODE 61, ASTM A182-F316, FLANGE HEAD	1/2	MFFAAWYFZZIARALS	1
	MISCELLANEOUS COMPONENTS			
6	FLANGE HYDRAULIC, 3000 PSI, SAE J518-1 CODE 61, ASTM A182-F316, SPLIT FLANGE CLAMP (ONE PAIR)	1/2	MWABAWYHZZIARALS	1

OTHER THAN SHOP MATERIALS - ERECTION ONLY

PT NO	DESCRIPTION	NPD (IN)	CMDTY CODE	QTY
7	TUBE REDUCING UNION TEE, SWGLKE END, SWAGELOK, ASTM A182-F316	1X1/2	MWBCZZCGZZAKWALS	1
8	TUBE UNION, SWGLKE END, SWAGELOK PART NO. , ASTM A182-F316	1	MWADZZCGZZAKWALS	1
9	TUBE REDUCER (FRACTIONAL TO METRIC), SWGLKE END x PE, SWAGELOK PART NO.SS-810-R-12M, ASTM A182-F316	1/2	MWAOZZYDZZAKWALS	1
10	GASKET, ORING, 0.125" THK, AS 568, VITON HS90 (DUROMETER90)	1/2	MKBZZACIAQXANUS	1
11	SCREW, DRILLED HEX SOCKET HEAD CAP (2 HOLES), ASME B18.3, ASTM A193-B7, 18-UNC - 1.5 in. Length	5/16	MBCZZBRZZIALBEUZ	4
	INSTRUMENTS			
12	SPRINKLER ASSEMBLY BODY SET FOR HI-FOG® 1000 SERIES SPRINKLER AND SPRAY HEADS, A1000S1CT00D12S10 SPRINKLER ASSEMBLY BODY SET	1/2		1
	PIPE SUPPORTS			
13	PIPE SUPPORT	1	**64A0A-PS-35	1
14	PIPE SUPPORT	1	**64A0A-PS-36	1
15	PIPE SUPPORT	1	**64A0A-PS-37	1
16	PIPE SUPPORT	1	**64A0A-PS-38	1
17	PIPE SUPPORT	1/2	**64A0A-PS-45	1
	MISCELLANEOUS COMPONENTS			
18	LOCK WIRE, 1mm DIA x 1m LENGTH, ASTM A580-304	1/2	MWRZZZZBAIASXAPU	1

SPOOLS

**64A0A-62, **64A0A-63, **64A0A-64, **64A0A-65

SEE SHT 4

ACTUAL QUANTITY OF WASHERS SHALL BE CONSIDERED AS HALF OF THE QUANTITY SHOWN ON ISOMETRICS

NOTES:
1. THE X AND Y COORDINATES ARE WITH RESPECT TO THE INTERSECTION OF THE CENTERLINE OF THE GAS TURBINE (GT) AND THE INLET MANIFOLD (IM). THE Z (EL) COORDINATE IS WITH RESPECT TO THE TOP OF CONCRETE, WHICH IS 15'-9" [4800mm] BELOW THE CENTERLINE OF THE GT AND IS 0'-0".
2. ALL PIPE SUPPORT COORDINATES ARE WITH RESPECT TO THE CENTERLINE OF THE INSTALLED SUPPORTED PIPE.
3. THE MARK 'FFW' (FIELD FIT WELD) IN THE DRAWING INDICATES EXCESS LENGTH OF APPROX. 4 INCHES IDENTIFIED AS LOOSE IN THE DRAWING TO BE CUT AND ADJUSTED AT THE FIELD.
4. ALL WELDING TO BE DONE AND INSPECTED PER LATEST APPLICABLE EDITION OF ASME B31.1 AND SECTION IX AVAILABLE ON THE PROJECT AWARD DATE.
5. PIPING SHALL BE INSTALLED PER THE INDICATED SLOPE.
6. FOR ANY FURTHER INFORMATION REQUIRED DURING INSTALLATION, CONSULT MITSUBISHI POWER TFA (TECHNICAL FIELD ADVISOR) AT SITE.
7. XXX## ### INDICATES PIPING TERMINAL POINT.

ADDITIONAL NOTE:
TUBE REDUCER (FRACTIONAL TO METRIC) SHALL BE IDENTIFIED BY COLOR OR TAGGING FOR QUICK NOTATION IN THE FIELD.

WELD NO	WELD TYPE	WELD CAT	NPD	QTY
3	BW	S	1/2"	1
4	BW	S	1/2"	1
2	BW	S	1"	1

PIPING SPEC:	SDQ	PAINTING SPEC: PER PROJECT SPECIFICATION	REVISION RECORD				
FLUID CODE:	HW, FIRE FIGHTING WATER	INTERIOR SURFACE TREATMENT SPEC: LU-00129	REV. #	DESCRIPTION	BY	CHK	APP	DATE
MAX. PRESS.:	2030.00 psig / 14.00 MPag							
TEST PRESS.:	3045.00 psig / 20.99 MPag							
MAX. TEMP:	100 F / 38 C							
INSULATION SPEC:	NOT INSULATED							

WELD TYPE IDENTIFIER:
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WELD CATEGORY IDENTIFICATION:
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FW - FIELD WELD
FFW - FIELD FIT WELD

SYMBOLOGY:
WELD NUMBER
PART NUMBER (PT NO)
< > CUT LIST PIECE NUMBER

SYMBOL IDENTIFIER:
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G - GASKET
B - BOLT
N - NUT
W - WASHER
O - O-RING
L - LOCK WIRE

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MITSUBISHI POWER

ORIGIN:	MITSUBISHI POWER-LAKE MARY	DATE:	MITSUBISHI POWER DRAWING NUMBER:	REV. NO.:	SYSTEM CODE:
PREPARED BY:	E.BALATLI	07/23/2024	-	0	64A0A
REVIEWED BY:	M.FAZLIOGLU	07/24/2024	FIRE FIGHTING SYSTEM PIPING		
APPROVED BY:	N.KOZAN	07/25/2024	INTERNAL TO GT PACKAGE		
			FOR FABRICATION & ERECTION		
			GRAND RIVER ENERGY CENTER UNIT 4		
			GRAND RIVER DAM AUTHORITY		
			M501JAC GAS TURBINE		
PROJECT CODE:	K-GREC1	P&ID:			
PIPELINE:	FFS_SPACE-IN_MD3.1_2	PAGE NUMBER:	3 OF 4		

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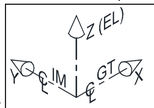

SEE SHT 3

CUT PIPE LENGTH

PIECE NO	LENGTH (FT-INS)	NPD (IN)	REMARKS	END ONE	END TWO	ITEM CODE
<6>	7 3/8"	1/2		SQ.CUT	BEVEL	MTAAZZBPZZZZZXAQ
<7>	3'1 3/16"	1/2	PLD BEND(2)	SQ.CUT	SQ.CUT	MTAAZZBPZZZZZXAQ
<8>	7'1 13/16"	1	PLD BEND(1)	SQ.CUT	SQ.CUT	MTAAZZBPZZZZZXAQ
<9>	7'11 3/8"	1	PLD BEND(1)	SQ.CUT	BEVEL	MTAAZZBPZZZZZXAQ

ACTUAL QUANTITY OF WASHERS SHALL BE CONSIDERED AS HALF OF THE QUANTITY SHOWN ON ISOMETRICS

NOTES:
1. THE X AND Y COORDINATES ARE WITH RESPECT TO THE INTERSECTION OF THE CENTERLINE OF THE GAS TURBINE (GT) AND THE INLET MANIFOLD (IM). THE Z (EL) COORDINATE IS WITH RESPECT TO THE TOP OF CONCRETE, WHICH IS 15'-9" [4800mm] BELOW THE CENTERLINE OF THE GT AND IS 0'-0".
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6. FOR ANY FURTHER INFORMATION REQUIRED DURING INSTALLATION, CONSULT MITSUBISHI POWER TFA (TECHNICAL FIELD ADVISOR) AT SITE.
7. ⟨XXX## ###⟩ INDICATES PIPING TERMINAL POINT.

ORIGIN:	MITSUBISHI POWER-LAKE MARY	DATE:	MITSUBISHI POWER DRAWING NUMBER:	REV. NO.:	SYSTEM CODE:
PREPARED BY:	E.BALATLI	07/23/2024	-	0	64A0A
REVIEWED BY:	M.FAZLIOGLU	07/24/2024	FIRE FIGHTING SYSTEM PIPING		
APPROVED BY:	N.KOZAN	07/25/2024	INTERNAL TO GT PACKAGE		
			FOR FABRICATION & ERECTION		
			GRAND RIVER ENERGY CENTER UNIT 4		
			GRAND RIVER DAM AUTHORITY		
			M501JAC GAS TURBINE		
			PROJECT CODE: K-GREC1	P&ID:	
			PIPELINE: FFS_SPACE-IN_MD3.1_2	PAGE NUMBER: 4 OF 4	

PIPING SPEC:	SDQ
FLUID CODE:	HW, FIRE FIGHTING WATER
MAX. PRESS.:	2030.00 psig / 14.00 MPag
TEST PRESS.:	3045.00 psig / 20.99 MPag
MAX. TEMP:	100 F / 38 C
INSULATION SPEC:	NOT INSULATED

PAINTING SPEC: PER PROJECT SPECIFICATION
INTERIOR SURFACE TREATMENT SPEC: LU-00129

REVISION RECORD

REV. #	DESCRIPTION	BY	CHK	APP	DATE

WELD TYPE IDENTIFIER:
BW - BUTT WELD
SW - SOCKET WELD
SOF - SLIP-ON FLANGE
SOB - SET-ON BRANCH
LET - OLET, LATROLET OR HALF COUPLING
TW - TACK WELD
MW - MITER WELD

WELD CATEGORY IDENTIFICATION:
S - SHOP
FW - FIELD WELD
FFW - FIELD FIT WELD

SYMBOLOGY:
WELD NUMBER
PART NUMBER (PT NO)
< > CUT LIST PIECE NUMBER

SYMBOL IDENTIFIER:
F - FLANGE
G - GASKET
B - BOLT
N - NUT
W - WASHER
O - O-RING
L - LOCK WIRE

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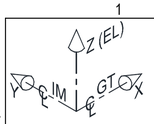
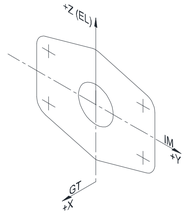
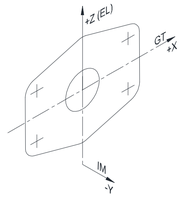
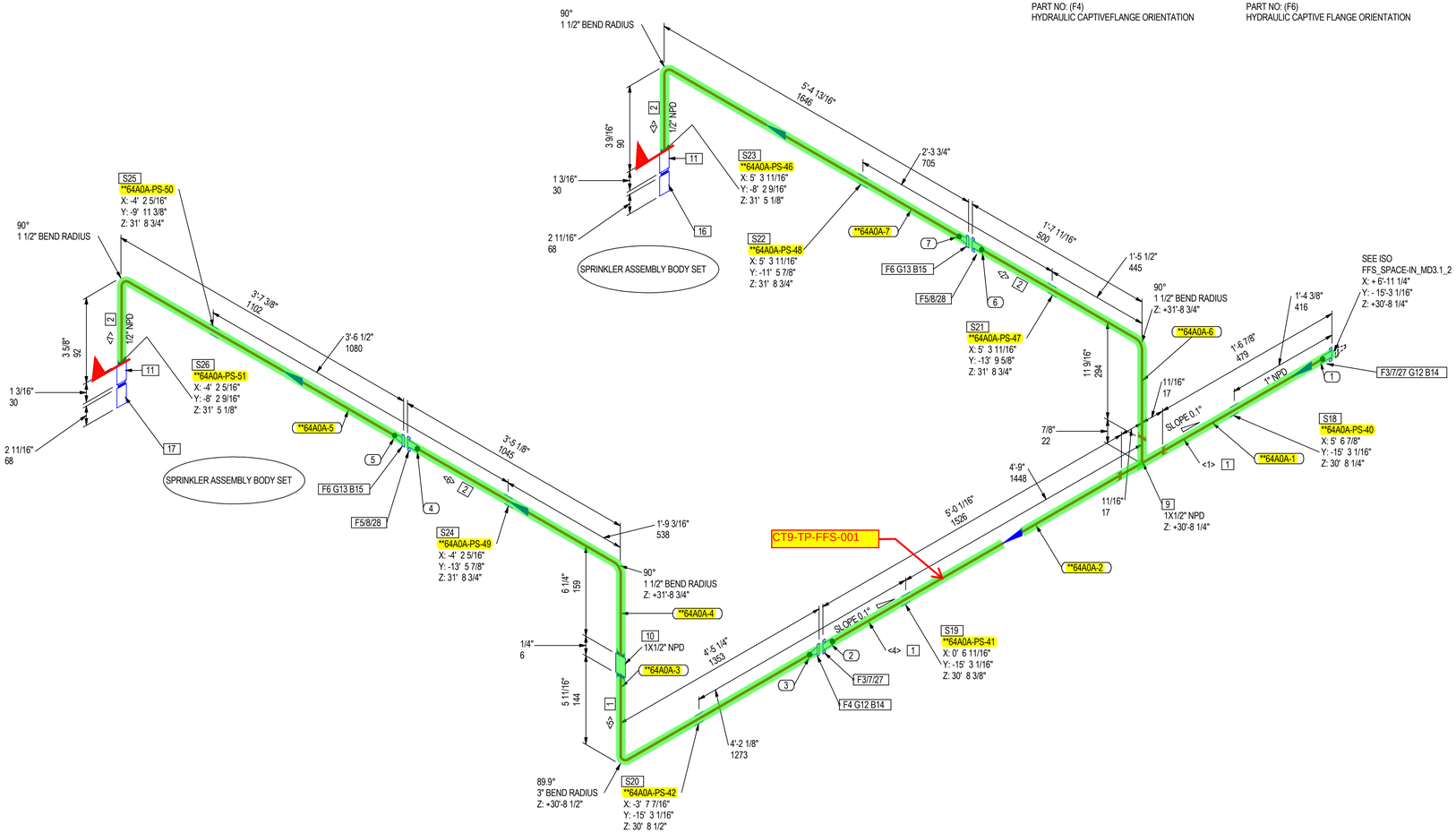




PART NO: (F4)
HYDRAULIC CAPTIVEFLANGE ORIENTATION



PART NO: (F6)
HYDRAULIC CAPTIVE FLANGE ORIENTATION



ADDITIONAL NOTE:
TUBE REDUCER (FRACTIONAL TO METRIC) SHALL BE IDENTIFIED BY COLOR OR TAGGING FOR QUICK NOTATION IN THE FIELD.

SHOP MATERIALS - FABRICATION ONLY

PT NO	DESCRIPTION	NPD (IN)	CMDTY CODE	QTY
1	TUBE, 0.083 in THK, PE, ASTM A213-TP316, SEAMLESS	1	MTAAZZBPZZZZZXAQ	11.4'
2	TUBE, 0.049 in THK, PE, ASTM A213-TP316, SEAMLESS	1/2	MTAAZZBPZZZZZXAQ	15.9'
3	FLANGE HYDRAULIC, 3000 PSI, GROOVED END x SWE, SAE J518-1 CODE 61, ASTM A182-F316, FLANGE HEAD	1	MFFAAWYFZZIARALS	2
4	FLANGE HYDRAULIC, 3000 PSI, FFFE x SWE, SAE J518-1 CODE 61, ASTM A182-F316, CAPTIVE FLANGE	1	MFFBAWYGZZIARALS	1
5	FLANGE HYDRAULIC, 3000 PSI, GROOVED END x SWE, SAE J518-1 CODE 61, ASTM A182-F316, FLANGE HEAD	1/2	MFFAAWYFZZIARALS	2
6	FLANGE HYDRAULIC, 3000 PSI, FFFE x SWE, SAE J518-1 CODE 61, ASTM A182-F316, CAPTIVE FLANGE	1/2	MFFBAWYGZZIARALS	2
	MISCELLANEOUS COMPONENTS			
7	FLANGE HYDRAULIC, 3000 PSI, SAE J518-1 CODE 61, ASTM A182-F316, SPLIT FLANGE CLAMP (ONE PAIR)	1	MWABAWYHZZIARALS	2
8	FLANGE HYDRAULIC, 3000 PSI, SAE J518-1 CODE 61, ASTM A182-F316, SPLIT FLANGE CLAMP (ONE PAIR)	1/2	MWABAWYHZZIARALS	2

OTHER THAN SHOP MATERIALS - ERECTION ONLY

PT NO	DESCRIPTION	NPD (IN)	CMDTY CODE	QTY
9	TUBE REDUCING UNION TEE, SWGLKE END, SWAGELOK, ASTM A182-F316	1X1/2	MWBCZZCGZZAKWALS	1
10	REDUCING TUBE UNION, SWGLKE END, SWAGELOK PART NO. SS-1610-6-8, ASTM A182-F316	1X1/2	MWARZZCGZZAKWALS	1
11	TUBE REDUCER (FRACTIONAL TO METRIC), SWGLKE END x PE, SWAGELOK PART NO.SS-810-R-12M, ASTM A182-F316	1/2	MWAOZZYDZZAKWALS	2
12	GASKET, ORING, 0.125" THK, AS 568, VITON HS90 (DUROMETER90)	1	MKBZZACIAQXANUS	2
13	GASKET, ORING, 0.125" THK, AS 568, VITON HS90 (DUROMETER90)	1/2	MKBZZACIAQXANUS	2
14	SCREW, DRILLED HEX SOCKET HEAD CAP (2 HOLES), ASME B18.3, ASTM A193-B7, 16-UNC - 1.75 in. Length	3/8	MBCZZABZZIALBEUZ	8
15	SCREW, DRILLED HEX SOCKET HEAD CAP (2 HOLES), ASME B18.3, ASTM A193-B7, 18-UNC - 1.5 in. Length	5/16	MBCZZBRZZIALBEUZ	8
	INSTRUMENTS			
16	SPRINKLER ASSEMBLY BODY SET FOR HI-FOG® 1000 SERIES SPRINKLER AND SPRAY HEADS, A1000S1CT00D12S10 SPRINKLER ASSEMBLY BODY SET	1/2		1
17	SPRINKLER ASSEMBLY BODY SET FOR HI-FOG® 1000 SERIES SPRINKLER AND SPRAY HEADS, A1000S1CT00D12S10 SPRINKLER ASSEMBLY BODY SET	1/2		1

SEE SHT 2

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7. XXX## ### INDICATES PIPING TERMINAL POINT.

WELD NO	WELD TYPE	WELD CAT	NPD	QTY
1	BW	S	1"	1
6	BW	S	1/2"	1
7	BW	S	1/2"	1
2	BW	S	1"	1
3	BW	S	1"	1
4	BW	S	1/2"	1
5	BW	S	1/2"	1

PIPING SPEC:	SDQ	PAINTING SPEC: PER PROJECT SPECIFICATION	REVISION RECORD				
FLUID CODE:	HW, FIRE FIGHTING WATER	INTERIOR SURFACE TREATMENT SPEC: LU-00129	REV. #	DESCRIPTION	BY	CHK	APP	DATE
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MITSUBISHI POWER

ORIGIN:	MITSUBISHI POWER-LAKE MARY	DATE:	MITSUBISHI POWER DRAWING NUMBER:	REV. NO.:	SYSTEM CODE:
PREPARED BY:	E.BALATLI	07/23/2024	-	0	64A0A
REVIEWED BY:	M.FAZLIOGLU	07/24/2024	FIRE FIGHTING SYSTEM PIPING		
APPROVED BY:	N.KOZAN	07/25/2024	INTERNAL TO GT PACKAGE		
			FOR FABRICATION & ERECTION		
			GRAND RIVER ENERGY CENTER UNIT 4		
			GRAND RIVER DAM AUTHORITY		
			M501JAC GAS TURBINE		
PROJECT CODE:	K-GREC1	P&ID:			
PIPELINE:	FFS_SPACE-AROUND-IM_1	PAGE NUMBER:	1 OF 2		

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PLANT ENGINEERING

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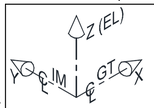

SEE SHT 1

PIPE SUPPORTS

18	PIPE SUPPORT	1	**64A0A-PS-40	1
19	PIPE SUPPORT	1	**64A0A-PS-41	1
20	PIPE SUPPORT	1	**64A0A-PS-42	1
21	PIPE SUPPORT	1/2	**64A0A-PS-47	1
22	PIPE SUPPORT	1/2	**64A0A-PS-48	1
23	NOZZLE MOUNTING ADAPTER SUPPORT	1/2	**64A0A-PS-46	1
24	PIPE SUPPORT	1/2	**64A0A-PS-49	1
25	PIPE SUPPORT	1/2	**64A0A-PS-50	1
26	NOZZLE MOUNTING ADAPTER SUPPORT	1/2	**64A0A-PS-51	1

MISCELLANEOUS COMPONENTS

27	LOCK WIRE, 1mm DIA x 1m LENGTH, ASTM A580-304	1	MWRZZZZBAIASXAPU	2
28	LOCK WIRE, 1mm DIA x 1m LENGTH, ASTM A580-304	1/2	MWRZZZZBAIASXAPU	2

SPOOLS

**64A0A-1, **64A0A-2, **64A0A-3, **64A0A-4, **64A0A-5, **64A0A-6, **64A0A-7

CUT PIPE LENGTH

PIECE NO	LENGTH (FT-INS)	NPD (IN)	REMARKS	END ONE	END TWO	ITEM CODE
<1>	1'6 5/8"	1		SQ.CUT	SQ.CUT	MTAAZZBPZZZZZXAQ
<2>	2'6 3/8"	1/2	PLD BEND(1)	SQ.CUT	SQ.CUT	MTAAZZBPZZZZZXAQ
<3>	5'7 1/2"	1/2	PLD BEND(1)	BEVEL	SQ.CUT	MTAAZZBPZZZZZXAQ
<4>	4'11 13/16"	1		SQ.CUT	SQ.CUT	MTAAZZBPZZZZZXAQ
<5>	4'9 3/8"	1	PLD BEND(1)	BEVEL	SQ.CUT	MTAAZZBPZZZZZXAQ
<6>	3'10 9/16"	1/2	PLD BEND(1)	SQ.CUT	SQ.CUT	MTAAZZBPZZZZZXAQ
<7>	3'10 1/16"	1/2	PLD BEND(1)	BEVEL	SQ.CUT	MTAAZZBPZZZZZXAQ

ACTUAL QUANTITY OF WASHERS SHALL BE CONSIDERED AS HALF OF THE QUANTITY SHOWN ON ISOMETRICS

NOTES:
1. THE X AND Y COORDINATES ARE WITH RESPECT TO THE INTERSECTION OF THE CENTERLINE OF THE GAS TURBINE (GT) AND THE INLET MANIFOLD (IM). THE Z (EL) COORDINATE IS WITH RESPECT TO THE TOP OF CONCRETE, WHICH IS 15'-9" [4800mm] BELOW THE CENTERLINE OF THE GT AND IS 0'-0".
2. ALL PIPE SUPPORT COORDINATES ARE WITH RESPECT TO THE CENTERLINE OF THE INSTALLED SUPPORTED PIPE.
3. THE MARK 'FFW' (FIELD FIT WELD) IN THE DRAWING INDICATES EXCESS LENGTH OF APPROX. 4 INCHES IDENTIFIED AS LOOSE IN THE DRAWING TO BE CUT AND ADJUSTED AT THE FIELD.
4. ALL WELDING TO BE DONE AND INSPECTED PER LATEST APPLICABLE EDITION OF ASME B31.1 AND SECTION IX AVAILABLE ON THE PROJECT AWARD DATE.
5. PIPING SHALL BE INSTALLED PER THE INDICATED SLOPE.
6. FOR ANY FURTHER INFORMATION REQUIRED DURING INSTALLATION, CONSULT MITSUBISHI POWER TFA (TECHNICAL FIELD ADVISOR) AT SITE.
7. ⟨XXX## ###⟩ INDICATES PIPING TERMINAL POINT.

ORIGIN:	MITSUBISHI POWER-LAKE MARY	DATE:	MITSUBISHI POWER DRAWING NUMBER:	REV. NO.:	SYSTEM CODE:
PREPARED BY:	E.BALATLI	07/23/2024	-	0	64A0A
REVIEWED BY:	M.FAZLIOGLU	07/24/2024	FIRE FIGHTING SYSTEM PIPING		
APPROVED BY:	N.KOZAN	07/25/2024	INTERNAL TO GT PACKAGE		
			FOR FABRICATION & ERECTION		
			GRAND RIVER ENERGY CENTER UNIT 4		
			GRAND RIVER DAM AUTHORITY		
			M501JAC GAS TURBINE		
			PROJECT CODE: K-GREC1	P&ID:	
			PIPELINE: FFS_SPACE-AROUND-IM_1	PAGE NUMBER: 2 OF 2	

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SL2

MITSUBISHI POWER ENGINEERING | PLANT ENGINEERING

DEPARTMENT: | SECTION:

DRAWING DISTRIBUTION LIST

PIPING SPEC:	SDQ
FLUID CODE:	HW, FIRE FIGHTING WATER
MAX. PRESS.:	2030.00 psig / 14.00 MPag
TEST PRESS.:	3045.00 psig / 20.99 MPag
MAX. TEMP:	100 F / 38 C
INSULATION SPEC:	NOT INSULATED

PAINTING SPEC: PER PROJECT SPECIFICATION

INTERIOR SURFACE TREATMENT SPEC: LU-00129

REV. #	DESCRIPTION	BY	CHK	APP	DATE

WELD TYPE IDENTIFIER:
BW - BUTT WELD
SW - SOCKET WELD
SOF - SLIP-ON FLANGE
SOB - SET-ON BRANCH
LET - OLET, LATROLET OR HALF COUPLING
TW - TACK WELD
MW - MITER WELD

WELD CATEGORY IDENTIFICATION:
S - SHOP
FW - FIELD WELD
FFW - FIELD FIT WELD

SYMBOLOGY:
WELD NUMBER
PART NUMBER (PT NO)
< > CUT LIST PIECE NUMBER

SYMBOL IDENTIFIER:
F - FLANGE
G - GASKET
B - BOLT
N - NUT
W - WASHER
O - O-RING
L - LOCK WIRE

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MITSUBISHI POWER

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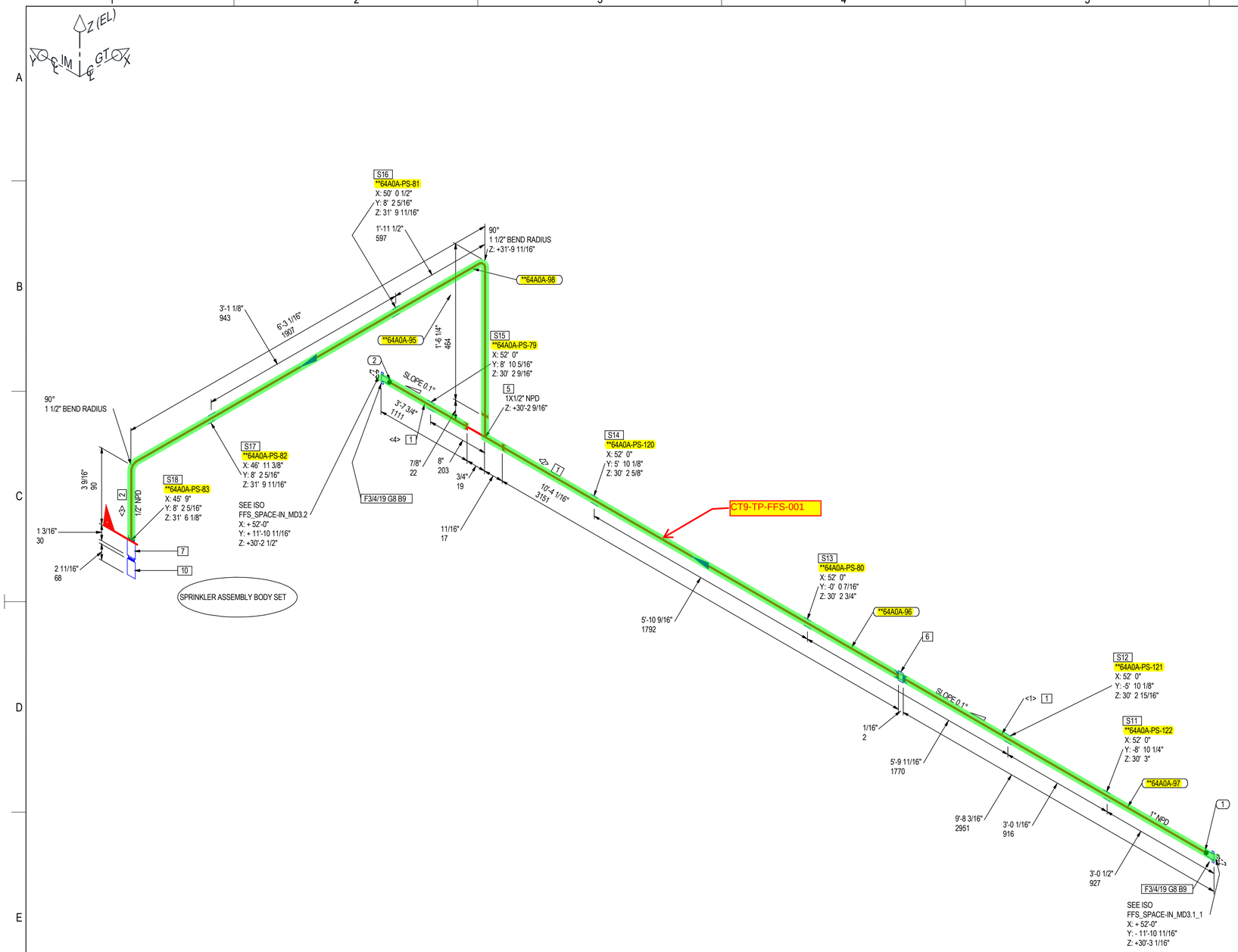

SHOP MATERIALS - FABRICATION ONLY

PT NO	DESCRIPTION	NPD (IN)	CMDTY CODE	QTY
1	TUBE, 0.083 in THK, PE, ASTM A213-TP316, SEAMLESS	1	MTAAZZBPZZZZZXAQ	23.7'
2	TUBE, 0.049 in THK, PE, ASTM A213-TP316, SEAMLESS	1/2	MTAAZZBPZZZZZXAQ	8.0'
3	FLANGE HYDRAULIC, 3000 PSI, GROOVED END x SWE, SAE J518-1 CODE 61, ASTM A182-F316, FLANGE HEAD	1	MFFAAWYFZZIARALS	2
	MISCELLANEOUS COMPONENTS			
4	FLANGE HYDRAULIC, 3000 PSI, SAE J518-1 CODE 61, ASTM A182-F316, SPLIT FLANGE CLAMP (ONE PAIR)	1	MWABAWYHZZIARALS	2

OTHER THAN SHOP MATERIALS - ERECTION ONLY

PT NO	DESCRIPTION	NPD (IN)	CMDTY CODE	QTY
5	TUBE REDUCING UNION TEE, SWGLKE END, SWAGELOK, ASTM A182-F316	1X1/2	MWBCZZCGZZAKWALS	1
6	TUBE UNION, SWGLKE END, SWAGELOK PART NO. , ASTM A182-F316	1	MWADZZCGZZAKWALS	1
7	TUBE REDUCER (FRACTIONAL TO METRIC), SWGLKE END x PE, SWAGELOK PART NO.SS-810-R-12M, ASTM A182-F316	1/2	MWAOZZYDZZAKWALS	1
8	GASKET, ORING, 0.125" THK, AS 568, VITON HS90 (DUROMETER90)	1	MKBZZACIAQXANUS	2
9	SCREW, DRILLED HEX SOCKET HEAD CAP (2 HOLES), ASME B18.3, ASTM A193-B7, 16-UNC - 1.75 in. Length	3/8	MBCZZABZZIALBEUZ	8
	INSTRUMENTS			
10	SPRINKLER ASSEMBLY BODY SET FOR HI-FOG® 1000 SERIES SPRINKLER AND SPRAY HEADS, A1000S1CT00D12S10 SPRINKLER ASSEMBLY BODY SET	1/2		1
	PIPE SUPPORTS			
11	PIPE SUPPORT	1	**64A0A-PS-122	1
12	PIPE SUPPORT	1	**64A0A-PS-121	1
13	PIPE SUPPORT	1	**64A0A-PS-80	1
14	PIPE SUPPORT	1	**64A0A-PS-120	1
15	PIPE SUPPORT	1	**64A0A-PS-79	1
16	PIPE SUPPORT	1/2	**64A0A-PS-81	1
17	PIPE SUPPORT	1/2	**64A0A-PS-82	1
18	NOZZLE MOUNTING ADAPTER SUPPORT	1/2	**64A0A-PS-83	1
	MISCELLANEOUS COMPONENTS			
19	LOCK WIRE, 1mm DIA x 1m LENGTH, ASTM A580-304	1	MWRZZZZBAIASXAPU	2

SPOOLS

**64A0A-95, **64A0A-96, **64A0A-97, **64A0A-98

SEE SHT 2

ACTUAL QUANTITY OF WASHERS SHALL BE CONSIDERED AS HALF OF THE QUANTITY SHOWN ON ISOMETRICS

NOTES:
1. THE X AND Y COORDINATES ARE WITH RESPECT TO THE INTERSECTION OF THE CENTERLINE OF THE GAS TURBINE (GT) AND THE INLET MANIFOLD (IM). THE Z (EL) COORDINATE IS WITH RESPECT TO THE TOP OF CONCRETE, WHICH IS 15'-9" [4800mm] BELOW THE CENTERLINE OF THE GT AND IS 0'-0".
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4. ALL WELDING TO BE DONE AND INSPECTED PER LATEST APPLICABLE EDITION OF ASME B31.1 AND SECTION IX AVAILABLE ON THE PROJECT AWARD DATE.
5. PIPING SHALL BE INSTALLED PER THE INDICATED SLOPE.
6. FOR ANY FURTHER INFORMATION REQUIRED DURING INSTALLATION, CONSULT MITSUBISHI POWER TFA (TECHNICAL FIELD ADVISOR) AT SITE.
7. <XXX## ###> INDICATES PIPING TERMINAL POINT.

ADDITIONAL NOTE:
TUBE REDUCER (FRACTIONAL TO METRIC) SHALL BE IDENTIFIED BY COLOR OR TAGGING FOR QUICK NOTATION IN THE FIELD.

WELD NO	WELD TYPE	WELD CAT	NPD	QTY
1	BW	S	1"	1
2	BW	S	1"	1

PIPING SPEC:	SDQ	PAINTING SPEC: PER PROJECT SPECIFICATION	REVISION RECORD				
FLUID CODE:	HW, FIRE FIGHTING WATER	INTERIOR SURFACE TREATMENT SPEC: LU-00129	REV. #	DESCRIPTION	BY	CHK	APP	DATE
MAX. PRESS.:	2030.00 psig / 14.00 MPag							
TEST PRESS.:	3045.00 psig / 20.99 MPag							
MAX. TEMP:	100 F / 38 C							
INSULATION SPEC:	NOT INSULATED							

WELD TYPE IDENTIFIER:
BW - BUTT WELD
SW - SOCKET WELD
SOF - SLIP-ON FLANGE
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TW - TACK WELD
MW - MITER WELD

WELD CATEGORY IDENTIFICATION:
S - SHOP
FW - FIELD WELD
FFW - FIELD FIT WELD

SYMBOLOGY:
WELD NUMBER
PART NUMBER (PT NO)
< > CUT LIST PIECE NUMBER

SYMBOL IDENTIFIER:
F - FLANGE
G - GASKET
B - BOLT
N - NUT
W - WASHER
O - O-RING
L - LOCK WIRE

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MITSUBISHI POWER

ORIGIN:	MITSUBISHI POWER-LAKE MARY	DATE:	MITSUBISHI POWER DRAWING NUMBER:	REV. NO.:	SYSTEM CODE:
PREPARED BY:	E.BALATLI	07/23/2024	-	0	64A0A
REVIEWED BY:	M.FAZLIOGLU	07/24/2024	FIRE FIGHTING SYSTEM PIPING		
APPROVED BY:	N.KOZAN	07/25/2024	INTERNAL TO GT PACKAGE		
			FOR FABRICATION & ERECTION		
			GRAND RIVER ENERGY CENTER UNIT 4		
			GRAND RIVER DAM AUTHORITY		
			M501JAC GAS TURBINE		
PROJECT CODE:	K-GREC1	P&ID:			
PIPELINE:	FFS_SPACE-IN_RP-7_2	PAGE NUMBER:	1 OF 2		

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MITSUBISHI POWER ENGINEERING
PLANT ENGINEERING

DEPARTMENT:
SECTION:

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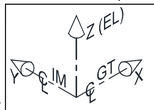

SEE SHT 1

CUT PIPE LENGTH

PIECE NO	LENGTH (FT-INS)	NPD (IN)	REMARKS	END ONE	END TWO	ITEM CODE
<1>	9'8"	1		SQ.CUT	SQ.CUT	MTAAZZBPZZZZZXAQ
<2>	10'4 1/16"	1		SQ.CUT	SQ.CUT	MTAAZZBPZZZZZXAQ
<3>	7'11 5/8"	1/2	PLD BEND(2)	SQ.CUT	SQ.CUT	MTAAZZBPZZZZZXAQ
<4>	3'7 1/2"	1		SQ.CUT	SQ.CUT	MTAAZZBPZZZZZXAQ

ACTUAL QUANTITY OF WASHERS SHALL BE CONSIDERED AS HALF OF THE QUANTITY SHOWN ON ISOMETRICS

NOTES:
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7. ⟨XXX## ###⟩ INDICATES PIPING TERMINAL POINT.

ORIGIN:	MITSUBISHI POWER-LAKE MARY	DATE:	MITSUBISHI POWER DRAWING NUMBER:	REV. NO.:	SYSTEM CODE:
PREPARED BY:	E.BALATLI	07/23/2024	-	0	64A0A
REVIEWED BY:	M.FAZLIOGLU	07/24/2024	FIRE FIGHTING SYSTEM PIPING		
APPROVED BY:	N.KOZAN	07/25/2024	INTERNAL TO GT PACKAGE		
			FOR FABRICATION & ERECTION		
			GRAND RIVER ENERGY CENTER UNIT 4		
			GRAND RIVER DAM AUTHORITY		
			M501JAC GAS TURBINE		
			PROJECT CODE: K-GREC1	P&ID:	
			PIPELINE: FFS_SPACE-IN_RP-7_2	PAGE NUMBER: 2 OF 2	

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MITSUBISHI POWER ENGINEERING
PLANT ENGINEERING

DEPARTMENT:
SECTION:

DRAWING DISTRIBUTION LIST

PIPING SPEC:	SDQ
FLUID CODE:	HW, FIRE FIGHTING WATER
MAX. PRESS.:	2030.00 psig / 14.00 MPag
TEST PRESS.:	3045.00 psig / 20.99 MPag
MAX. TEMP:	100 F / 38 C
INSULATION SPEC:	NOT INSULATED

PAINTING SPEC: PER PROJECT SPECIFICATION
INTERIOR SURFACE TREATMENT SPEC: LU-00129

REVISION RECORD

REV. #	DESCRIPTION	BY	CHK	APP	DATE

WELD TYPE IDENTIFIER:
BW - BUTT WELD
SW - SOCKET WELD
SOF - SLIP-ON FLANGE
SOB - SET-ON BRANCH
LET - OLET, LATROLET OR HALF COUPLING
TW - TACK WELD
MW - MITER WELD

WELD CATEGORY IDENTIFICATION:
● S - SHOP
⊗ FW - FIELD WELD
⊗ FFW - FIELD FIT WELD

SYMBOLOGY:
WELD NUMBER
PART NUMBER (PT NO)
< > CUT LIST PIECE NUMBER

SYMBOL IDENTIFIER:
F - FLANGE
G - GASKET
B - BOLT
N - NUT
W - WASHER
O - O-RING
L - LOCK WIRE

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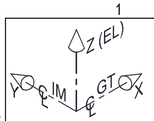

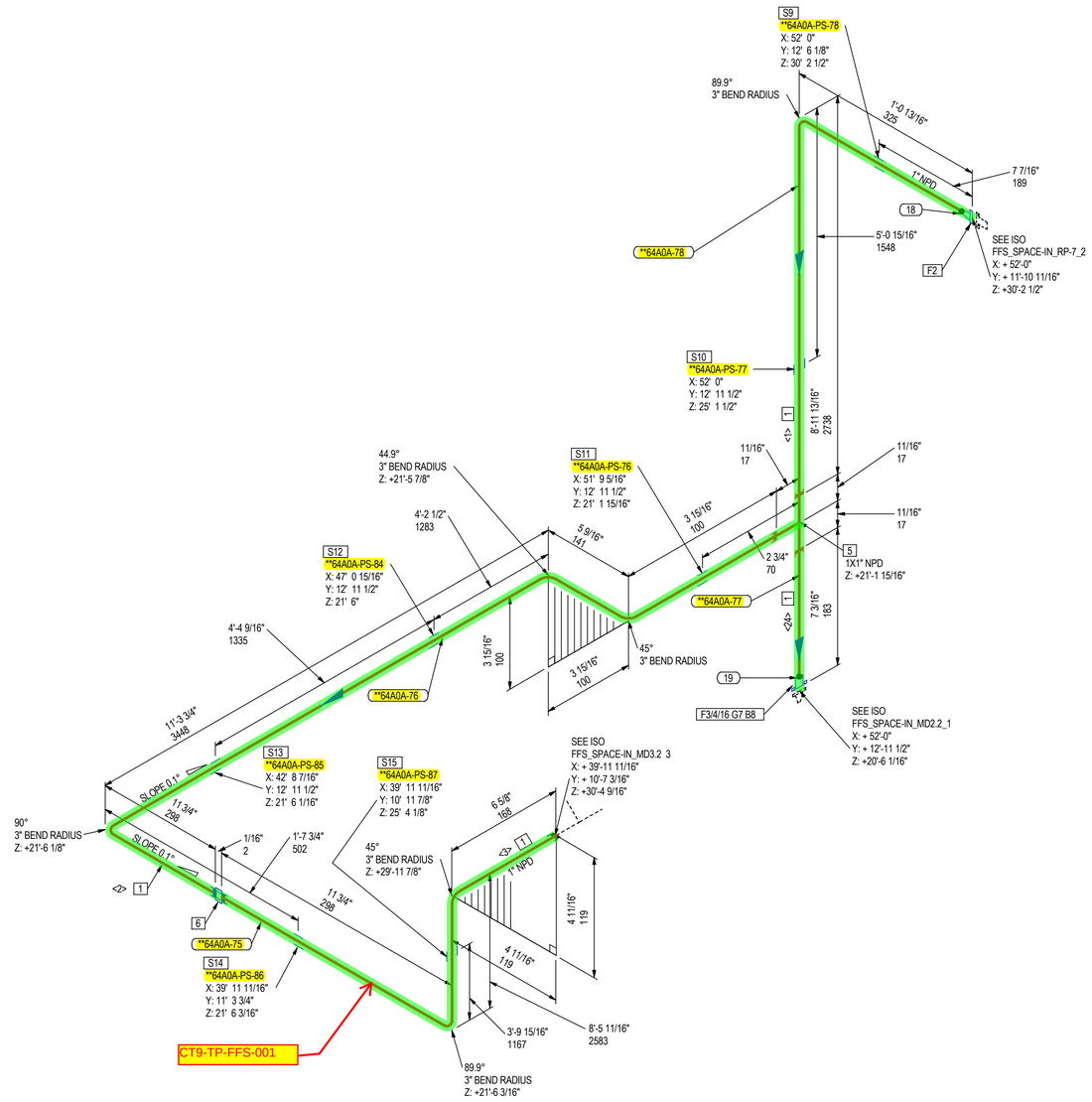

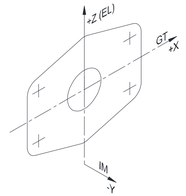

PART NO: (F2)
HYDRAULIC CAPTIVE FLANGE ORIENTATION

SHOP MATERIALS - FABRICATION ONLY

PT NO	DESCRIPTION	NPD (IN)	CMDTY CODE	QTY
1	TUBE, 0.083 in THK, PE, ASTM A213-TP316, SEAMLESS	1	MTAAZZBPZZZZZXAQ	33.4'
2	FLANGE HYDRAULIC, 3000 PSI, FFFE x SWE, SAE J518-1 CODE 61, ASTM A182-F316, CAPTIVE FLANGE	1	MFFBAWYGZZIARALS	1
3	FLANGE HYDRAULIC, 3000 PSI, GROOVED END x SWE, SAE J518-1 CODE 61, ASTM A182-F316, FLANGE HEAD	1	MFFAAWYFZZIARALS	1
	MISCELLANEOUS COMPONENTS			
4	FLANGE HYDRAULIC, 3000 PSI, SAE J518-1 CODE 61, ASTM A182-F316, SPLIT FLANGE CLAMP (ONE PAIR)	1	MWABAWYHZZIARALS	1

OTHER THAN SHOP MATERIALS - ERECTION ONLY

PT NO	DESCRIPTION	NPD (IN)	CMDTY CODE	QTY
5	TUBE UNION TEE, SWGLKE END, SWAGELOK PART NO. , ASTM A182-F316	1X1	MWAEZZCGZZAKWALS	1
6	TUBE UNION, SWGLKE END, SWAGELOK PART NO. , ASTM A182-F316	1	MWADZZCGZZAKWALS	1
7	GASKET, ORING, 0.125" THK, AS 568, VITON HS90 (DUROMETER90)	1	MKBZZACIAQXANUS	1
8	SCREW, DRILLED HEX SOCKET HEAD CAP (2 HOLES), ASME B18.3, ASTM A193-B7, 16-UNC - 1.75 in. Length	3/8	MBCZZABZZIALBEUZ	4
	PIPE SUPPORTS			
9	PIPE SUPPORT	1	**64A0A-PS-78	1
10	PIPE SUPPORT	1	**64A0A-PS-77	1
11	PIPE SUPPORT	1	**64A0A-PS-76	1
12	PIPE SUPPORT	1	**64A0A-PS-84	1
13	PIPE SUPPORT	1	**64A0A-PS-85	1
14	PIPE SUPPORT	1	**64A0A-PS-86	1
15	PIPE SUPPORT	1	**64A0A-PS-87	1
	MISCELLANEOUS COMPONENTS			
16	LOCK WIRE, 1mm DIA x 1m LENGTH, ASTM A580-304	1	MWRZZZZBAIASXAPU	1

SPOOLS

**64A0A-75, **64A0A-76, **64A0A-77, **64A0A-78

SEE SHT 2

ACTUAL QUANTITY OF WASHERS SHALL BE CONSIDERED AS HALF OF THE QUANTITY SHOWN ON ISOMETRICS

NOTES:
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7. XXX## ### INDICATES PIPING TERMINAL POINT.

WELD NO	WELD TYPE	WELD CAT	NPD	QTY
18	BW	S	1"	1
19	BW	S	1"	1

PIPING SPEC:	SDQ	PAINTING SPEC: PER PROJECT SPECIFICATION	REVISION RECORD				
FLUID CODE:	HW, FIRE FIGHTING WATER	INTERIOR SURFACE TREATMENT SPEC: LU-00129	REV. #	DESCRIPTION	BY	CHK	APP	DATE
MAX. PRESS.:	2030.00 psig / 14.00 MPag							
TEST PRESS.:	3045.00 psig / 20.99 MPag							
MAX. TEMP:	100 F / 38 C							
INSULATION SPEC:	NOT INSULATED							

WELD TYPE IDENTIFIER:
BW - BUTT WELD
SW - SOCKET WELD
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FW - FIELD WELD
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SYMBOLOGY:
WELD NUMBER
PART NUMBER (PT NO)
< > CUT LIST PIECE NUMBER

SYMBOL IDENTIFIER:
F - FLANGE
G - GASKET
B - BOLT
N - NUT
W - WASHER
O - O-RING
L - LOCK WIRE

ORIGIN:	MITSUBISHI POWER-LAKE MARY	DATE:	MITSUBISHI POWER DRAWING NUMBER:	REV. NO.:	SYSTEM CODE:
PREPARED BY:	E.BALATLI	07/23/2024	-	0	64A0A
REVIEWED BY:	M.FAZLIOGLU	07/24/2024	FIRE FIGHTING SYSTEM PIPING		
APPROVED BY:	N.KOZAN	07/25/2024	INTERNAL TO GT PACKAGE		
			FOR FABRICATION & ERECTION		
			GRAND RIVER ENERGY CENTER UNIT 4		
			GRAND RIVER DAM AUTHORITY		
			M501JAC GAS TURBINE		
PROJECT CODE:	K-GREC1	P&ID:			
PIPELINE:	FFS_SPACE-IN_MD3.2	PAGE NUMBER:	1 OF 11		

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MITSUBISHI POWER ENGINEERING
PLANT ENGINEERING

DEPARTMENT:
SECTION:

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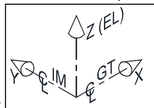

SEE SHT 1

CUT PIPE LENGTH

PIECE NO	LENGTH (FT-INS)	NPD (IN)	REMARKS	END ONE	END TWO	ITEM CODE
<1>	9'11 1/16"	1	PLD BEND(1)	BEVEL	SQ.CUT	MTAAZZBPZZZZZXAQ
<2>	12'11 1/2"	1	PLD BEND(3)	SQ.CUT	SQ.CUT	MTAAZZBPZZZZZXAQ
<3>	9'10 5/8"	1	PLD BEND(2)	SQ.CUT	SQ.CUT	MTAAZZBPZZZZZXAQ
<24>	6 15/16"	1		SQ.CUT	SQ.CUT	MTAAZZBPZZZZZXAQ

ACTUAL QUANTITY OF WASHERS SHALL BE CONSIDERED AS HALF OF THE QUANTITY SHOWN ON ISOMETRICS

NOTES:
1. THE X AND Y COORDINATES ARE WITH RESPECT TO THE INTERSECTION OF THE CENTERLINE OF THE GAS TURBINE (GT) AND THE INLET MANIFOLD (IM). THE Z (EL) COORDINATE IS WITH RESPECT TO THE TOP OF CONCRETE, WHICH IS 15'-9" [4800mm] BELOW THE CENTERLINE OF THE GT AND IS 0'-0".
2. ALL PIPE SUPPORT COORDINATES ARE WITH RESPECT TO THE CENTERLINE OF THE INSTALLED SUPPORTED PIPE.
3. THE MARK 'FFW' (FIELD FIT WELD) IN THE DRAWING INDICATES EXCESS LENGTH OF APPROX. 4 INCHES IDENTIFIED AS LOOSE IN THE DRAWING TO BE CUT AND ADJUSTED AT THE FIELD.
4. ALL WELDING TO BE DONE AND INSPECTED PER LATEST APPLICABLE EDITION OF ASME B31.1 AND SECTION IX AVAILABLE ON THE PROJECT AWARD DATE.
5. PIPING SHALL BE INSTALLED PER THE INDICATED SLOPE.
6. FOR ANY FURTHER INFORMATION REQUIRED DURING INSTALLATION, CONSULT MITSUBISHI POWER TFA (TECHNICAL FIELD ADVISOR) AT SITE.
7. ⟨XXX## ###⟩ INDICATES PIPING TERMINAL POINT.

ORIGIN:	MITSUBISHI POWER-LAKE MARY	DATE:	MITSUBISHI POWER DRAWING NUMBER:	REV. NO.:	SYSTEM CODE:
PREPARED BY:	E.BALATLI	07/23/2024	-	0	64A0A
REVIEWED BY:	M.FAZLIOGLU	07/24/2024	FIRE FIGHTING SYSTEM PIPING		
APPROVED BY:	N.KOZAN	07/25/2024	INTERNAL TO GT PACKAGE		
			FOR FABRICATION & ERECTION		
			GRAND RIVER ENERGY CENTER UNIT 4		
			GRAND RIVER DAM AUTHORITY		
			M501JAC GAS TURBINE		
			PROJECT CODE: K-GREC1	P&ID:	
			PIPELINE: FFS_SPACE-IN_MD3.2	PAGE NUMBER: 2 OF 11	

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MITSUBISHI POWER ENGINEERING
PLANT ENGINEERING

DEPARTMENT:
SECTION:

DRAWING DISTRIBUTION LIST

PIPING SPEC:	SDQ
FLUID CODE:	HW, FIRE FIGHTING WATER
MAX. PRESS.:	2030.00 psig / 14.00 MPag
TEST PRESS.:	3045.00 psig / 20.99 MPag
MAX. TEMP:	100 F / 38 C
INSULATION SPEC:	NOT INSULATED

PAINTING SPEC: PER PROJECT SPECIFICATION
INTERIOR SURFACE TREATMENT SPEC: LU-00129

REVISION RECORD

REV. #	DESCRIPTION	BY	CHK	APP	DATE

WELD TYPE IDENTIFIER:
BW - BUTT WELD
SW - SOCKET WELD
SOF - SLIP-ON FLANGE
SOB - SET-ON BRANCH
LET - OLET, LATROLET OR HALF COUPLING
TW - TACK WELD
MW - MITER WELD

WELD CATEGORY IDENTIFICATION:
● S - SHOP
⊠ FW - FIELD WELD
⊠ FFW - FIELD FIT WELD

SYMBOLOGY:
▭ WELD NUMBER
▭ PART NUMBER (PT NO)
< > CUT LIST PIECE NUMBER

SYMBOL IDENTIFIER:
F - FLANGE
G - GASKET
B - BOLT
N - NUT
W - WASHER
O - O-RING
L - LOCK WIRE

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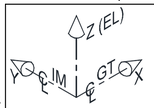
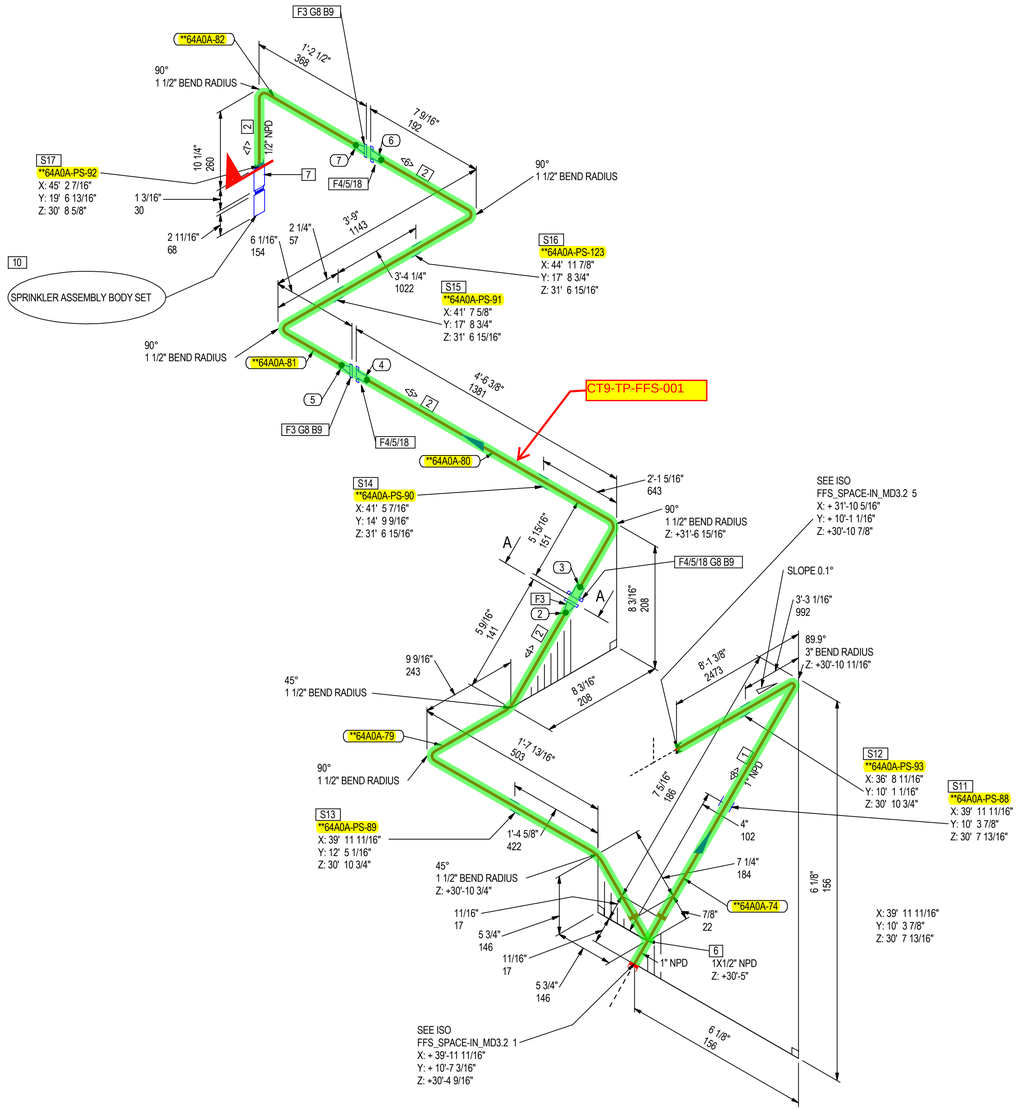
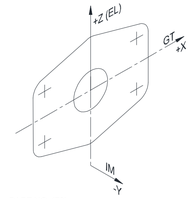
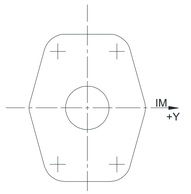






PART NO: (F3)
HYDRAULIC CAPTIVE FLANGE ORIENTATION



VIEW A - A
HYDRAULIC CAPTIVE FLANGE ORIENTATION

SHOP MATERIALS - FABRICATION ONLY

PT NO	DESCRIPTION	NPD (IN)	CMDTY CODE	QTY
1	TUBE, 0.083 in THK, PE, ASTM A213-TP316, SEAMLESS	1	MTAAZZBPZZZZZXAQ	8.7'
2	TUBE, 0.049 in THK, PE, ASTM A213-TP316, SEAMLESS	1/2	MTAAZZBPZZZZZXAQ	15.1'
3	FLANGE HYDRAULIC, 3000 PSI, FFFE x SWE, SAE J518-1 CODE 61, ASTM A182-F316, CAPTIVE FLANGE	1/2	MFFBAWYGZZIARALS	3
4	FLANGE HYDRAULIC, 3000 PSI, GROOVED END x SWE, SAE J518-1 CODE 61, ASTM A182-F316, FLANGE HEAD	1/2	MFFAAWYFZZIARALS	3
	MISCELLANEOUS COMPONENTS			
5	FLANGE HYDRAULIC, 3000 PSI, SAE J518-1 CODE 61, ASTM A182-F316, SPLIT FLANGE CLAMP (ONE PAIR)	1/2	MWABAWYHZZIARALS	3

OTHER THAN SHOP MATERIALS - ERECTION ONLY

PT NO	DESCRIPTION	NPD (IN)	CMDTY CODE	QTY
6	TUBE REDUCING UNION TEE, SWGLKE END, SWAGELOK, ASTM A182-F316	1X1/2	MWBCZZCGZZAKWALS	1
7	TUBE REDUCER (FRACTIONAL TO METRIC), SWGLKE END x PE, SWAGELOK PART NO.SS-810-R-12M, ASTM A182-F316	1/2	MWAOZZYDZZAKWALS	1
8	GASKET, ORING, 0.125" THK, AS 568, VITON HS90 (DUROMETER90)	1/2	MKBZZACIAQXANUS	3
9	SCREW, DRILLED HEX SOCKET HEAD CAP (2 HOLES), ASME B18.3, ASTM A193-B7, 18-UNC - 1.5 in. Length	5/16	MBCZZBRZZIALBEUZ	12
	INSTRUMENTS			
10	SPRINKLER ASSEMBLY BODY SET FOR HI-FOG® 1000 SERIES SPRINKLER AND SPRAY HEADS, A1000S1CT00D12S10 SPRINKLER ASSEMBLY BODY SET	1/2		1
	PIPE SUPPORTS			
11	PIPE SUPPORT	1	**64A0A-PS-88	1
12	PIPE SUPPORT	1	**64A0A-PS-93	1
13	PIPE SUPPORT	1/2	**64A0A-PS-89	1
14	PIPE SUPPORT	1/2	**64A0A-PS-90	1
15	PIPE SUPPORT	1/2	**64A0A-PS-91	1
16	PIPE SUPPORT	1/2	**64A0A-PS-123	1
17	NOZZLE MOUNTING ADAPTER SUPPORT	1/2	**64A0A-PS-92	1
	MISCELLANEOUS COMPONENTS			
18	LOCK WIRE, 1mm DIA x 1m LENGTH, ASTM A580-304	1/2	MWRZZZZBAIASXAPU	3
	SPOOLS			

**64A0A-74, **64A0A-79, **64A0A-80, **64A0A-81, **64A0A-82

SEE SHT 4

ACTUAL QUANTITY OF WASHERS SHALL BE CONSIDERED AS HALF OF THE QUANTITY SHOWN ON ISOMETRICS

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7. 〈XXX## ###〉 INDICATES PIPING TERMINAL POINT.

ADDITIONAL NOTE:
TUBE REDUCER (FRACTIONAL TO METRIC) SHALL BE IDENTIFIED BY COLOR OR TAGGING FOR QUICK NOTATION IN THE FIELD.

WELD NO	WELD TYPE	WELD CAT	NPD	QTY
2	BW	S	1/2"	1
3	BW	S	1/2"	1
4	BW	S	1/2"	1
5	BW	S	1/2"	1
6	BW	S	1/2"	1
7	BW	S	1/2"	1

PIPING SPEC:	SDQ
FLUID CODE:	HW, FIRE FIGHTING WATER
MAX. PRESS.:	2030.00 psig / 14.00 MPag
TEST PRESS.:	3045.00 psig / 20.99 MPag
MAX. TEMP:	100 F / 38 C
INSULATION SPEC:	NOT INSULATED

PAINTING SPEC: PER PROJECT SPECIFICATION
INTERIOR SURFACE TREATMENT SPEC: LU-00129

REVISION RECORD					
REV. #	DESCRIPTION	BY	CHK	APP	DATE

WELD TYPE IDENTIFIER:
BW - BUTT WELD
SW - SOCKET WELD
SOF - SLIP-ON FLANGE
SOB - SET-ON BRANCH
LET - OLET, LATROLET OR HALF COUPLING
TW - TACK WELD
MW - MITER WELD

WELD CATEGORY IDENTIFICATION:
S - SHOP
FW - FIELD WELD
FFW - FIELD FIT WELD

SYMBOLOGY:
WELD NUMBER
PART NUMBER (PT NO)
< > CUT LIST PIECE NUMBER

SYMBOL IDENTIFIER:
F - FLANGE
G - GASKET
B - BOLT
N - NUT
W - WASHER
O - O-RING
L - LOCK WIRE

ORIGIN:	MITSUBISHI POWER-LAKE MARY	DATE:	MITSUBISHI POWER DRAWING NUMBER:	REV. NO.:	SYSTEM CODE:
PREPARED BY:	E.BALATLI	07/23/2024	-	0	64A0A
REVIEWED BY:	M.FAZLIOGLU	07/24/2024	FIRE FIGHTING SYSTEM PIPING		
APPROVED BY:	N.KOZAN	07/25/2025	INTERNAL TO GT PACKAGE		
			FOR FABRICATION & ERECTION		
			GRAND RIVER ENERGY CENTER UNIT 4		
			GRAND RIVER DAM AUTHORITY		
			M501JAC GAS TURBINE		
PROJECT CODE:	K-GREC1	P&ID:			
PIPELINE:	FFS_SPACE-IN_MD3.2	PAGE NUMBER:	3 OF 11		

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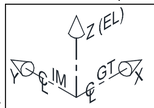

SEE SHT 3

CUT PIPE LENGTH

PIECE NO	LENGTH (FT-INS)	NPD (IN)	REMARKS	END ONE	END TWO	ITEM CODE
<4>	3'5 3/16"	1/2	PLD BEND(3)	SQ.CUT	BEVEL	MTAAZZBPZZZZZXAQ
<5>	4'11 1/4"	1/2	PLD BEND(1)	SQ.CUT	SQ.CUT	MTAAZZBPZZZZZXAQ
<6>	4'8 13/16"	1/2	PLD BEND(2)	BEVEL	SQ.CUT	MTAAZZBPZZZZZXAQ
<7>	1'11 15/16"	1/2	PLD BEND(1)	BEVEL	SQ.CUT	MTAAZZBPZZZZZXAQ
<8>	8'7 3/8"	1	PLD BEND(1)	SQ.CUT	SQ.CUT	MTAAZZBPZZZZZXAQ

ACTUAL QUANTITY OF WASHERS SHALL BE CONSIDERED AS HALF OF THE QUANTITY SHOWN ON ISOMETRICS

NOTES:
1. THE X AND Y COORDINATES ARE WITH RESPECT TO THE INTERSECTION OF THE CENTERLINE OF THE GAS TURBINE (GT) AND THE INLET MANIFOLD (IM). THE Z (EL) COORDINATE IS WITH RESPECT TO THE TOP OF CONCRETE, WHICH IS 15'-9" [4800mm] BELOW THE CENTERLINE OF THE GT AND IS 0'-0".
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7. XXX## ### INDICATES PIPING TERMINAL POINT.

ORIGIN:	MITSUBISHI POWER-LAKE MARY	DATE:	MITSUBISHI POWER DRAWING NUMBER:	REV. NO.:	SYSTEM CODE:
PREPARED BY:	E.BALATLI	07/23/2024	-	0	64A0A
REVIEWED BY:	M.FAZLIOGLU	07/24/2024	FIRE FIGHTING SYSTEM PIPING		
APPROVED BY:	N.KOZAN	07/25/2024	INTERNAL TO GT PACKAGE		
			FOR FABRICATION & ERECTION		
			GRAND RIVER ENERGY CENTER UNIT 4		
			GRAND RIVER DAM AUTHORITY		
			M501JAC GAS TURBINE		
			PROJECT CODE: K-GREC1	P&ID:	
			PIPELINE: FFS_SPACE-IN_MD3.2	PAGE NUMBER: 4 OF 11	

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MITSUBISHI POWER

PIPING SPEC:	SDQ	PAINTING SPEC: PER PROJECT SPECIFICATION	
FLUID CODE:	HW, FIRE FIGHTING WATER	INTERIOR SURFACE TREATMENT SPEC: LU-00129	
MAX. PRESS.:	2030.00 psig / 14.00 MPag		
TEST PRESS.:	3045.00 psig / 20.99 MPag		
MAX. TEMP:	100 F / 38 C		
INSULATION SPEC:	NOT INSULATED		

REVISION RECORD

REV. #	DESCRIPTION	BY	CHK	APP	DATE

WELD TYPE IDENTIFIER:
BW - BUTT WELD
SW - SOCKET WELD
SOF - SLIP-ON FLANGE
SOB - SET-ON BRANCH
LET - OLET, LATROLET OR HALF COUPLING
TW - TACK WELD
MW - MITER WELD

WELD CATEGORY IDENTIFICATION:
S - SHOP
FW - FIELD WELD
FFW - FIELD FIT WELD

SYMBOLOGY:
WELD NUMBER
PART NUMBER (PT NO)
< > CUT LIST PIECE NUMBER

SYMBOL IDENTIFIER:
F - FLANGE
G - GASKET
B - BOLT
N - NUT
W - WASHER
O - O-RING
L - LOCK WIRE

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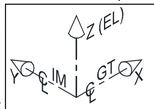
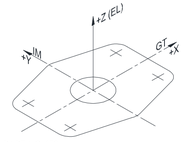
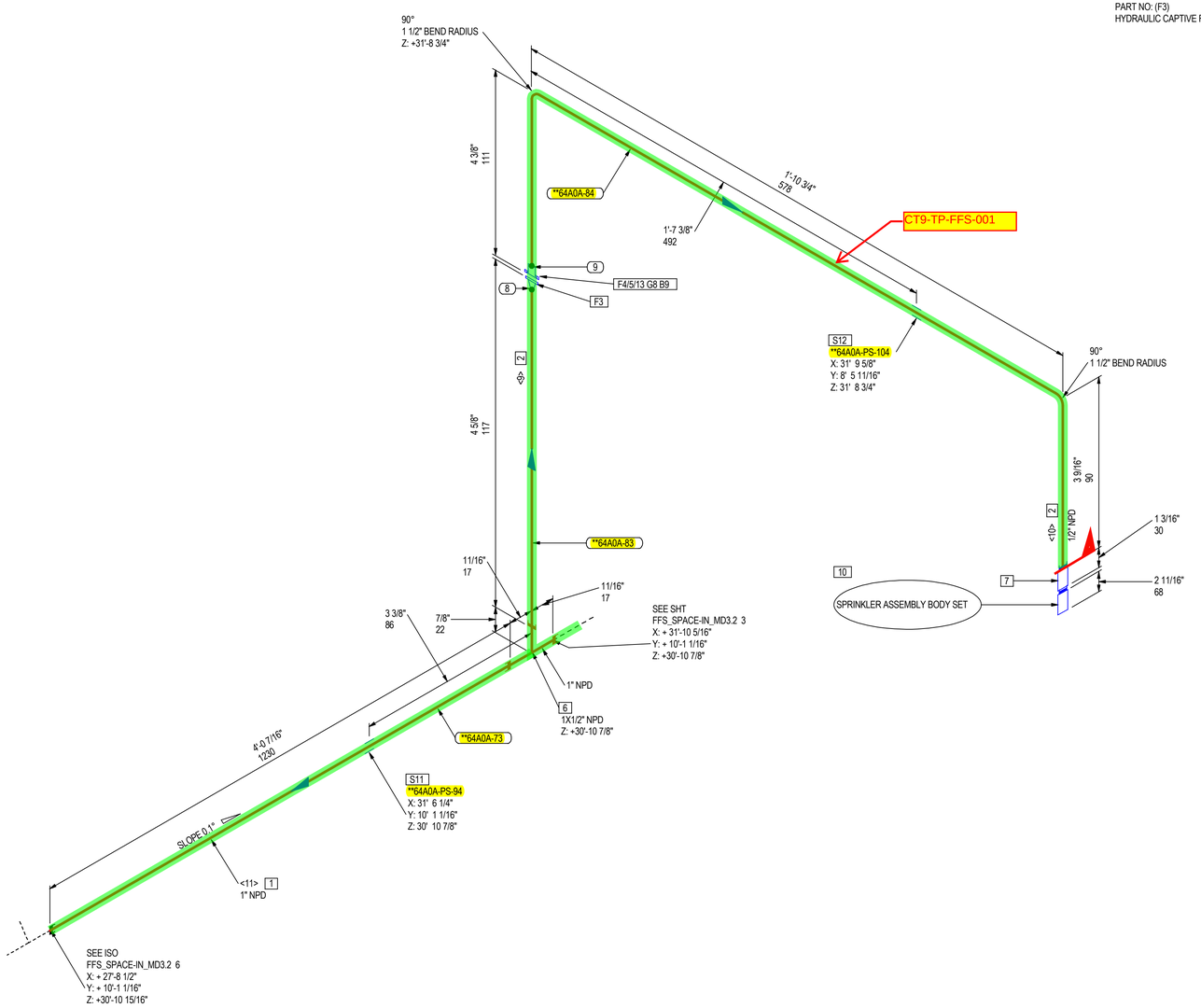




PART NO: (F3)
HYDRAULIC CAPTIVE FLANGE ORIENTATION



ADDITIONAL NOTE:
TUBE REDUCER (FRACTIONAL TO METRIC) SHALL BE IDENTIFIED BY COLOR OR TAGGING FOR QUICK NOTATION IN THE FIELD.

WELD NO	WELD TYPE	WELD CAT	NPD	QTY
8	BW	S	1/2"	1
9	BW	S	1/2"	1

SHOP MATERIALS - FABRICATION ONLY

PT NO	DESCRIPTION	NPD (IN)	CMDTY CODE	QTY
1	TUBE, 0.083 in THK, PE, ASTM A213-TP316, SEAMLESS	1	MTAAZZBPZZZZZXAQ	4.1'
2	TUBE, 0.049 in THK, PE, ASTM A213-TP316, SEAMLESS	1/2	MTAAZZBPZZZZZXAQ	2.8'
3	FLANGE HYDRAULIC, 3000 PSI, FFFE x SWE, SAE J518-1 CODE 61, ASTM A182-F316, CAPTIVE FLANGE	1/2	MFFBAWYGZZIARALS	1
4	FLANGE HYDRAULIC, 3000 PSI, GROOVED END x SWE, SAE J518-1 CODE 61, ASTM A182-F316, FLANGE HEAD	1/2	MFFAAWYFZZIARALS	1
	MISCELLANEOUS COMPONENTS			
5	FLANGE HYDRAULIC, 3000 PSI, SAE J518-1 CODE 61, ASTM A182-F316, SPLIT FLANGE CLAMP (ONE PAIR)	1/2	MWABAWYHZZIARALS	1

OTHER THAN SHOP MATERIALS - ERECTION ONLY

PT NO	DESCRIPTION	NPD (IN)	CMDTY CODE	QTY
6	TUBE REDUCING UNION TEE, SWGLKE END, SWAGELOK, ASTM A182-F316	1X1/2	MWBCZZCGZZAKWALS	1
7	TUBE REDUCER (FRACTIONAL TO METRIC), SWGLKE END x PE, SWAGELOK PART NO.SS-810-R-12M, ASTM A182-F316	1/2	MWAOZZYDZZAKWALS	1
8	GASKET, ORING, 0.125" THK, AS 568, VITON HS90 (DUROMETER90)	1/2	MKBZZACIAQXANUS	1
9	SCREW, DRILLED HEX SOCKET HEAD CAP (2 HOLES), ASME B18.3, ASTM A193-B7, 18-UNC - 1.5 in. Length	5/16	MBCZZBRZZIALBEUZ	4
	INSTRUMENTS			
10	SPRINKLER ASSEMBLY BODY SET FOR HI-FOG® 1000 SERIES SPRINKLER AND SPRAY HEADS, A1000S1CT00D12S10 SPRINKLER ASSEMBLY BODY SET	1/2		1
	PIPE SUPPORTS			
11	PIPE SUPPORT	1	**64A0A-PS-94	1
12	PIPE SUPPORT	1/2	**64A0A-PS-104	1
	MISCELLANEOUS COMPONENTS			
13	LOCK WIRE, 1mm DIA x 1m LENGTH, ASTM A580-304	1/2	MWRZZZZBAIASXAPU	1
	SPOOLS			
	**64A0A-73, **64A0A-83, **64A0A-84			

CUT PIPE LENGTH

PIECE NO	LENGTH (FT-INS)	NPD (IN)	REMARKS	END ONE	END TWO	ITEM CODE
<9>	4 3/8"	1/2		SQ.CUT	BEVEL	MTAAZZBPZZZZZXAQ
<10>	2'5 3/16"	1/2	PLD BEND(2)	SQ.CUT	SQ.CUT	MTAAZZBPZZZZZXAQ
<11>	4'0 7/16"	1		SQ.CUT	SQ.CUT	MTAAZZBPZZZZZXAQ

ACTUAL QUANTITY OF WASHERS SHALL BE CONSIDERED AS HALF OF THE QUANTITY SHOWN ON ISOMETRICS

NOTES:
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7. ⟨XXX## ###⟩ INDICATES PIPING TERMINAL POINT.

PIPING SPEC:	SDQ	PAINTING SPEC: PER PROJECT SPECIFICATION	REVISION RECORD					
FLUID CODE:	HW, FIRE FIGHTING WATER	INTERIOR SURFACE TREATMENT SPEC: LU-00129	REV. #	DESCRIPTION	BY	CHK	APP	DATE
MAX. PRESS.:	2030.00 psig / 14.00 MPag							
TEST PRESS.:	3045.00 psig / 20.99 MPag							
MAX. TEMP:	100 F / 38 C							
INSULATION SPEC:	NOT INSULATED							

WELD TYPE IDENTIFIER:
BW - BUTT WELD
SW - SOCKET WELD
SOF - SLIP-ON FLANGE
SOB - SET-ON BRANCH
LET - OLET, LATROLET OR HALF COUPLING
TW - TACK WELD
MW - MITER WELD

WELD CATEGORY IDENTIFICATION:
S - SHOP
FW - FIELD WELD
FFW - FIELD FIT WELD

SYMBOLOGY:
WELD NUMBER
PART NUMBER (PT NO)
< > CUT LIST PIECE NUMBER

SYMBOL IDENTIFIER:
F - FLANGE
G - GASKET
B - BOLT
N - NUT
W - WASHER
O - O-RING
L - LOCK WIRE

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MITSUBISHI POWER

ORIGIN:	MITSUBISHI POWER-LAKE MARY	DATE:	MITSUBISHI POWER DRAWING NUMBER:	REV. NO.:	SYSTEM CODE:
PREPARED BY:	E.BALATLI	07/23/2024	-	0	64A0A
REVIEWED BY:	M.FAZLIOGLU	07/24/2024	FIRE FIGHTING SYSTEM PIPING		
APPROVED BY:	N.KOZAN	07/25/2024	INTERNAL TO GT PACKAGE		
			FOR FABRICATION & ERECTION		
			GRAND RIVER ENERGY CENTER UNIT 4		
			GRAND RIVER DAM AUTHORITY		
			M501JAC GAS TURBINE		
PROJECT CODE:	K-GREC1	P&ID:			
PIPELINE:		PAGE NUMBER:	FFS_SPACE-IN_MD3.2	5 OF 11	

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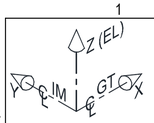
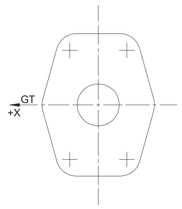
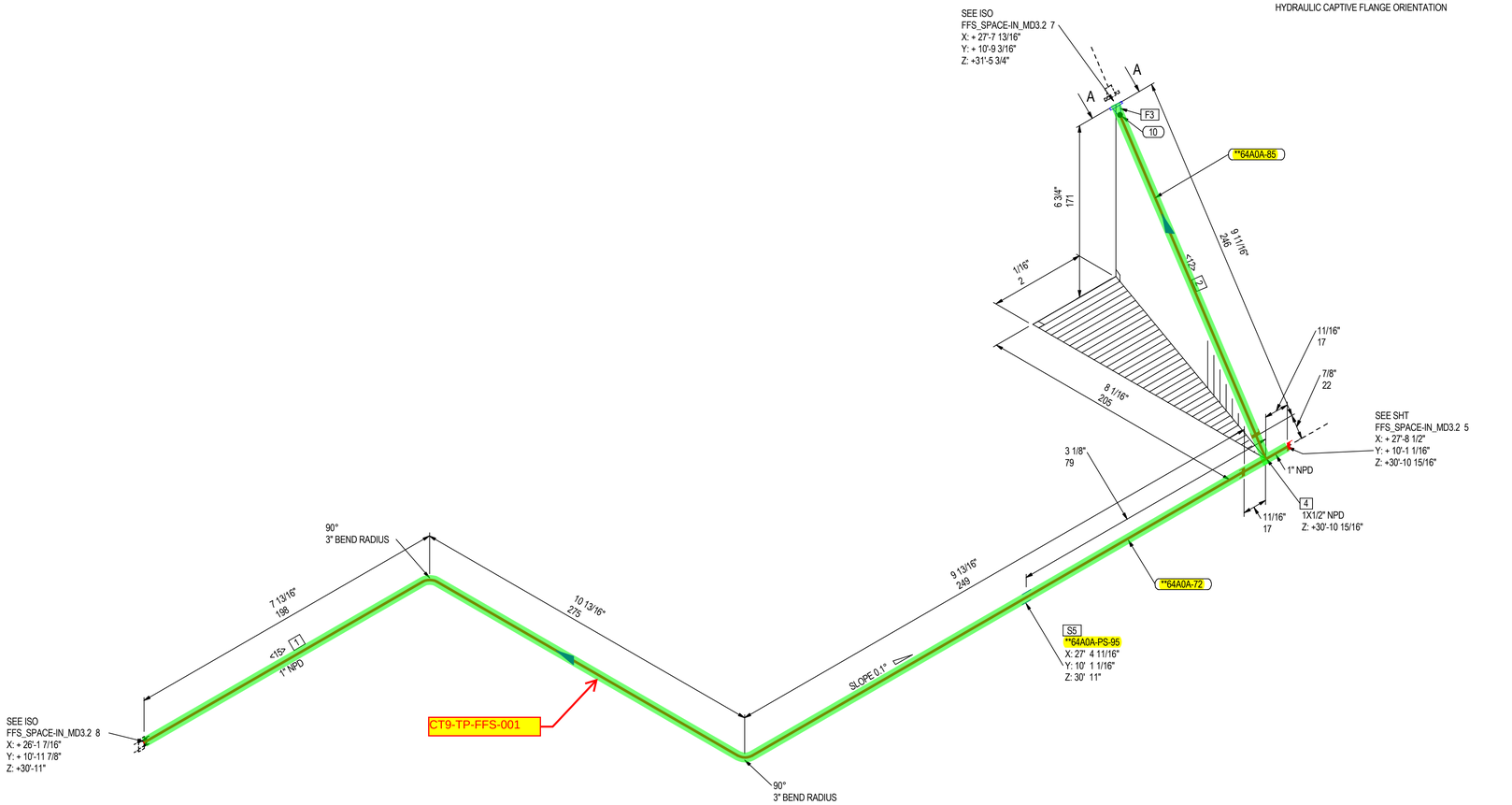




VIEW A-A
HYDRAULIC CAPTIVE FLANGE ORIENTATION



SHOP MATERIALS - FABRICATION ONLY

PT NO	DESCRIPTION	NPD (IN)	CMDTY CODE	QTY
1	TUBE, 0.083 in THK, PE, ASTM A213-TP316, SEAMLESS	1	MTAAZZBPZZZZZXAQ	2.2'
2	TUBE, 0.049 in THK, PE, ASTM A213-TP316, SEAMLESS	1/2	MTAAZZBPZZZZZXAQ	0.8'
3	FLANGE HYDRAULIC, 3000 PSI, FFFE x SWE, SAE J518-1 CODE 61, ASTM A182-F316, CAPTIVE FLANGE	1/2	MFFBAWYGZZIARALS	1

OTHER THAN SHOP MATERIALS - ERECTION ONLY

PT NO	DESCRIPTION	NPD (IN)	CMDTY CODE	QTY
4	TUBE REDUCING UNION TEE, SWGLKE END, SWAGELOK, ASTM A182-F316	1X1/2	MWBCZZCGZZAKWALS	1
	PIPE SUPPORTS			
5	PIPE SUPPORT	1	**64A0A-PS-95	1
	SPOOLS			
	**64A0A-72, **64A0A-85			

CUT PIPE LENGTH

PIECE NO	LENGTH (FT-INS)	NPD (IN)	REMARKS	END ONE	END TWO	ITEM CODE
<12>	9 7/16"	1/2		SQ.CUT	BEVEL	MTAAZZBPZZZZZXAQ
<15>	2'1 7/8"	1	PLD BEND(2)	SQ.CUT	SQ.CUT	MTAAZZBPZZZZZXAQ

WELD NO	WELD TYPE	WELD CAT	NPD	QTY
10	BW	S	1/2"	1

ACTUAL QUANTITY OF WASHERS SHALL BE CONSIDERED AS HALF OF THE QUANTITY SHOWN ON ISOMETRICS

NOTES:
1. THE X AND Y COORDINATES ARE WITH RESPECT TO THE INTERSECTION OF THE CENTERLINE OF THE GAS TURBINE (GT) AND THE INLET MANIFOLD (IM). THE Z (EL) COORDINATE IS WITH RESPECT TO THE TOP OF CONCRETE, WHICH IS 15'-9" [4800mm] BELOW THE CENTERLINE OF THE GT AND IS 0'-0".
2. ALL PIPE SUPPORT COORDINATES ARE WITH RESPECT TO THE CENTERLINE OF THE INSTALLED SUPPORTED PIPE.
3. THE MARK 'FFW' (FIELD FIT WELD) IN THE DRAWING INDICATES EXCESS LENGTH OF APPROX. 4 INCHES IDENTIFIED AS LOOSE IN THE DRAWING TO BE CUT AND ADJUSTED AT THE FIELD.
4. ALL WELDING TO BE DONE AND INSPECTED PER LATEST APPLICABLE EDITION OF ASME B31.1 AND SECTION IX AVAILABLE ON THE PROJECT AWARD DATE.
5. PIPING SHALL BE INSTALLED PER THE INDICATED SLOPE.
6. FOR ANY FURTHER INFORMATION REQUIRED DURING INSTALLATION, CONSULT MITSUBISHI POWER TFA (TECHNICAL FIELD ADVISOR) AT SITE.
7. XXX## ### INDICATES PIPING TERMINAL POINT.

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TEST PRESS.:	3045.00 psig / 20.99 MPag							
MAX. TEMP:	100 F / 38 C							
INSULATION SPEC:	NOT INSULATED							

WELD TYPE IDENTIFIER:
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SYMBOLOGY:
WELD NUMBER
PART NUMBER (PT NO)
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APPROVED BY:	N.KOZAN	07/25/2024	INTERNAL TO GT PACKAGE		
			FOR FABRICATION & ERECTION		
			GRAND RIVER ENERGY CENTER UNIT 4		
			GRAND RIVER DAM AUTHORITY		
			M501JAC GAS TURBINE		
			PROJECT CODE: K-GREC1	P&ID:	
			PIPELINE:	PAGE NUMBER:	
			FFS_SPACE-IN_MD3.2	6 OF 11	

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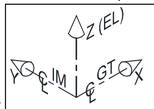

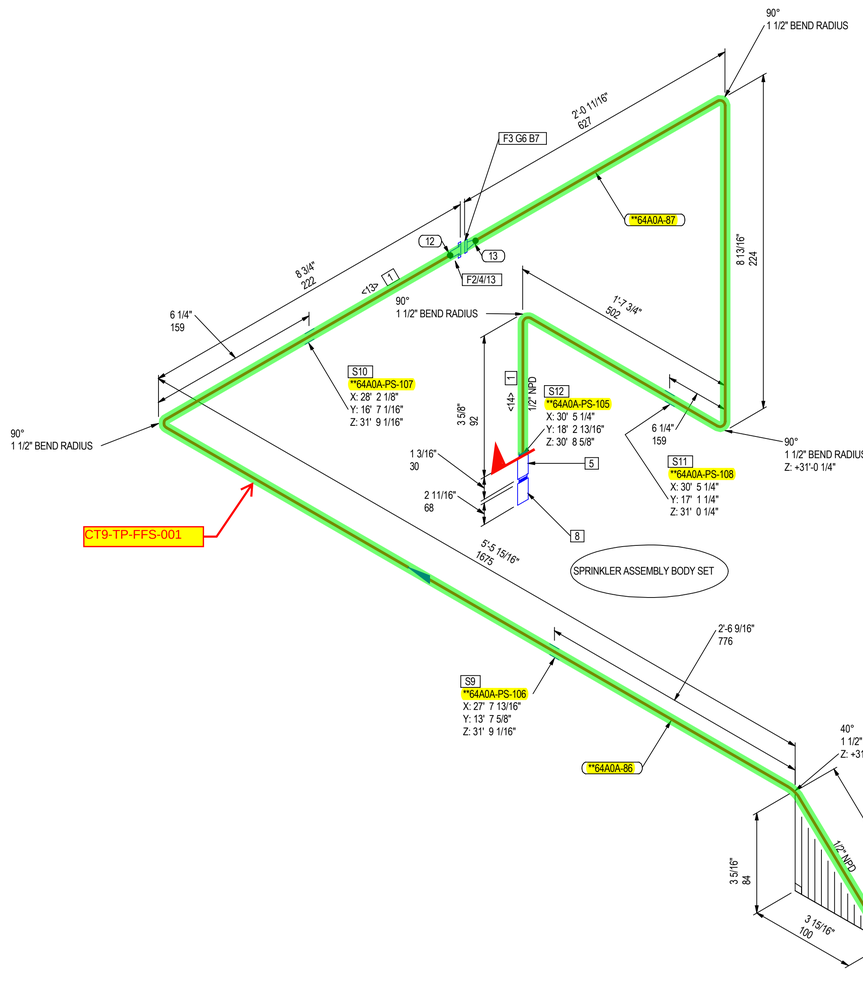

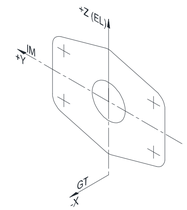

PART NO: (F2)
HYDRAULIC CAPTIVE FLANGE ORIENTATION

SHOP MATERIALS - FABRICATION ONLY

PT NO	DESCRIPTION	NPD (IN)	CMDTY CODE	QTY
1	TUBE, 0.049 in THK, PE, ASTM A213-TP316, SEAMLESS	1/2	MTAAZZBPZZZZZXAQ	11.2'
2	FLANGE HYDRAULIC, 3000 PSI, GROOVED END x SWE, SAE J518-1 CODE 61, ASTM A182-F316, FLANGE HEAD	1/2	MFFAAWYFZZIARALS	2
3	FLANGE HYDRAULIC, 3000 PSI, FFFE x SWE, SAE J518-1 CODE 61, ASTM A182-F316, CAPTIVE FLANGE	1/2	MFFBAWYGZZIARALS	1
	MISCELLANEOUS COMPONENTS			
4	FLANGE HYDRAULIC, 3000 PSI, SAE J518-1 CODE 61, ASTM A182-F316, SPLIT FLANGE CLAMP (ONE PAIR)	1/2	MWABAWYHZZIARALS	2

OTHER THAN SHOP MATERIALS - ERECTION ONLY

PT NO	DESCRIPTION	NPD (IN)	CMDTY CODE	QTY
5	TUBE REDUCER (FRACTIONAL TO METRIC), SWGLKE END x PE, SWAGELOK PART NO.SS-810-R-12M, ASTM A182-F316	1/2	MWAOZZYDZZAKWALS	1
6	GASKET, ORING, 0.125" THK, AS 568, VITON HS90 (DUROMETER90)	1/2	MKBZZACIAQXANUS	2
7	SCREW, DRILLED HEX SOCKET HEAD CAP (2 HOLES), ASME B18.3, ASTM A193-B7, 18-UNC - 1.5 in. Length	5/16	MBCZZBRZZIALBEUZ	8
	INSTRUMENTS			
8	SPRINKLER ASSEMBLY BODY SET FOR HI-FOG® 1000 SERIES SPRINKLER AND SPRAY HEADS, A1000S1CT00D12S10 SPRINKLER ASSEMBLY BODY SET	1/2		1
	PIPE SUPPORTS			
9	PIPE SUPPORT	1/2	**64A0A-PS-106	1
10	PIPE SUPPORT	1/2	**64A0A-PS-107	1
11	PIPE SUPPORT	1/2	**64A0A-PS-108	1
12	NOZZLE MOUNTING ADAPTER SUPPORT	1/2	**64A0A-PS-105	1
	MISCELLANEOUS COMPONENTS			
13	LOCK WIRE, 1mm DIA x 1m LENGTH, ASTM A580-304	1/2	MWRZZZZBAIASXAPU	2

SPOOLS

**64A0A-86, **64A0A-87

CUT PIPE LENGTH

PIECE NO	LENGTH (FT-INS)	NPD (IN)	REMARKS	END ONE	END TWO	ITEM CODE
<13>	6'6 9/16"	1/2	PLD BEND(2)	SQ.CUT	SQ.CUT	MTAAZZBPZZZZZXAQ
<14>	4'6 11/16"	1/2	PLD BEND(3)	BEVEL	SQ.CUT	MTAAZZBPZZZZZXAQ

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NOTES:
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5. PIPING SHALL BE INSTALLED PER THE INDICATED SLOPE.
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7. ⟨XXX## ###⟩ INDICATES PIPING TERMINAL POINT.

ADDITIONAL NOTE:
TUBE REDUCER (FRACTIONAL TO METRIC) SHALL BE IDENTIFIED BY COLOR OR TAGGING FOR QUICK NOTATION IN THE FIELD.

WELD NO	WELD TYPE	WELD CAT	NPD	QTY
11	BW	S	1/2"	1
12	BW	S	1/2"	1
13	BW	S	1/2"	1

PIPING SPEC:	SDQ	PAINTING SPEC: PER PROJECT SPECIFICATION	REVISION RECORD				
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SYMBOLOGY:
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PART NUMBER (PT NO)
< > CUT LIST PIECE NUMBER

SYMBOL IDENTIFIER:
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ORIGIN:	MITSUBISHI POWER-LAKE MARY	DATE:	MITSUBISHI POWER DRAWING NUMBER: / REV. NO.: / SYSTEM CODE:
PREPARED BY:	E.BALATLI	07/23/2024	- / 0 / 64A0A
REVIEWED BY:	M.FAZLIOGLU	07/24/2024	FIRE FIGHTING SYSTEM PIPING
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			FOR FABRICATION & ERECTION
			GRAND RIVER ENERGY CENTER UNIT 4
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PIPELINE:		PAGE NUMBER:	FFS_SPACE-IN_MD3.2 7 OF 11

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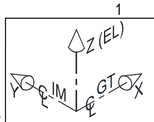
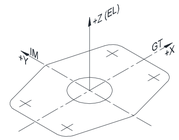
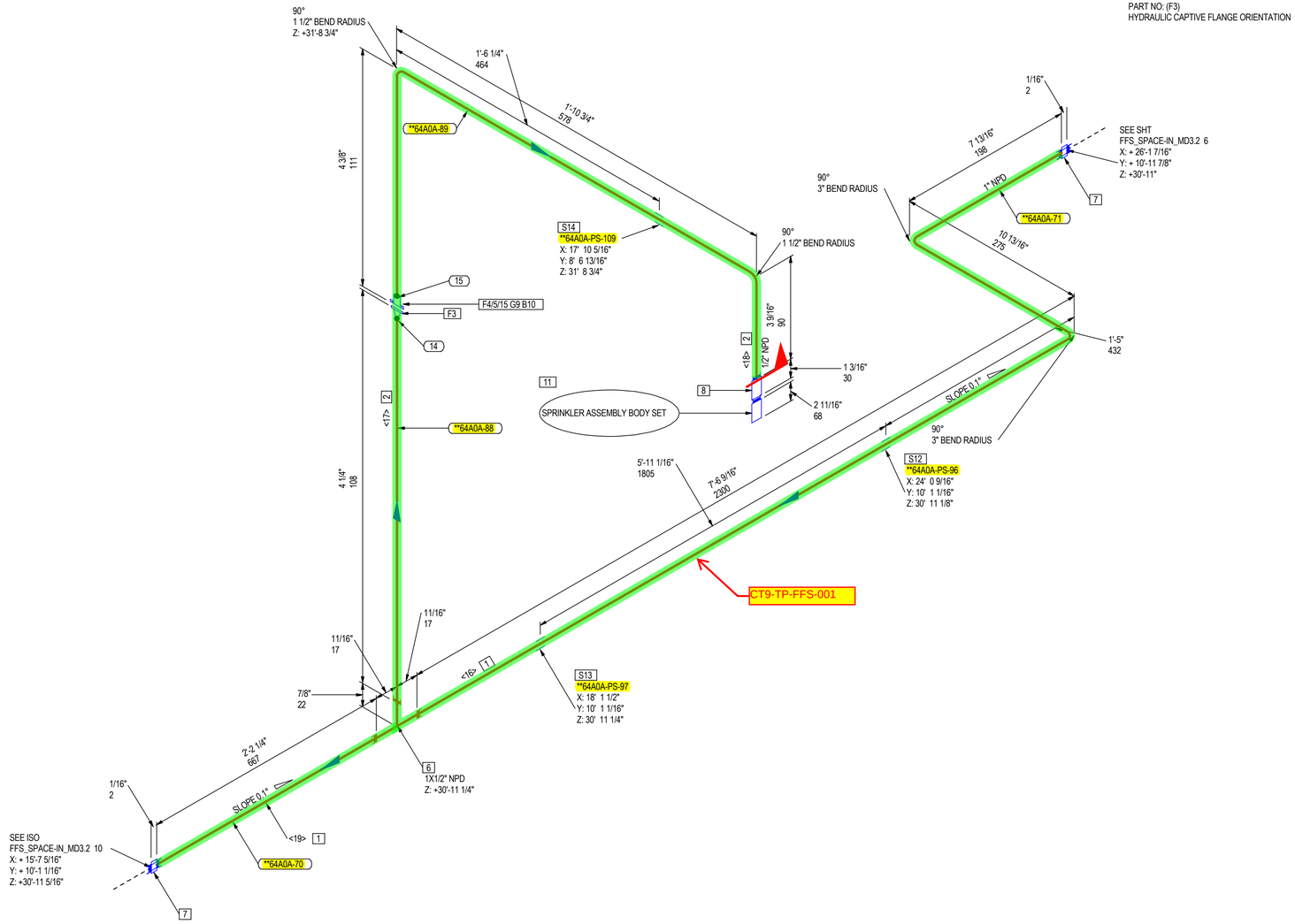




PART NO: (F3)
HYDRAULIC CAPTIVE FLANGE ORIENTATION



ADDITIONAL NOTE:
TUBE REDUCER (FRACTIONAL TO METRIC) SHALL BE IDENTIFIED BY COLOR OR TAGGING FOR QUICK NOTATION IN THE FIELD.

WELD NO	WELD TYPE	WELD CAT	NPD	QTY
14	BW	S	1/2"	1
15	BW	S	1/2"	1

SHOP MATERIALS - FABRICATION ONLY

PT NO	DESCRIPTION	NPD (IN)	CMDTY CODE	QTY
1	TUBE, 0.083 in THK, PE, ASTM A213-TP316, SEAMLESS	1	MTAAZZBPZZZZZXAQ	11.1'
2	TUBE, 0.049 in THK, PE, ASTM A213-TP316, SEAMLESS	1/2	MTAAZZBPZZZZZXAQ	2.8'
3	FLANGE HYDRAULIC, 3000 PSI, FFFE x SWE, SAE J518-1 CODE 61, ASTM A182-F316, CAPTIVE FLANGE	1/2	MFFBAWYGZZIARALS	1
4	FLANGE HYDRAULIC, 3000 PSI, GROOVED END x SWE, SAE J518-1 CODE 61, ASTM A182-F316, FLANGE HEAD	1/2	MFFAAWYFZZIARALS	1
	MISCELLANEOUS COMPONENTS			
5	FLANGE HYDRAULIC, 3000 PSI, SAE J518-1 CODE 61, ASTM A182-F316, SPLIT FLANGE CLAMP (ONE PAIR)	1/2	MWABAWYHZZIARALS	1

OTHER THAN SHOP MATERIALS - ERECTION ONLY

PT NO	DESCRIPTION	NPD (IN)	CMDTY CODE	QTY
6	TUBE REDUCING UNION TEE, SWGLKE END, SWAGELOK, ASTM A182-F316	1X1/2	MWBCZZCGZZAKWALS	1
7	TUBE UNION, SWGLKE END, SWAGELOK PART NO. , ASTM A182-F316	1	MWADZZCGZZAKWALS	2
8	TUBE REDUCER (FRACTIONAL TO METRIC), SWGLKE END x PE, SWAGELOK PART NO.SS-810-R-12M, ASTM A182-F316	1/2	MWAOZZYDZZAKWALS	1
9	GASKET, ORING, 0.125" THK, AS 568, VITON HS90 (DUROMETER90)	1/2	MKBZZACIAQXANUS	1
10	SCREW, DRILLED HEX SOCKET HEAD CAP (2 HOLES), ASME B18.3, ASTM A193-B7, 18-UNC - 1.5 in. Length	5/16	MBCZZBRZZIALBEUZ	4
	INSTRUMENTS			
11	SPRINKLER ASSEMBLY BODY SET FOR HI-FOG® 1000 SERIES SPRINKLER AND SPRAY HEADS, A1000S1CT00D12S10 SPRINKLER ASSEMBLY BODY SET	1/2		1
	PIPE SUPPORTS			
12	PIPE SUPPORT	1	**64A0A-PS-96	1
13	PIPE SUPPORT	1	**64A0A-PS-97	1
14	PIPE SUPPORT	1/2	**64A0A-PS-109	1
	MISCELLANEOUS COMPONENTS			
15	LOCK WIRE, 1mm DIA x 1m LENGTH, ASTM A580-304	1/2	MWRZZZZBAIASXAPU	1
	SPOOLS			

**64A0A-70, **64A0A-71, **64A0A-88, **64A0A-89

SEE SHT 9

ACTUAL QUANTITY OF WASHERS SHALL BE CONSIDERED AS HALF OF THE QUANTITY SHOWN ON ISOMETRICS

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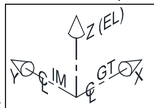

SEE SHT 8

CUT PIPE LENGTH

PIECE NO	LENGTH (FT-INS)	NPD (IN)	REMARKS	END ONE	END TWO	ITEM CODE
<16>	8'10 5/8"	1	PLD BEND(2)	SQ.CUT	SQ.CUT	MTAAZZBPZZZZZXAQ
<17>	4"	1/2		SQ.CUT	BEVEL	MTAAZZBPZZZZZXAQ
<18>	2'5 3/16"	1/2	PLD BEND(2)	SQ.CUT	SQ.CUT	MTAAZZBPZZZZZXAQ
<19>	2'2 1/4"	1		SQ.CUT	SQ.CUT	MTAAZZBPZZZZZXAQ

ACTUAL QUANTITY OF WASHERS SHALL BE CONSIDERED AS HALF OF THE QUANTITY SHOWN ON ISOMETRICS

NOTES:
1. THE X AND Y COORDINATES ARE WITH RESPECT TO THE INTERSECTION OF THE CENTERLINE OF THE GAS TURBINE (GT) AND THE INLET MANIFOLD (IM). THE Z (EL) COORDINATE IS WITH RESPECT TO THE TOP OF CONCRETE, WHICH IS 15'-9" [4800mm] BELOW THE CENTERLINE OF THE GT AND IS 0'-0".
2. ALL PIPE SUPPORT COORDINATES ARE WITH RESPECT TO THE CENTERLINE OF THE INSTALLED SUPPORTED PIPE.
3. THE MARK 'FFW' (FIELD FIT WELD) IN THE DRAWING INDICATES EXCESS LENGTH OF APPROX. 4 INCHES IDENTIFIED AS LOOSE IN THE DRAWING TO BE CUT AND ADJUSTED AT THE FIELD.
4. ALL WELDING TO BE DONE AND INSPECTED PER LATEST APPLICABLE EDITION OF ASME B31.1 AND SECTION IX AVAILABLE ON THE PROJECT AWARD DATE.
5. PIPING SHALL BE INSTALLED PER THE INDICATED SLOPE.
6. FOR ANY FURTHER INFORMATION REQUIRED DURING INSTALLATION, CONSULT MITSUBISHI POWER TFA (TECHNICAL FIELD ADVISOR) AT SITE.
7. ⟨XXX## ###⟩ INDICATES PIPING TERMINAL POINT.

ORIGIN:	MITSUBISHI POWER-LAKE MARY	DATE:	MITSUBISHI POWER DRAWING NUMBER:	REV. NO.:	SYSTEM CODE:
PREPARED BY:	E.BALATLI	07/23/2024	-	0	64A0A
REVIEWED BY:	M.FAZLIOGLU	07/24/2024	FIRE FIGHTING SYSTEM PIPING		
APPROVED BY:	N.KOZAN	07/25/2024	INTERNAL TO GT PACKAGE		
			FOR FABRICATION & ERECTION		
			GRAND RIVER ENERGY CENTER UNIT 4		
			GRAND RIVER DAM AUTHORITY		
			M501JAC GAS TURBINE		
			PROJECT CODE: K-GREC1	P&ID:	
			PIPELINE: FFS_SPACE-IN_MD3.2	PAGE NUMBER: 9 OF 11	

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SL2

MITSUBISHI POWER ENGINEERING
PLANT ENGINEERING

DEPARTMENT:
SECTION:

DRAWING DISTRIBUTION LIST

PIPING SPEC:	SDQ
FLUID CODE:	HW, FIRE FIGHTING WATER
MAX. PRESS.:	2030.00 psig / 14.00 MPag
TEST PRESS.:	3045.00 psig / 20.99 MPag
MAX. TEMP:	100 F / 38 C
INSULATION SPEC:	NOT INSULATED

PAINTING SPEC: PER PROJECT SPECIFICATION
INTERIOR SURFACE TREATMENT SPEC: LU-00129

REVISION RECORD

REV. #	DESCRIPTION	BY	CHK	APP	DATE

WELD TYPE IDENTIFIER:
BW - BUTT WELD
SW - SOCKET WELD
SOF - SLIP-ON FLANGE
SOB - SET-ON BRANCH
LET - OLET, LATROLET OR HALF COUPLING
TW - TACK WELD
MW - MITER WELD

WELD CATEGORY IDENTIFICATION:
S - SHOP
FW - FIELD WELD
FFW - FIELD FIT WELD

SYMBOLOGY:
WELD NUMBER
PART NUMBER (PT NO)
< > CUT LIST PIECE NUMBER

SYMBOL IDENTIFIER:
F - FLANGE
G - GASKET
B - BOLT
N - NUT
W - WASHER
O - O-RING
L - LOCK WIRE

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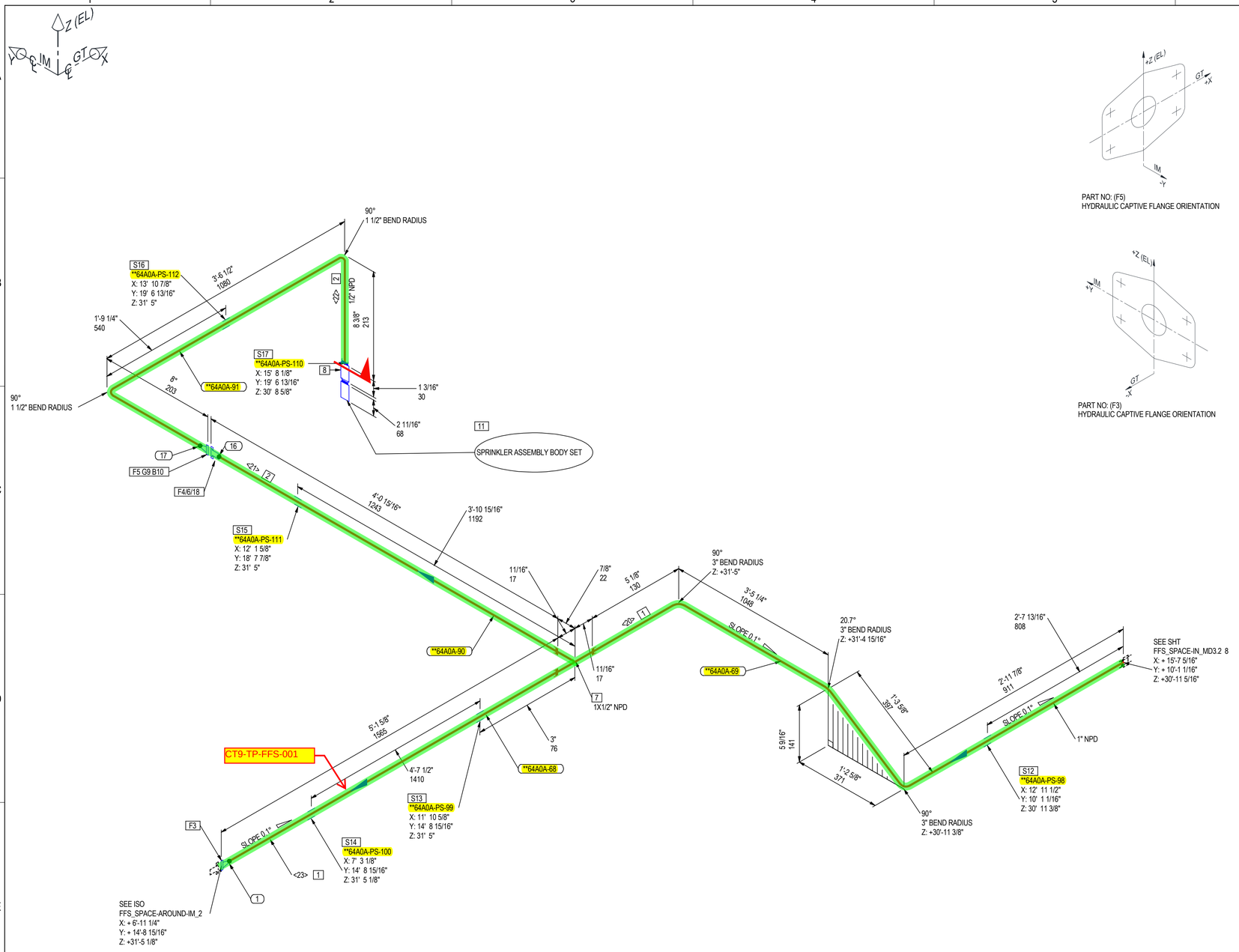

PART NO: (F5)
HYDRAULIC CAPTIVE FLANGE ORIENTATION

PART NO: (F3)
HYDRAULIC CAPTIVE FLANGE ORIENTATION

SHOP MATERIALS - FABRICATION ONLY

PT NO	DESCRIPTION	NPD (IN)	CMDTY CODE	QTY
1	TUBE, 0.083 in THK, PE, ASTM A213-TP316, SEAMLESS	1	MTAAZZBPZZZZZXAQ	13.1'
2	TUBE, 0.049 in THK, PE, ASTM A213-TP316, SEAMLESS	1/2	MTAAZZBPZZZZZXAQ	8.9'
3	FLANGE HYDRAULIC, 3000 PSI, FFFE x SWE, SAE J518-1 CODE 61, ASTM A182-F316, CAPTIVE FLANGE	1	MFFBAWYGZZIARALS	1
4	FLANGE HYDRAULIC, 3000 PSI, GROOVED END x SWE, SAE J518-1 CODE 61, ASTM A182-F316, FLANGE HEAD	1/2	MFFAAWYFZZIARALS	1
5	FLANGE HYDRAULIC, 3000 PSI, FFFE x SWE, SAE J518-1 CODE 61, ASTM A182-F316, CAPTIVE FLANGE	1/2	MFFBAWYGZZIARALS	1
	MISCELLANEOUS COMPONENTS			
6	FLANGE HYDRAULIC, 3000 PSI, SAE J518-1 CODE 61, ASTM A182-F316, SPLIT FLANGE CLAMP (ONE PAIR)	1/2	MWABAWYHZZIARALS	1

OTHER THAN SHOP MATERIALS - ERECTION ONLY

PT NO	DESCRIPTION	NPD (IN)	CMDTY CODE	QTY
7	TUBE REDUCING UNION TEE, SWGLKE END, SWAGELOK, ASTM A182-F316	1X1/2	MWBCZZCGZZAKWALS	1
8	TUBE REDUCER (FRACTIONAL TO METRIC), SWGLKE END x PE, SWAGELOK PART NO.SS-810-R-12M, ASTM A182-F316	1/2	MWAOZZYDZZAKWALS	1
9	GASKET, ORING, 0.125" THK, AS 568, VITON HS90 (DUROMETER90)	1/2	MKBZZACIAQXANUS	1
10	SCREW, DRILLED HEX SOCKET HEAD CAP (2 HOLES), ASME B18.3, ASTM A193-B7, 18-UNC - 1.5 in. Length	5/16	MBCZZBRZZIALBEUZ	4
	INSTRUMENTS			
11	SPRINKLER ASSEMBLY BODY SET FOR HI-FOG® 1000 SERIES SPRINKLER AND SPRAY HEADS, A1000S1CT00D12S10 SPRINKLER ASSEMBLY BODY SET	1/2		1
	PIPE SUPPORTS			
12	PIPE SUPPORT	1	**64A0A-PS-98	1
13	PIPE SUPPORT	1	**64A0A-PS-99	1
14	PIPE SUPPORT	1	**64A0A-PS-100	1
15	PIPE SUPPORT	1/2	**64A0A-PS-111	1
16	PIPE SUPPORT	1/2	**64A0A-PS-112	1
17	NOZZLE MOUNTING ADAPTER SUPPORT	1/2	**64A0A-PS-110	1
	MISCELLANEOUS COMPONENTS			
18	LOCK WIRE, 1mm DIA x 1m LENGTH, ASTM A580-304	1/2	MWRZZZZBAIASXAPU	1

SPOOLS

**64A0A-68, **64A0A-69, **64A0A-90, **64A0A-91

SEE SHT 11

ACTUAL QUANTITY OF WASHERS SHALL BE CONSIDERED AS HALF OF THE QUANTITY SHOWN ON ISOMETRICS

NOTES:
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6. FOR ANY FURTHER INFORMATION REQUIRED DURING INSTALLATION, CONSULT MITSUBISHI POWER TFA (TECHNICAL FIELD ADVISOR) AT SITE.
7. <XXX## ###> INDICATES PIPING TERMINAL POINT.

ADDITIONAL NOTE:
TUBE REDUCER (FRACTIONAL TO METRIC) SHALL BE IDENTIFIED BY COLOR OR TAGGING FOR QUICK NOTATION IN THE FIELD.

WELD NO	WELD TYPE	WELD CAT	NPD	QTY
16	BW	S	1/2"	1
17	BW	S	1/2"	1
1	BW	S	1"	1

PIPING SPEC:	SDQ	PAINTING SPEC: PER PROJECT SPECIFICATION	REVISION RECORD				
FLUID CODE:	HW, FIRE FIGHTING WATER	INTERIOR SURFACE TREATMENT SPEC: LU-00129	REV. #	DESCRIPTION	BY	CHK	APP	DATE
MAX. PRESS.:	2030.00 psig / 14.00 MPag							
TEST PRESS.:	3045.00 psig / 20.99 MPag							
MAX. TEMP:	100 F / 38 C							
INSULATION SPEC:	NOT INSULATED							

WELD TYPE IDENTIFIER:
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FFW - FIELD FIT WELD

SYMBOLOGY:
WELD NUMBER
PART NUMBER (PT NO)
< > CUT LIST PIECE NUMBER

SYMBOL IDENTIFIER:
F - FLANGE
G - GASKET
B - BOLT
N - NUT
W - WASHER
O - O-RING
L - LOCK WIRE

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MITSUBISHI POWER

ORIGIN:	MITSUBISHI POWER-LAKE MARY	DATE:	MITSUBISHI POWER DRAWING NUMBER:	REV. NO.:	SYSTEM CODE:
PREPARED BY:	E.BALATLI	07/23/2024	-	0	64A0A
REVIEWED BY:	M.FAZLIOGLU	07/24/2024	FIRE FIGHTING SYSTEM PIPING		
APPROVED BY:	N.KOZAN	07/25/2024	INTERNAL TO GT PACKAGE		
			FOR FABRICATION & ERECTION		
			GRAND RIVER ENERGY CENTER UNIT 4		
			GRAND RIVER DAM AUTHORITY		
			M501JAC GAS TURBINE		
PROJECT CODE:	K-GREC1	P&ID:			
PIPELINE:		PAGE NUMBER:	FFS_SPACE-IN_MD3.2	10OF 11	

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PLANT ENGINEERING
DEPARTMENT:
SECTION:
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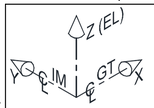

SEE SHT 10

CUT PIPE LENGTH

PIECE NO	LENGTH (FT-INS)	NPD (IN)	REMARKS	END ONE	END TWO	ITEM CODE
<20>	7'11 5/16"	1	PLD BEND(3)	SQ.CUT	SQ.CUT	MTAAZZBPZZZZZXAQ
<21>	4'0 3/4"	1/2		SQ.CUT	SQ.CUT	MTAAZZBPZZZZZXAQ
<22>	4'9 3/8"	1/2	PLD BEND(2)	BEVEL	SQ.CUT	MTAAZZBPZZZZZXAQ
<23>	5'1 5/16"	1		SQ.CUT	BEVEL	MTAAZZBPZZZZZXAQ

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7. ⟨XXX## ###⟩ INDICATES PIPING TERMINAL POINT.

ORIGIN:	MITSUBISHI POWER-LAKE MARY	DATE:	MITSUBISHI POWER DRAWING NUMBER:	REV. NO.:	SYSTEM CODE:
PREPARED BY:	E.BALATLI	07/23/2024	-	0	64A0A
REVIEWED BY:	M.FAZLIOGLU	07/24/2024	FIRE FIGHTING SYSTEM PIPING		
APPROVED BY:	N.KOZAN	07/25/2024	INTERNAL TO GT PACKAGE		
			FOR FABRICATION & ERECTION		
			GRAND RIVER ENERGY CENTER UNIT 4		
			GRAND RIVER DAM AUTHORITY		
			M501JAC GAS TURBINE		
			PROJECT CODE: K-GREC1	P&ID:	
			PIPELINE: FFS_SPACE-IN_MD3.2	PAGE NUMBER: 11OF 11	

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MITSUBISHI POWER ENGINEERING
PLANT ENGINEERING

DEPARTMENT:
SECTION:

DRAWING DISTRIBUTION LIST

PIPING SPEC:	SDQ
FLUID CODE:	HW, FIRE FIGHTING WATER
MAX. PRESS.:	2030.00 psig / 14.00 MPag
TEST PRESS.:	3045.00 psig / 20.99 MPag
MAX. TEMP:	100 F / 38 C
INSULATION SPEC:	NOT INSULATED

PAINTING SPEC: PER PROJECT SPECIFICATION
INTERIOR SURFACE TREATMENT SPEC: LU-00129

REVISION RECORD

REV. #	DESCRIPTION	BY	CHK	APP	DATE

WELD TYPE IDENTIFIER:
BW - BUTT WELD
SW - SOCKET WELD
SOF - SLIP-ON FLANGE
SOB - SET-ON BRANCH
LET - OLET, LATROLET OR HALF COUPLING
TW - TACK WELD
MW - MITER WELD

WELD CATEGORY IDENTIFICATION:
S - SHOP
FW - FIELD WELD
FFW - FIELD FIT WELD

SYMBOLOGY:
WELD NUMBER
PART NUMBER (PT NO)
< > CUT LIST PIECE NUMBER

SYMBOL IDENTIFIER:
F - FLANGE
G - GASKET
B - BOLT
N - NUT
W - WASHER
O - O-RING
L - LOCK WIRE

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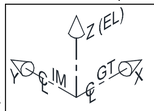

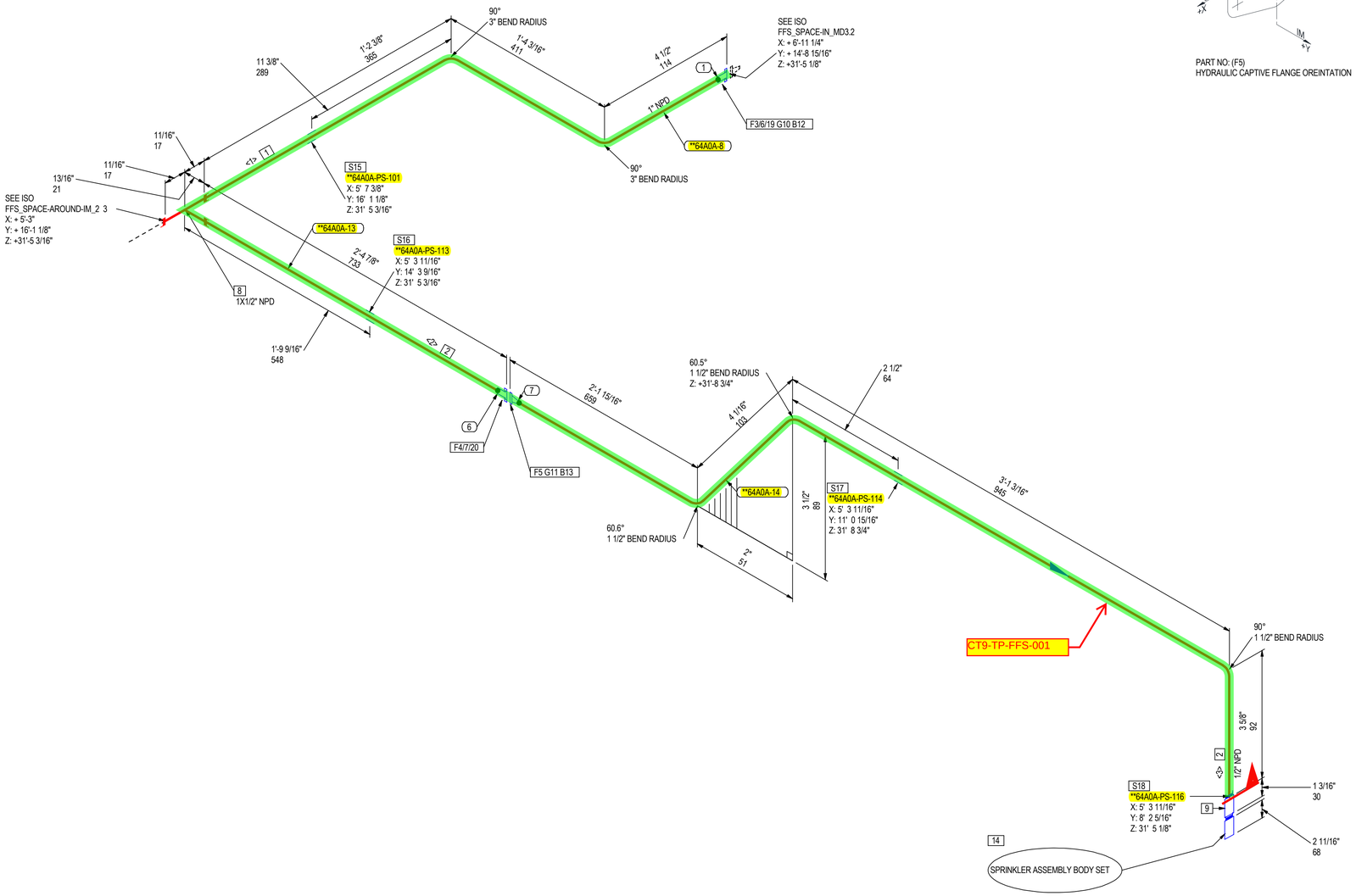

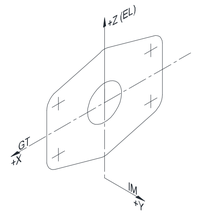

PART NO: (F5)
HYDRAULIC CAPTIVE FLANGE OREINTATION

SHOP MATERIALS - FABRICATION ONLY

PT NO	DESCRIPTION	NPD (IN)	CMDTY CODE	QTY
1	TUBE, 0.083 in THK, PE, ASTM A213-TP316, SEAMLESS	1	MTAAZZBPZZZZZXAQ	2.7'
2	TUBE, 0.049 in THK, PE, ASTM A213-TP316, SEAMLESS	1/2	MTAAZZBPZZZZZXAQ	8.2'
3	FLANGE HYDRAULIC, 3000 PSI, GROOVED END x SWE, SAE J518-1 CODE 61, ASTM A182-F316, FLANGE HEAD	1	MFFAAWYFZZIARALS	1
4	FLANGE HYDRAULIC, 3000 PSI, GROOVED END x SWE, SAE J518-1 CODE 61, ASTM A182-F316, FLANGE HEAD	1/2	MFFAAWYFZZIARALS	1
5	FLANGE HYDRAULIC, 3000 PSI, FFFE x SWE, SAE J518-1 CODE 61, ASTM A182-F316, CAPTIVE FLANGE	1/2	MFFBAWYGZZIARALS	1
	MISCELLANEOUS COMPONENTS			
6	FLANGE HYDRAULIC, 3000 PSI, SAE J518-1 CODE 61, ASTM A182-F316, SPLIT FLANGE CLAMP (ONE PAIR)	1	MWABAWYHZZIARALS	1
7	FLANGE HYDRAULIC, 3000 PSI, SAE J518-1 CODE 61, ASTM A182-F316, SPLIT FLANGE CLAMP (ONE PAIR)	1/2	MWABAWYHZZIARALS	1

OTHER THAN SHOP MATERIALS - ERECTION ONLY

PT NO	DESCRIPTION	NPD (IN)	CMDTY CODE	QTY
8	TUBE REDUCING UNION TEE, SWGLKE END, SWAGELOK, ASTM A182-F316	1X1/2	MWBCZZCGZZAKWALS	1
9	TUBE REDUCER (FRACTIONAL TO METRIC), SWGLKE END x PE, SWAGELOK PART NO.SS-810-R-12M, ASTM A182-F316	1/2	MWAOZZYDZZAKWALS	1
10	GASKET, ORING, 0.125" THK, AS 568, VITON HS90 (DUROMETER90)	1	MKBZZACIAQXANUS	1
11	GASKET, ORING, 0.125" THK, AS 568, VITON HS90 (DUROMETER90)	1/2	MKBZZACIAQXANUS	1
12	SCREW, DRILLED HEX SOCKET HEAD CAP (2 HOLES), ASME B18.3, ASTM A193-B7, 16-UNC - 1.75 in. Length	3/8	MBCZZABZZIALBEUZ	4
13	SCREW, DRILLED HEX SOCKET HEAD CAP (2 HOLES), ASME B18.3, ASTM A193-B7, 18-UNC - 1.5 in. Length	5/16	MBCZZBRZZIALBEUZ	4
	INSTRUMENTS			
14	SPRINKLER ASSEMBLY BODY SET FOR HI-FOG® 1000 SERIES SPRINKLER AND SPRAY HEADS, A1000S1CT00D12S10 SPRINKLER ASSEMBLY BODY SET	1/2		1
	PIPE SUPPORTS			
15	PIPE SUPPORT	1	**64A0A-PS-101	1
16	PIPE SUPPORT	1/2	**64A0A-PS-113	1
17	PIPE SUPPORT	1/2	**64A0A-PS-114	1
18	NOZZLE MOUNTING ADAPTER SUPPORT	1/2	**64A0A-PS-116	1

SEE SHT 2

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WELD NO	WELD TYPE	WELD CAT	NPD	QTY
1	BW	S	1"	1
6	BW	S	1/2"	1
7	BW	S	1/2"	1

PIPING SPEC:	SDQ	PAINTING SPEC: PER PROJECT SPECIFICATION	REVISION RECORD				
FLUID CODE:	HW, FIRE FIGHTING WATER	INTERIOR SURFACE TREATMENT SPEC: LU-00129	REV. #	DESCRIPTION	BY	CHK	APP	DATE
MAX. PRESS.:	2030.00 psig / 14.00 MPag							
TEST PRESS.:	3045.00 psig / 20.99 MPag							
MAX. TEMP:	100 F / 38 C							
INSULATION SPEC:	NOT INSULATED							

WELD TYPE IDENTIFIER:
BW - BUTT WELD
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WELD CATEGORY IDENTIFICATION:
S - SHOP
FW - FIELD WELD
FFW - FIELD FIT WELD

SYMBOLOGY:
WELD NUMBER
PART NUMBER (PT NO)
< > CUT LIST PIECE NUMBER

SYMBOL IDENTIFIER:
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B - BOLT
N - NUT
W - WASHER
O - O-RING
L - LOCK WIRE

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MITSUBISHI POWER

ORIGIN:	MITSUBISHI POWER-LAKE MARY	DATE:	MITSUBISHI POWER DRAWING NUMBER:	REV. NO.:	SYSTEM CODE:
PREPARED BY:	E.BALATLI	07/23/2024	-	0	64A0A
REVIEWED BY:	M.FAZLIOGLU	07/24/2024	FIRE FIGHTING SYSTEM PIPING		
APPROVED BY:	N.KOZAN	07/25/2024	INTERNAL TO GT PACKAGE		
			FOR FABRICATION & ERECTION		
			GRAND RIVER ENERGY CENTER UNIT 4		
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			M501JAC GAS TURBINE		
PROJECT CODE:	K-GREC1	P&ID:			
PIPELINE:	FFS_SPACE-AROUND-IM_2	PAGE NUMBER:	1 OF 4		

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PLANT ENGINEERING

DEPARTMENT:
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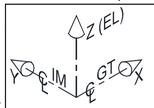

SEE SHT 1

MISCELLANEOUS COMPONENTS

19	LOCK WIRE, 1mm DIA x 1m LENGTH, ASTM A580-304	1	MWRZZZZBAIASXAPU	1
20	LOCK WIRE, 1mm DIA x 1m LENGTH, ASTM A580-304	1/2	MWRZZZZBAIASXAPU	1

SPOOLS

**64A0A-8, **64A0A-13, **64A0A-14

CUT PIPE LENGTH

PIECE NO	LENGTH (FT-INS)	NPD (IN)	REMARKS	END ONE	END TWO	ITEM CODE
<1>	2'8 1/4"	1	PLD BEND(2)	SQ.CUT	SQ.CUT	MTAAZZBPZZZZZXAQ
<2>	2'4 5/8"	1/2		SQ.CUT	SQ.CUT	MTAAZZBPZZZZZXAQ
<3>	5'9 1/2"	1/2	PLD BEND(3)	BEVEL	SQ.CUT	MTAAZZBPZZZZZXAQ

ACTUAL QUANTITY OF WASHERS SHALL BE CONSIDERED AS HALF OF THE QUANTITY SHOWN ON ISOMETRICS

NOTES:
1. THE X AND Y COORDINATES ARE WITH RESPECT TO THE INTERSECTION OF THE CENTERLINE OF THE GAS TURBINE (GT) AND THE INLET MANIFOLD (IM). THE Z (EL) COORDINATE IS WITH RESPECT TO THE TOP OF CONCRETE, WHICH IS 15'-9" [4800mm] BELOW THE CENTERLINE OF THE GT AND IS 0'-0".
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4. ALL WELDING TO BE DONE AND INSPECTED PER LATEST APPLICABLE EDITION OF ASME B31.1 AND SECTION IX AVAILABLE ON THE PROJECT AWARD DATE.
5. PIPING SHALL BE INSTALLED PER THE INDICATED SLOPE.
6. FOR ANY FURTHER INFORMATION REQUIRED DURING INSTALLATION, CONSULT MITSUBISHI POWER TFA (TECHNICAL FIELD ADVISOR) AT SITE.
7. XXX## ### INDICATES PIPING TERMINAL POINT.

ORIGIN:	MITSUBISHI POWER-LAKE MARY	DATE:	MITSUBISHI POWER DRAWING NUMBER:	REV. NO.:	SYSTEM CODE:
PREPARED BY:	E.BALATLI	07/23/2024	-	0	64A0A
REVIEWED BY:	M.FAZLIOGLU	07/24/2024	FIRE FIGHTING SYSTEM PIPING		
APPROVED BY:	N.KOZAN	07/25/2024	INTERNAL TO GT PACKAGE		

FOR FABRICATION & ERECTION
GRAND RIVER ENERGY CENTER UNIT 4
GRAND RIVER DAM AUTHORITY
M501JAC GAS TURBINE

PROJECT CODE: K-GREC1 P&ID:
PIPELINE: FFS_SPACE-AROUND-IM_2 PAGE NUMBER: 2 OF 4

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MITSUBISHI POWER

PIPING SPEC:	SDQ	PAINTING SPEC: PER PROJECT SPECIFICATION	REVISION RECORD				
FLUID CODE:	HW, FIRE FIGHTING WATER	INTERIOR SURFACE TREATMENT SPEC: LU-00129	REV. #	DESCRIPTION	BY	CHK	APP	DATE
MAX. PRESS.:	2030.00 psig / 14.00 MPag							
TEST PRESS.:	3045.00 psig / 20.99 MPag							
MAX. TEMP:	100 F / 38 C							
INSULATION SPEC:	NOT INSULATED							

WELD TYPE IDENTIFIER:
BW - BUTT WELD
SW - SOCKET WELD
SOF - SLIP-ON FLANGE
SOB - SET-ON BRANCH
LET - OLET, LATROLET OR HALF COUPLING
TW - TACK WELD
MW - MITER WELD

WELD CATEGORY IDENTIFICATION:
S - SHOP
FW - FIELD WELD
FFW - FIELD FIT WELD

SYMBOLOGY:
WELD NUMBER
PART NUMBER (PT NO)
< > CUT LIST PIECE NUMBER

SYMBOL IDENTIFIER:
F - FLANGE
G - GASKET
B - BOLT
N - NUT
W - WASHER
O - O-RING
L - LOCK WIRE

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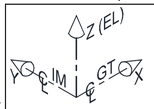

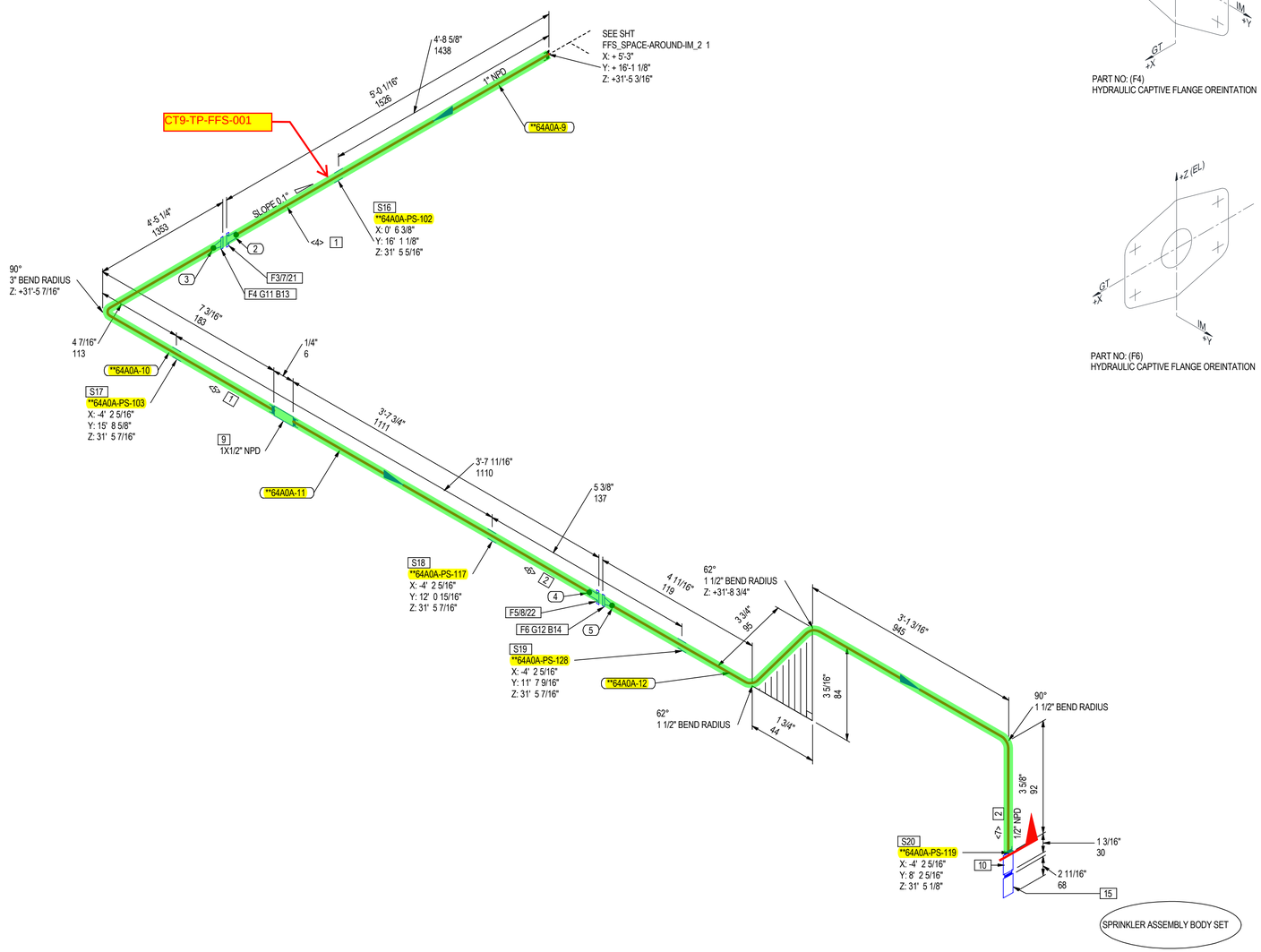

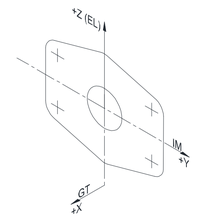

PART NO: (F4)
HYDRAULIC CAPTIVE FLANGE OREINTATION

PART NO: (F6)
HYDRAULIC CAPTIVE FLANGE OREINTATION

SHOP MATERIALS - FABRICATION ONLY

PT NO	DESCRIPTION	NPD (IN)	CMDTY CODE	QTY
1	TUBE, 0.083 in THK, PE, ASTM A213-TP316, SEAMLESS	1	MTAAZZBPZZZZZXAQ	9.9'
2	TUBE, 0.049 in THK, PE, ASTM A213-TP316, SEAMLESS	1/2	MTAAZZBPZZZZZXAQ	7.7"
3	FLANGE HYDRAULIC, 3000 PSI, GROOVED END x SWE, SAE J518-1 CODE 61, ASTM A182-F316, FLANGE HEAD	1	MFFAAWYFZZIARALS	1
4	FLANGE HYDRAULIC, 3000 PSI, FFFE x SWE, SAE J518-1 CODE 61, ASTM A182-F316, CAPTIVE FLANGE	1	MFFBAWYGZZIARALS	1
5	FLANGE HYDRAULIC, 3000 PSI, GROOVED END x SWE, SAE J518-1 CODE 61, ASTM A182-F316, FLANGE HEAD	1/2	MFFAAWYFZZIARALS	1
6	FLANGE HYDRAULIC, 3000 PSI, FFFE x SWE, SAE J518-1 CODE 61, ASTM A182-F316, CAPTIVE FLANGE	1/2	MFFBAWYGZZIARALS	1
	MISCELLANEOUS COMPONENTS			
7	FLANGE HYDRAULIC, 3000 PSI, SAE J518-1 CODE 61, ASTM A182-F316, SPLIT FLANGE CLAMP (ONE PAIR)	1	MWABAWYHZZIARALS	1
8	FLANGE HYDRAULIC, 3000 PSI, SAE J518-1 CODE 61, ASTM A182-F316, SPLIT FLANGE CLAMP (ONE PAIR)	1/2	MWABAWYHZZIARALS	1

OTHER THAN SHOP MATERIALS - ERECTION ONLY

PT NO	DESCRIPTION	NPD (IN)	CMDTY CODE	QTY
9	REDUCING TUBE UNION, SWGLKE END, SWAGELOK PART NO. SS-1610-6-8, ASTM A182-F316	1X1/2	MWARZZCGZZAKWALS	1
10	TUBE REDUCER (FRACTIONAL TO METRIC), SWGLKE END x PE, SWAGELOK PART NO.SS-810-R-12M, ASTM A182-F316	1/2	MWAOZZYDZZAKWALS	1
11	GASKET, ORING, 0.125" THK, AS 568, VITON HS90 (DUROMETER90)	1	MKBZZACIAQXANUS	1
12	GASKET, ORING, 0.125" THK, AS 568, VITON HS90 (DUROMETER90)	1/2	MKBZZACIAQXANUS	1
13	SCREW, DRILLED HEX SOCKET HEAD CAP (2 HOLES), ASME B18.3, ASTM A193-B7, 16-UNC - 1.75 in. Length	3/8	MBCZZABZZIALBEUZ	4
14	SCREW, DRILLED HEX SOCKET HEAD CAP (2 HOLES), ASME B18.3, ASTM A193-B7, 18-UNC - 1.5 in. Length	5/16	MBCZZBRZZIALBEUZ	4
	INSTRUMENTS			
15	SPRINKLER ASSEMBLY BODY SET FOR HI-FOG® 1000 SERIES SPRINKLER AND SPRAY HEADS, A1000S1CT00D12S10 SPRINKLER ASSEMBLY BODY SET	1/2		1
	PIPE SUPPORTS			
16	PIPE SUPPORT	1	**64A0A-PS-102	1
17	PIPE SUPPORT	1	**64A0A-PS-103	1
18	PIPE SUPPORT	1/2	**64A0A-PS-117	1
19	PIPE SUPPORT	1/2	**64A0A-PS-128	1

SEE SHT 4

ACTUAL QUANTITY OF WASHERS SHALL BE CONSIDERED AS HALF OF THE QUANTITY SHOWN ON ISOMETRICS

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6. FOR ANY FURTHER INFORMATION REQUIRED DURING INSTALLATION, CONSULT MITSUBISHI POWER TFA (TECHNICAL FIELD ADVISOR) AT SITE.
7. 〈XXX## ###〉 INDICATES PIPING TERMINAL POINT.

ADDITIONAL NOTE:
TUBE REDUCER (FRACTIONAL TO METRIC) SHALL BE IDENTIFIED BY COLOR OR TAGGING FOR QUICK NOTATION IN THE FIELD.

WELD NO	WELD TYPE	WELD CAT	NPD	QTY
2	BW	S	1"	1
3	BW	S	1"	1
4	BW	S	1/2"	1
5	BW	S	1/2"	1

PIPING SPEC:	SDQ	PAINTING SPEC: PER PROJECT SPECIFICATION	REVISION RECORD					
FLUID CODE:	HW, FIRE FIGHTING WATER	INTERIOR SURFACE TREATMENT SPEC: LU-00129	REV. #	DESCRIPTION	BY	CHK	APP	DATE
MAX. PRESS.:	2030.00 psig / 14.00 MPag							
TEST PRESS.:	3045.00 psig / 20.99 MPag							
MAX. TEMP:	100 F / 38 C							
INSULATION SPEC:	NOT INSULATED							

WELD TYPE IDENTIFIER:
BW - BUTT WELD
SW - SOCKET WELD
SOF - SLIP-ON FLANGE
SOB - SET-ON BRANCH
LET - OLET, LATROLET OR HALF COUPLING
TW - TACK WELD
MW - MITER WELD

WELD CATEGORY IDENTIFICATION:
S - SHOP
FW - FIELD WELD
FFW - FIELD FIT WELD

SYMBOLOGY:
WELD NUMBER
PART NUMBER (PT NO)
< > CUT LIST PIECE NUMBER

SYMBOL IDENTIFIER:
F - FLANGE
G - GASKET
B - BOLT
N - NUT
W - WASHER
O - O-RING
L - LOCK WIRE

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MITSUBISHI POWER

ORIGIN:	MITSUBISHI POWER-LAKE MARY	DATE:	MITSUBISHI POWER DRAWING NUMBER:	REV. NO.:	SYSTEM CODE:
PREPARED BY:	E.BALATLI	07/23/2024	-	0	64A0A
REVIEWED BY:	M.FAZLIOGLU	07/24/2024	FIRE FIGHTING SYSTEM PIPING		
APPROVED BY:	N.KOZAN	07/25/2024	INTERNAL TO GT PACKAGE		
			FOR FABRICATION & ERECTION		
			GRAND RIVER ENERGY CENTER UNIT 4		
			GRAND RIVER DAM AUTHORITY		
			M501JAC GAS TURBINE		
			PROJECT CODE: K-GREC1	P&ID:	
			PIPELINE:	PAGE NUMBER:	
			FFS_SPACE-AROUND-IM_2	3 OF 4	

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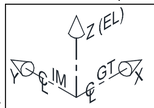

SEE SHT 3

20	NOZZLE MOUNTING ADAPTER SUPPORT	1/2	**64A0A-PS-119	1
	MISCELLANEOUS COMPONENTS			
21	LOCK WIRE, 1mm DIA x 1m LENGTH, ASTM A580-304	1	MWRZZZZBAIASXAPU	1
22	LOCK WIRE, 1mm DIA x 1m LENGTH, ASTM A580-304	1/2	MWRZZZZBAIASXAPU	1

SPOOLS

**64A0A-9, **64A0A-10, **64A0A-11, **64A0A-12

CUT PIPE LENGTH

PIECE NO	LENGTH (FT-INS)	NPD (IN)	REMARKS	END ONE	END TWO	ITEM CODE
<4>	4'11 13/16"	1		SQ.CUT	SQ.CUT	MTAAZZBPZZZZZXAQ
<5>	4'10 7/8"	1	PLD BEND(1)	BEVEL	SQ.CUT	MTAAZZBPZZZZZXAQ
<6>	3'7 1/2"	1/2		SQ.CUT	SQ.CUT	MTAAZZBPZZZZZXAQ
<7>	3'11 15/16"	1/2	PLD BEND(3)	BEVEL	SQ.CUT	MTAAZZBPZZZZZXAQ

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7. XXX## ### INDICATES PIPING TERMINAL POINT.

ORIGIN:	MITSUBISHI POWER-LAKE MARY	DATE:	MITSUBISHI POWER DRAWING NUMBER:	REV. NO.:	SYSTEM CODE:
PREPARED BY:	E.BALATLI	07/23/2024	-	0	64A0A
REVIEWED BY:	M.FAZLIOGLU	07/24/2024	FIRE FIGHTING SYSTEM PIPING		
APPROVED BY:	N.KOZAN	07/25/2024	INTERNAL TO GT PACKAGE		
			FOR FABRICATION & ERECTION		
			GRAND RIVER ENERGY CENTER UNIT 4		
			GRAND RIVER DAM AUTHORITY		
			M501JAC GAS TURBINE		
			PROJECT CODE: K-GREC1	P&ID:	
			PIPELINE: FFS_SPACE-AROUND-IM_2	PAGE NUMBER: 4 OF 4	

PIPING SPEC:	SDQ
FLUID CODE:	HW, FIRE FIGHTING WATER
MAX. PRESS.:	2030.00 psig / 14.00 MPag
TEST PRESS.:	3045.00 psig / 20.99 MPag
MAX. TEMP:	100 F / 38 C
INSULATION SPEC:	NOT INSULATED

PAINTING SPEC: PER PROJECT SPECIFICATION

INTERIOR SURFACE TREATMENT SPEC: LU-00129

REVISION RECORD

REV. #	DESCRIPTION	BY	CHK	APP	DATE

WELD TYPE IDENTIFIER:
BW - BUTT WELD
SW - SOCKET WELD
SOF - SLIP-ON FLANGE
SOB - SET-ON BRANCH
LET - OLET, LATROLET OR HALF COUPLING
TW - TACK WELD
MW - MITER WELD

WELD CATEGORY IDENTIFICATION:
S - SHOP
FW - FIELD WELD
FFW - FIELD FIT WELD

SYMBOLOGY:
WELD NUMBER
PART NUMBER (PT NO)
< > CUT LIST PIECE NUMBER

SYMBOL IDENTIFIER:
F - FLANGE
G - GASKET
B - BOLT
N - NUT
W - WASHER
O - O-RING
L - LOCK WIRE

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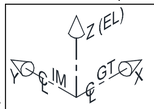

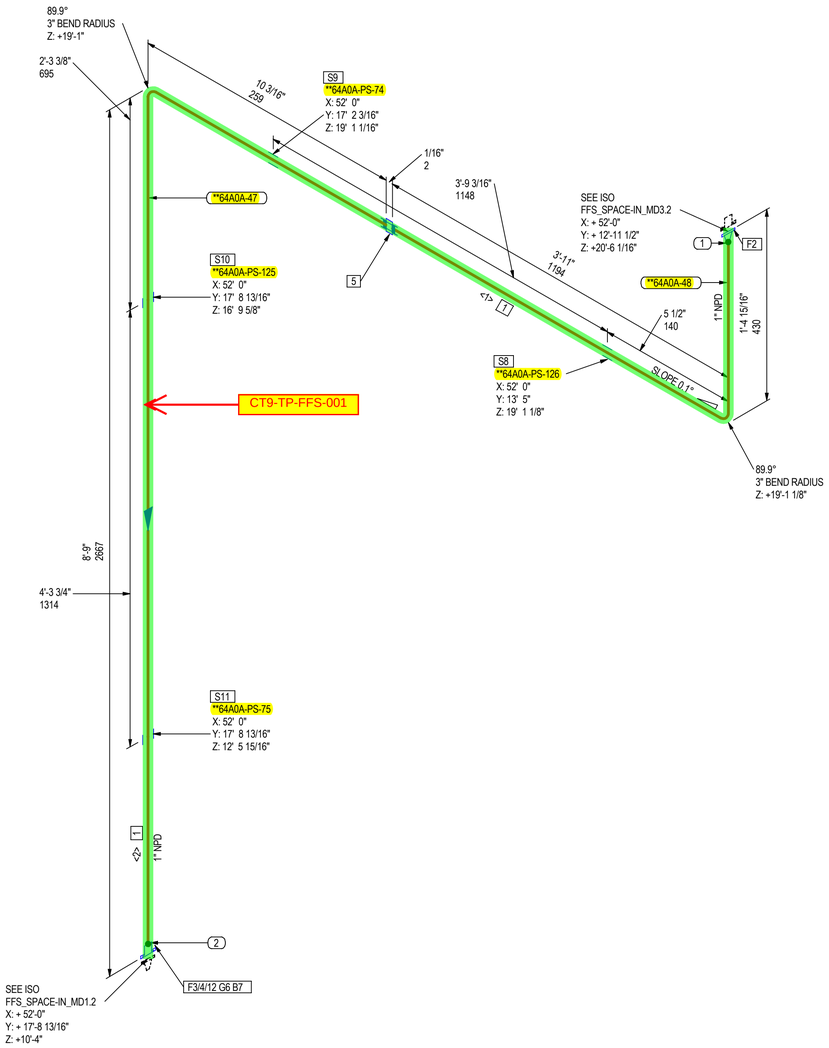

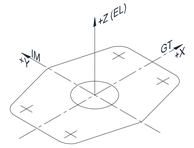

PART NO: (F2)
HYDRAULIC CAPTIVE FLANGE ORIENTATION

SHOP MATERIALS - FABRICATION ONLY

PT NO	DESCRIPTION	NPD (IN)	CMDTY CODE	QTY
1	TUBE, 0.083 in THK, PE, ASTM A213-TP316, SEAMLESS	1	MTAAZZBPZZZZZXAQ	14.7'
2	FLANGE HYDRAULIC, 3000 PSI, FFFE x SWE, SAE J518-1 CODE 61, ASTM A182-F316, CAPTIVE FLANGE	1	MFFBAWYGZZIARALS	1
3	FLANGE HYDRAULIC, 3000 PSI, GROOVED END x SWE, SAE J518-1 CODE 61, ASTM A182-F316, FLANGE HEAD	1	MFFAAWYFZZIARALS	1
	MISCELLANEOUS COMPONENTS			
4	FLANGE HYDRAULIC, 3000 PSI, SAE J518-1 CODE 61, ASTM A182-F316, SPLIT FLANGE CLAMP (ONE PAIR)	1	MWABAWYHZZIARALS	1

OTHER THAN SHOP MATERIALS - ERECTION ONLY

PT NO	DESCRIPTION	NPD (IN)	CMDTY CODE	QTY
5	TUBE UNION, SWGLKE END, SWAGELOK PART NO. , ASTM A182-F316	1	MWADZZCGZZAKWALS	1
6	GASKET, ORING, 0.125" THK, AS 568, VITON HS90 (DUROMETER90)	1	MKBZZACIAQXANUS	1
7	SCREW, DRILLED HEX SOCKET HEAD CAP (2 HOLES), ASME B18.3, ASTM A193-B7, 16-UNC - 1.75 in. Length	3/8	MBCZZABZZIALBEUZ	4
	PIPE SUPPORTS			
8	PIPE SUPPORT	1	**64A0A-PS-126	1
9	PIPE SUPPORT	1	**64A0A-PS-74	1
10	PIPE SUPPORT	1	**64A0A-PS-125	1
11	PIPE SUPPORT	1	**64A0A-PS-75	1
	MISCELLANEOUS COMPONENTS			
12	LOCK WIRE, 1mm DIA x 1m LENGTH, ASTM A580-304	1	MWRZZZZBAIASXAPU	1
	SPOOLS			
	**64A0A-47, **64A0A-48			

CUT PIPE LENGTH

PIECE NO	LENGTH (FT-INS)	NPD (IN)	REMARKS	END ONE	END TWO	ITEM CODE
<1>	5'2 3/8"	1	PLD BEND(1)	BEVEL	SQ.CUT	MTAAZZBPZZZZZXAQ
<2>	9'5 11/16"	1	PLD BEND(1)	SQ.CUT	SQ.CUT	MTAAZZBPZZZZZXAQ

ACTUAL QUANTITY OF WASHERS SHALL BE CONSIDERED AS HALF OF THE QUANTITY SHOWN ON ISOMETRICS

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7. <XXX## ###> INDICATES PIPING TERMINAL POINT.

WELD NO	WELD TYPE	WELD CAT	NPD	QTY
1	BW	S	1"	1
2	BW	S	1"	1

PIPING SPEC:	SDQ	PAINTING SPEC: PER PROJECT SPECIFICATION	REVISION RECORD					
FLUID CODE:	HW, FIRE FIGHTING WATER	INTERIOR SURFACE TREATMENT SPEC: LU-00129	REV. #	DESCRIPTION	BY	CHK	APP	DATE
MAX. PRESS.:	2030.00 psig / 14.00 MPag							
TEST PRESS.:	3045.00 psig / 20.99 MPag							
MAX. TEMP:	100 F / 38 C							
INSULATION SPEC:	NOT INSULATED							

WELD TYPE IDENTIFIER:
BW - BUTT WELD
SW - SOCKET WELD
SOF - SLIP-ON FLANGE
SOB - SET-ON BRANCH
LET - OLET, LATROLET OR HALF COUPLING
TW - TACK WELD
MW - MITER WELD

WELD CATEGORY IDENTIFICATION:
S - SHOP
FW - FIELD WELD
FFW - FIELD FIT WELD

SYMBOLOGY:
WELD NUMBER
PART NUMBER (PT NO)
< > CUT LIST PIECE NUMBER

SYMBOL IDENTIFIER:
F - FLANGE
G - GASKET
B - BOLT
N - NUT
W - WASHER
O - O-RING
L - LOCK WIRE

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MITSUBISHI POWER

ORIGIN:	MITSUBISHI POWER-LAKE MARY	DATE:	MITSUBISHI POWER DRAWING NUMBER: / REV. NO.: / SYSTEM CODE:
PREPARED BY:	E.BALATLI	07/23/2024	- / 0 / 64A0A
REVIEWED BY:	M.FAZLIOGLU	07/24/2024	FIRE FIGHTING SYSTEM PIPING
APPROVED BY:	N.KOZAN	07/25/2024	INTERNAL TO GT PACKAGE
			FOR FABRICATION & ERECTION
			GRAND RIVER ENERGY CENTER UNIT 4
			GRAND RIVER DAM AUTHORITY
			M501JAC GAS TURBINE
PROJECT CODE:	K-GREC1	P&ID:	
PIPELINE:		PAGE NUMBER:	
FFS_SPACE-IN_MD2.2_1		1 OF 1	

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PLANT ENGINEERING
DEPARTMENT:
SECTION:
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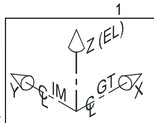
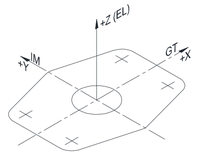
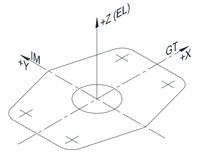
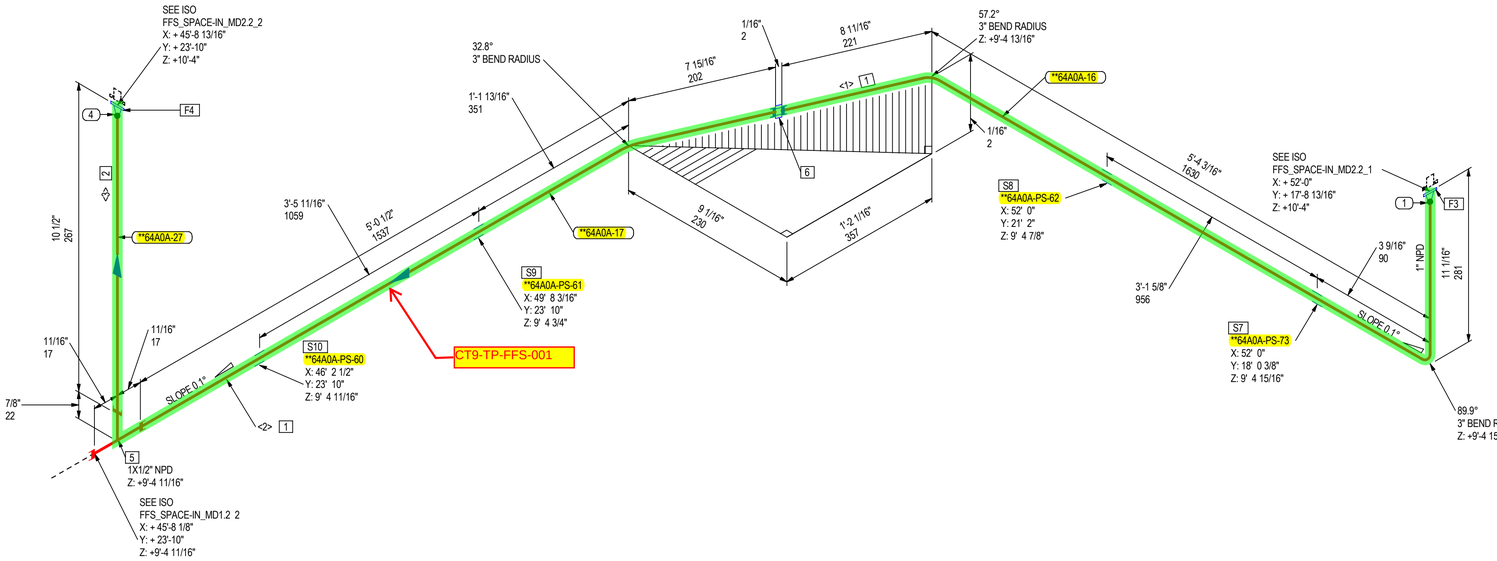




PART NO: (F3)
HYDRAULIC CAPTIVE FLANGE ORIENTATION



PART NO: (F4)
HYDRAULIC CAPTIVE FLANGE ORIENTATION



SHOP MATERIALS - FABRICATION ONLY

PT NO	DESCRIPTION	NPD (IN)	CMDTY CODE	QTY
1	TUBE, 0.083 in THK, PE, ASTM A213-TP316, SEAMLESS	1	MTAAZZBPZZZZZXAQ	12.6'
2	TUBE, 0.049 in THK, PE, ASTM A213-TP316, SEAMLESS	1/2	MTAAZZBPZZZZZXAQ	0.9'
3	FLANGE HYDRAULIC, 3000 PSI, FFFE x SWE, SAE J518-1 CODE 61, ASTM A182-F316, CAPTIVE FLANGE	1	MFFBAWYGZZIARALS	1
4	FLANGE HYDRAULIC, 3000 PSI, FFFE x SWE, SAE J518-1 CODE 61, ASTM A182-F316, CAPTIVE FLANGE	1/2	MFFBAWYGZZIARALS	1

OTHER THAN SHOP MATERIALS - ERECTION ONLY

PT NO	DESCRIPTION	NPD (IN)	CMDTY CODE	QTY
5	TUBE REDUCING UNION TEE, SWGLKE END, SWAGELOK, ASTM A182-F316	1X1/2	MWBCZZCGZZAKWALS	1
6	TUBE UNION, SWGLKE END, SWAGELOK PART NO. , ASTM A182-F316	1	MWADZZCGZZAKWALS	1
	PIPE SUPPORTS			
7	PIPE SUPPORT	1	**64A0A-PS-73	1
8	PIPE SUPPORT	1	**64A0A-PS-62	1
9	PIPE SUPPORT	1	**64A0A-PS-61	1
10	PIPE SUPPORT	1	**64A0A-PS-60	1

SPOOLS

**64A0A-16, **64A0A-17, **64A0A-27

CUT PIPE LENGTH

PIECE NO	LENGTH (FT-INS)	NPD (IN)	REMARKS	END ONE	END TWO	ITEM CODE
<1>	6'10 1/16"	1	PLD BEND(2)	BEVEL	SQ.CUT	MTAAZZBPZZZZZXAQ
<2>	5'8 3/8"	1	PLD BEND(1)	SQ.CUT	SQ.CUT	MTAAZZBPZZZZZXAQ
<3>	10 1/4"	1/2		SQ.CUT	BEVEL	MTAAZZBPZZZZZXAQ

ACTUAL QUANTITY OF WASHERS SHALL BE CONSIDERED AS HALF OF THE QUANTITY SHOWN ON ISOMETRICS

NOTES:
1. THE X AND Y COORDINATES ARE WITH RESPECT TO THE INTERSECTION OF THE CENTERLINE OF THE GAS TURBINE (GT) AND THE INLET MANIFOLD (IM). THE Z (EL) COORDINATE IS WITH RESPECT TO THE TOP OF CONCRETE, WHICH IS 15'-9" [4800mm] BELOW THE CENTERLINE OF THE GT AND IS 0'-0".
2. ALL PIPE SUPPORT COORDINATES ARE WITH RESPECT TO THE CENTERLINE OF THE INSTALLED SUPPORTED PIPE.
3. THE MARK 'FFW' (FIELD FIT WELD) IN THE DRAWING INDICATES EXCESS LENGTH OF APPROX. 4 INCHES IDENTIFIED AS LOOSE IN THE DRAWING TO BE CUT AND ADJUSTED AT THE FIELD.
4. ALL WELDING TO BE DONE AND INSPECTED PER LATEST APPLICABLE EDITION OF ASME B31.1 AND SECTION IX AVAILABLE ON THE PROJECT AWARD DATE.
5. PIPING SHALL BE INSTALLED PER THE INDICATED SLOPE.
6. FOR ANY FURTHER INFORMATION REQUIRED DURING INSTALLATION, CONSULT MITSUBISHI POWER TFA (TECHNICAL FIELD ADVISOR) AT SITE.
7. 〈XXX## ###〉 INDICATES PIPING TERMINAL POINT.

WELD NO	WELD TYPE	WELD CAT	NPD	QTY
1	BW	S	1"	1
4	BW	S	1/2"	1

PIPING SPEC:	SDQ	PAINTING SPEC: PER PROJECT SPECIFICATION	REVISION RECORD					
FLUID CODE:	HW, FIRE FIGHTING WATER	INTERIOR SURFACE TREATMENT SPEC: LU-00129	REV. #	DESCRIPTION	BY	CHK	APP	DATE
MAX. PRESS.:	2030.00 psig / 14.00 MPag							
TEST PRESS.:	3045.00 psig / 20.99 MPag							
MAX. TEMP:	100 F / 38 C							
INSULATION SPEC:	NOT INSULATED							

WELD TYPE IDENTIFIER:
BW - BUTT WELD
SW - SOCKET WELD
SOF - SLIP-ON FLANGE
SOB - SET-ON BRANCH
LET - OLET, LATROLET OR HALF COUPLING
TW - TACK WELD
MW - MITER WELD

WELD CATEGORY IDENTIFICATION:
S - SHOP
FW - FIELD WELD
FFW - FIELD FIT WELD

SYMBOLOGY:
WELD NUMBER
PART NUMBER (PT NO)
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SYMBOL IDENTIFIER:
F - FLANGE
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MITSUBISHI POWER

ORIGIN:	MITSUBISHI POWER-LAKE MARY	DATE:	MITSUBISHI POWER DRAWING NUMBER: / REV. NO.: / SYSTEM CODE:
PREPARED BY:	E.BALATLI	07/23/2024	- / 0 / 64A0A
REVIEWED BY:	M.FAZLIOGLU	07/24/2024	FIRE FIGHTING SYSTEM PIPING
APPROVED BY:	N.KOZAN	07/25/2024	INTERNAL TO GT PACKAGE
			FOR FABRICATION & ERECTION
			GRAND RIVER ENERGY CENTER UNIT 4
			GRAND RIVER DAM AUTHORITY
			M501JAC GAS TURBINE
PROJECT CODE:	K-GREC1	P&ID:	
PIPELINE:		PAGE NUMBER:	
FFS_SPACE-IN_MD1.2		1 OF 3	

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SECTION: PLANT ENGINEERING

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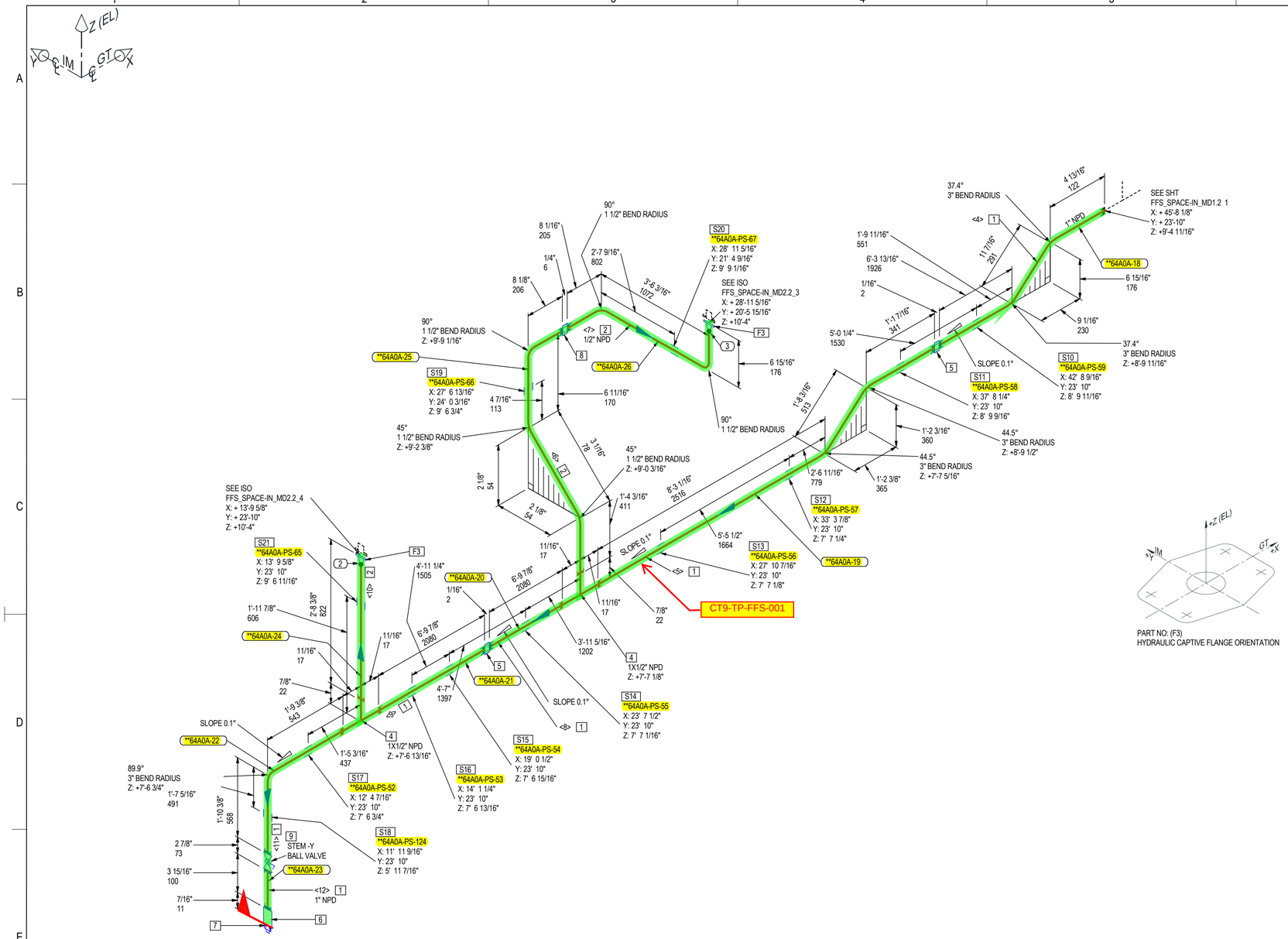

SHOP MATERIALS - FABRICATION ONLY

PT NO	DESCRIPTION	NPD (IN)	CMDTY CODE	QTY
1	TUBE, 0.083 in THK, PE, ASTM A213-TP316, SEAMLESS	1	MTAAZZBPZZZZZXAQ	36.3'
2	TUBE, 0.049 in THK, PE, ASTM A213-TP316, SEAMLESS	1/2	MTAAZZBPZZZZZXAQ	10.1'
3	FLANGE HYDRAULIC, 3000 PSI, FFFE x SWE, SAE J518-1 CODE 61, ASTM A182-F316, CAPTIVE FLANGE	1/2	MFFBAWYGZZIARALS	2

OTHER THAN SHOP MATERIALS - ERECTION ONLY

PT NO	DESCRIPTION	NPD (IN)	CMDTY CODE	QTY
4	TUBE REDUCING UNION TEE, SWGLKE END, SWAGELOK, ASTM A182-F316	1X1/2	MWBCZZCGZZAKWALS	2
5	TUBE UNION, SWGLKE END, SWAGELOK PART NO. , ASTM A182-F316	1	MWADZZCGZZAKWALS	2
6	TUBE FEMALE CONNECTOR, SWGLKE END x NPT, SWAGELOK, ASTM A182-F316	1	MWBDZZYAZZAKWALS	1
7	PLUG, CL3000, MTE, ASME B16.11, ASTM A182-F316, HEX HEAD	1	MMFIAWBNZZAAGALS	1
8	TUBE UNION, SWGLKE END, SWAGELOK PART NO. SS-810-6, ASTM A182-F316	1/2	MWADZZCGZZAKWALS	1
9	BALL VALVE, CL3000, SWGLKE END, MSS-SP-105, ASTM A182-F316, SWAGELOK	1	MVCKAWCGZZGLEALS	1
	PIPE SUPPORTS			
10	PIPE SUPPORT	1	**64A0A-PS-59	1
11	PIPE SUPPORT	1	**64A0A-PS-58	1
12	PIPE SUPPORT	1	**64A0A-PS-57	1
13	PIPE SUPPORT	1	**64A0A-PS-56	1
14	PIPE SUPPORT	1	**64A0A-PS-55	1
15	PIPE SUPPORT	1	**64A0A-PS-54	1
16	PIPE SUPPORT	1	**64A0A-PS-53	1
17	PIPE SUPPORT	1	**64A0A-PS-52	1
18	PIPE SUPPORT	1	**64A0A-PS-124	1
19	PIPE SUPPORT	1/2	**64A0A-PS-66	1
20	PIPE SUPPORT	1/2	**64A0A-PS-67	1
21	PIPE SUPPORT	1/2	**64A0A-PS-65	1

SPOOLS

**64A0A-18, **64A0A-19, **64A0A-20, **64A0A-21, **64A0A-22, **64A0A-23, **64A0A-24, **64A0A-25, **64A0A-26

SEE SHT 3

ACTUAL QUANTITY OF WASHERS SHALL BE CONSIDERED AS HALF OF THE QUANTITY SHOWN ON ISOMETRICS

NOTES:

1. THE X AND Y COORDINATES ARE WITH RESPECT TO THE INTERSECTION OF THE CENTERLINE OF THE GAS TURBINE (GT) AND THE INLET MANIFOLD (IM). THE Z (EL) COORDINATE IS WITH RESPECT TO THE TOP OF CONCRETE, WHICH IS 15'-9" [4800mm] BELOW THE CENTERLINE OF THE GT AND IS 0'-0".
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3. THE MARK 'FFW' (FIELD FIT WELD) IN THE DRAWING INDICATES EXCESS LENGTH OF APPROX. 4 INCHES IDENTIFIED AS LOOSE IN THE DRAWING TO BE CUT AND ADJUSTED AT THE FIELD.
4. ALL WELDING TO BE DONE AND INSPECTED PER LATEST APPLICABLE EDITION OF ASME B31.1 AND SECTION IX AVAILABLE ON THE PROJECT AWARD DATE.
5. PIPING SHALL BE INSTALLED PER THE INDICATED SLOPE.
6. FOR ANY FURTHER INFORMATION REQUIRED DURING INSTALLATION, CONSULT MITSUBISHI POWER TFA (TECHNICAL FIELD ADVISOR) AT SITE.
7. <XXX## ###> INDICATES PIPING TERMINAL POINT.

WELD NO	WELD TYPE	WELD CAT	NPD	QTY
3	BW	S	1/2"	1
2	BW	S	1/2"	1

PIPING SPEC:	SDQ	PAINTING SPEC: PER PROJECT SPECIFICATION	REVISION RECORD				
FLUID CODE:	HW, FIRE FIGHTING WATER	INTERIOR SURFACE TREATMENT SPEC: LU-00129	REV. #	DESCRIPTION	BY	CHK	APP	DATE
MAX. PRESS.:	2030.00 psig / 14.00 MPag							
TEST PRESS.:	3045.00 psig / 20.99 MPag							
MAX. TEMP:	100 F / 38 C							
INSULATION SPEC:	NOT INSULATED							

WELD TYPE IDENTIFIER:
BW - BUTT WELD
SW - SOCKET WELD
SOF - SLIP-ON FLANGE
SOB - SET-ON BRANCH
LET - OLET, LATROLET OR HALF COUPLING
TW - TACK WELD
MW - MITER WELD

WELD CATEGORY IDENTIFICATION:
● S - SHOP
⊗ FW - FIELD WELD
⊗ FFW - FIELD FIT WELD

SYMBOLOGY:
WELD NUMBER
PART NUMBER (PT NO)
< > CUT LIST PIECE NUMBER

SYMBOL IDENTIFIER:
F - FLANGE
G - GASKET
B - BOLT
N - NUT
W - WASHER
O - O-RING
L - LOCK WIRE

ORIGIN:	MITSUBISHI POWER-LAKE MARY	DATE:	MITSUBISHI POWER DRAWING NUMBER:	REV. NO.:	SYSTEM CODE:
PREPARED BY:	E.BALATLI	07/23/2024	-	0	64A0A
REVIEWED BY:	M.FAZLIOGLU	07/24/2024	FIRE FIGHTING SYSTEM PIPING		
APPROVED BY:	N.KOZAN	07/25/2024	INTERNAL TO GT PACKAGE		
			FOR FABRICATION & ERECTION		
			GRAND RIVER ENERGY CENTER UNIT 4		
			GRAND RIVER DAM AUTHORITY		
			M501JAC GAS TURBINE		
PROJECT CODE:	K-GREC1	P&ID:			
PIPELINE:	FFS_SPACE-IN_MD1.2	PAGE NUMBER:	2 OF 3		

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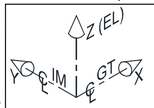

SEE SHT 2

CUT PIPE LENGTH

PIECE NO	LENGTH (FT-INS)	NPD (IN)	REMARKS	END ONE	END TWO	ITEM CODE
<4>	7'7 7/8"	1	PLD BEND(2)	SQ.CUT	SQ.CUT	MTAAZZBPZZZZZXAQ
<5>	11'0 7/16"	1	PLD BEND(2)	SQ.CUT	SQ.CUT	MTAAZZBPZZZZZXAQ
<6>	2'9 5/16"	1/2	PLD BEND(3)	SQ.CUT	SQ.CUT	MTAAZZBPZZZZZXAQ
<7>	4'7 3/4"	1/2	PLD BEND(2)	SQ.CUT	BEVEL	MTAAZZBPZZZZZXAQ
<8>	6'9 7/8"	1		SQ.CUT	SQ.CUT	MTAAZZBPZZZZZXAQ
<9>	6'9 7/8"	1		SQ.CUT	SQ.CUT	MTAAZZBPZZZZZXAQ
<10>	2'8 1/8"	1/2		SQ.CUT	BEVEL	MTAAZZBPZZZZZXAQ
<11>	3'6 1/2"	1	PLD BEND(1)	SQ.CUT	SQ.CUT	MTAAZZBPZZZZZXAQ
<12>	3 15/16"	1		SQ.CUT	SQ.CUT	MTAAZZBPZZZZZXAQ

ACTUAL QUANTITY OF WASHERS SHALL BE CONSIDERED AS HALF OF THE QUANTITY SHOWN ON ISOMETRICS

NOTES:
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7. ⟨XXX## ###⟩ INDICATES PIPING TERMINAL POINT.

ORIGIN:	MITSUBISHI POWER-LAKE MARY	DATE:	MITSUBISHI POWER DRAWING NUMBER:	REV. NO.:	SYSTEM CODE:
PREPARED BY:	E.BALATLI	07/23/2024	-	0	64A0A
REVIEWED BY:	M.FAZLIOGLU	07/24/2024	FIRE FIGHTING SYSTEM PIPING		
APPROVED BY:	N.KOZAN	07/25/2024	INTERNAL TO GT PACKAGE		
			FOR FABRICATION & ERECTION		
			GRAND RIVER ENERGY CENTER UNIT 4		
			GRAND RIVER DAM AUTHORITY		
			M501JAC GAS TURBINE		
			PROJECT CODE: K-GREC1	P&ID:	
			PIPELINE: FFS_SPACE-IN_MD1.2	PAGE NUMBER: 3 OF 3	

PIPING SPEC:	SDQ
FLUID CODE:	HW, FIRE FIGHTING WATER
MAX. PRESS.:	2030.00 psig / 14.00 MPag
TEST PRESS.:	3045.00 psig / 20.99 MPag
MAX. TEMP:	100 F / 38 C
INSULATION SPEC:	NOT INSULATED

PAINTING SPEC: PER PROJECT SPECIFICATION
INTERIOR SURFACE TREATMENT SPEC: LU-00129

REVISION RECORD					
REV. #	DESCRIPTION	BY	CHK	APP	DATE

WELD TYPE IDENTIFIER:
BW - BUTT WELD
SW - SOCKET WELD
SOF - SLIP-ON FLANGE
SOB - SET-ON BRANCH
LET - OLET, LATROLET OR HALF COUPLING
TW - TACK WELD
MW - MITER WELD

WELD CATEGORY IDENTIFICATION:
S - SHOP
FW - FIELD WELD
FFW - FIELD FIT WELD

SYMBOLOGY:
WELD NUMBER
PART NUMBER (PT NO)
< > CUT LIST PIECE NUMBER

SYMBOL IDENTIFIER:
F - FLANGE
G - GASKET
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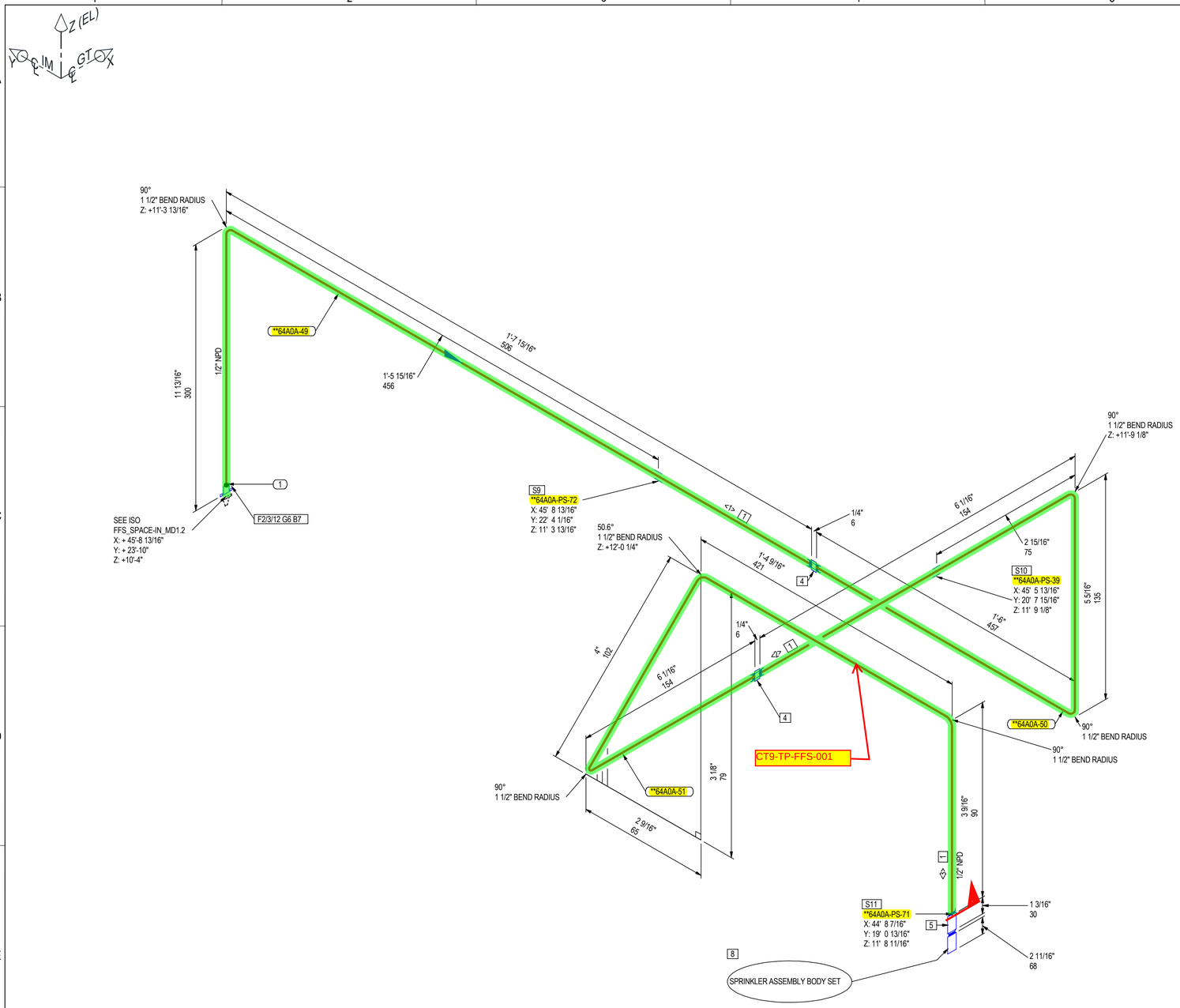

SHOP MATERIALS - FABRICATION ONLY

PT NO	DESCRIPTION	NPD (IN)	CMDTY CODE	QTY
1	TUBE, 0.049 in THK, PE, ASTM A213-TP316, SEAMLESS	1/2	MTAAZZBPZZZZZXAQ	7.4'
2	FLANGE HYDRAULIC, 3000 PSI, GROOVED END x SWE, SAE J518-1 CODE 61, ASTM A182-F316, FLANGE HEAD	1/2	MFFAAWYFZZIARALS	1
	MISCELLANEOUS COMPONENTS			
3	FLANGE HYDRAULIC, 3000 PSI, SAE J518-1 CODE 61, ASTM A182-F316, SPLIT FLANGE CLAMP (ONE PAIR)	1/2	MWABAWYHZZIARALS	1

OTHER THAN SHOP MATERIALS - ERECTION ONLY

PT NO	DESCRIPTION	NPD (IN)	CMDTY CODE	QTY
4	TUBE UNION, SWGLKE END, SWAGELOK PART NO. SS-810-6, ASTM A182-F316	1/2	MWADZZCGZZAKWALS	2
5	TUBE REDUCER (FRACTIONAL TO METRIC), SWGLKE END x PE, SWAGELOK PART NO.SS-810-R-12M, ASTM A182-F316	1/2	MWAOZZYDZZAKWALS	1
6	GASKET, ORING, 0.125" THK, AS 568, VITON HS90 (DUROMETER90)	1/2	MKBZZACIAQXANUS	1
7	SCREW, DRILLED HEX SOCKET HEAD CAP (2 HOLES), ASME B18.3, ASTM A193-B7, 18-UNC - 1.5 in. Length	5/16	MBCZZBRZZIALBEUZ	4
	INSTRUMENTS			
8	SPRINKLER ASSEMBLY BODY SET FOR HI-FOG® 1000 SERIES SPRINKLER AND SPRAY HEADS, A1000S1CT00D12S10 SPRINKLER ASSEMBLY BODY SET	1/2		1
	PIPE SUPPORTS			
9	PIPE SUPPORT	1/2	**64A0A-PS-72	1
10	PIPE SUPPORT	1/2	**64A0A-PS-39	1
11	NOZZLE MOUNTING ADAPTER SUPPORT	1/2	**64A0A-PS-71	1
	MISCELLANEOUS COMPONENTS			
12	LOCK WIRE, 1mm DIA x 1m LENGTH, ASTM A580-304	1/2	MWRZZZZBAIASXAPU	1

SPOOLS

**64A0A-49, **64A0A-50, **64A0A-51

CUT PIPE LENGTH

PIECE NO	LENGTH (FT-INS)	NPD (IN)	REMARKS	END ONE	END TWO	ITEM CODE
<1>	2'6 13/16"	1/2	PLD BEND(1)	SQ.CUT	SQ.CUT	MTAAZZBPZZZZZXAQ
<2>	2'4 1/16"	1/2	PLD BEND(2)	SQ.CUT	SQ.CUT	MTAAZZBPZZZZZXAQ
<3>	2'4 13/16"	1/2	PLD BEND(3)	SQ.CUT	SQ.CUT	MTAAZZBPZZZZZXAQ

ACTUAL QUANTITY OF WASHERS SHALL BE CONSIDERED AS HALF OF THE QUANTITY SHOWN ON ISOMETRICS

NOTES:
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2. ALL PIPE SUPPORT COORDINATES ARE WITH RESPECT TO THE CENTERLINE OF THE INSTALLED SUPPORTED PIPE.
3. THE MARK 'FFW' (FIELD FIT WELD) IN THE DRAWING INDICATES EXCESS LENGTH OF APPROX. 4 INCHES IDENTIFIED AS LOOSE IN THE DRAWING TO BE CUT AND ADJUSTED AT THE FIELD.
4. ALL WELDING TO BE DONE AND INSPECTED PER LATEST APPLICABLE EDITION OF ASME B31.1 AND SECTION IX AVAILABLE ON THE PROJECT AWARD DATE.
5. PIPING SHALL BE INSTALLED PER THE INDICATED SLOPE.
6. FOR ANY FURTHER INFORMATION REQUIRED DURING INSTALLATION, CONSULT MITSUBISHI POWER TFA (TECHNICAL FIELD ADVISOR) AT SITE.
7. <XXX## ###> INDICATES PIPING TERMINAL POINT.

ADDITIONAL NOTE:
TUBE REDUCER (FRACTIONAL TO METRIC) SHALL BE IDENTIFIED BY COLOR OR TAGGING FOR QUICK NOTATION IN THE FIELD.

WELD NO	WELD TYPE	WELD CAT	NPD	QTY
1	BW	S	1/2"	1

PIPING SPEC:	SDQ	PAINTING SPEC: PER PROJECT SPECIFICATION
FLUID CODE:	HW, FIRE FIGHTING WATER	INTERIOR SURFACE TREATMENT SPEC: LU-00129
MAX. PRESS.:	2030.00 psig / 14.00 MPag	
TEST PRESS.:	3045.00 psig / 20.99 MPag	
MAX. TEMP:	100 F / 38 C	
INSULATION SPEC:	NOT INSULATED	

REVISION RECORD					
REV. #	DESCRIPTION	BY	CHK	APP	DATE

WELD TYPE IDENTIFIER:
BW - BUTT WELD
SW - SOCKET WELD
SOF - SLIP-ON FLANGE
SOB - SET-ON BRANCH
LET - OLET, LATROLET OR HALF COUPLING
TW - TACK WELD
MW - MITER WELD

WELD CATEGORY IDENTIFICATION:
S - SHOP
FW - FIELD WELD
FFW - FIELD FIT WELD

SYMBOLOGY:
WELD NUMBER
PART NUMBER (PT NO)
< > CUT LIST PIECE NUMBER

SYMBOL IDENTIFIER:
F - FLANGE
G - GASKET
B - BOLT
N - NUT
W - WASHER
O - O-RING
L - LOCK WIRE

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MITSUBISHI POWER

ORIGIN:	MITSUBISHI POWER-LAKE MARY	DATE:	MITSUBISHI POWER DRAWING NUMBER:	REV. NO.:	SYSTEM CODE:
PREPARED BY:	E.BALATLI	07/23/2024	-	0	64A0A
REVIEWED BY:	M.FAZLIOGLU	07/24/2024	FIRE FIGHTING SYSTEM PIPING		
APPROVED BY:	N.KOZAN	07/25/2024	INTERNAL TO GT PACKAGE		
			FOR FABRICATION & ERECTION		
			GRAND RIVER ENERGY CENTER UNIT 4		
			GRAND RIVER DAM AUTHORITY		
			M501JAC GAS TURBINE		
PROJECT CODE:	K-GREC1	P&ID:			
PIPELINE:	FFS_SPACE-IN_MD2.2_2	PAGE NUMBER:	1 OF 1		

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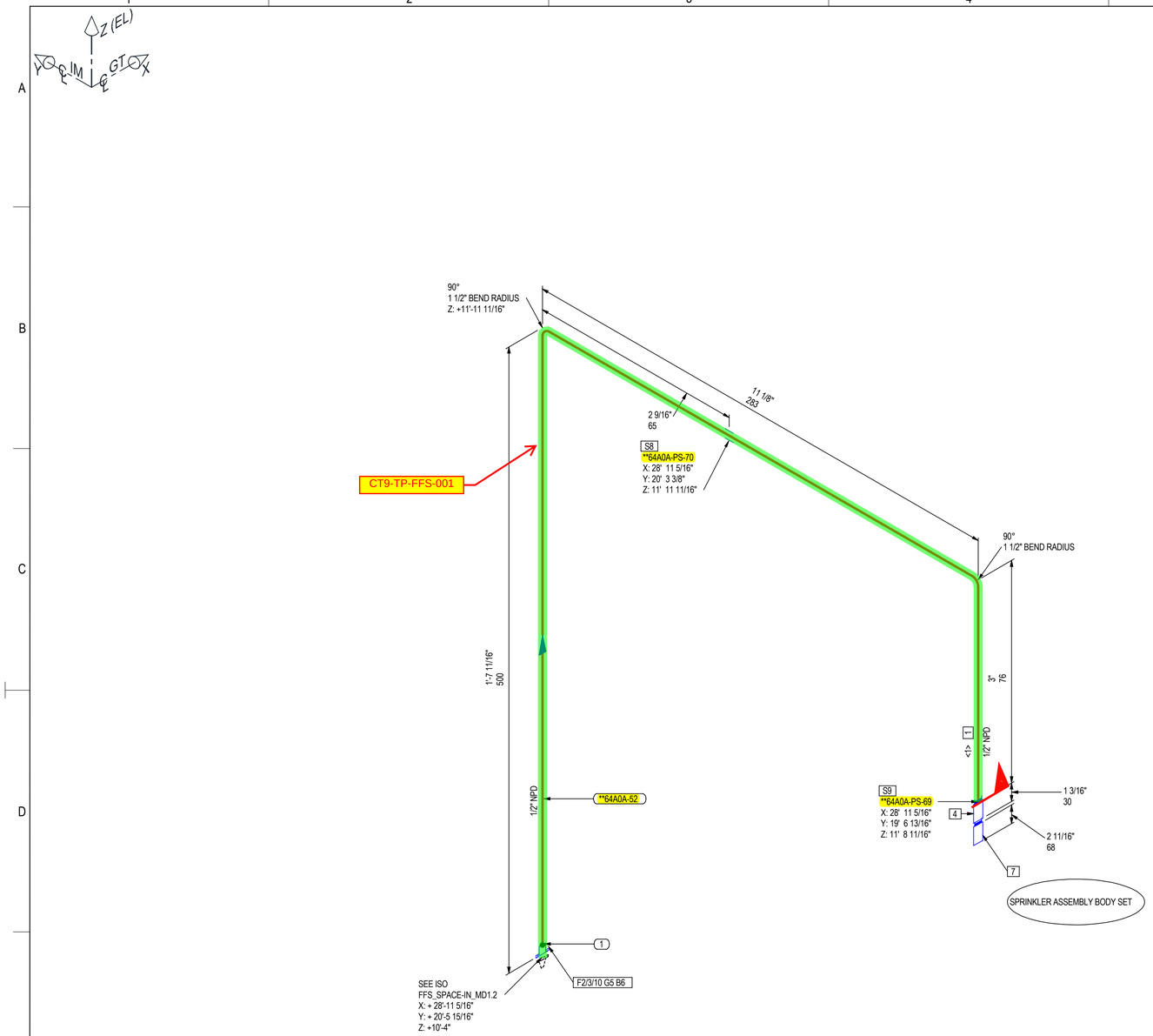

SHOP MATERIALS - FABRICATION ONLY

PT NO	DESCRIPTION	NPD (IN)	CMDTY CODE	QTY
1	TUBE, 0.049 in THK, PE, ASTM A213-TP316, SEAMLESS	1/2	MTAAZZBPZZZZZXAQ	2.7'
2	FLANGE HYDRAULIC, 3000 PSI, GROOVED END x SWE, SAE J518-1 CODE 61, ASTM A182-F316, FLANGE HEAD	1/2	MFFAAWYFZZIARALS	1
	MISCELLANEOUS COMPONENTS			
3	FLANGE HYDRAULIC, 3000 PSI, SAE J518-1 CODE 61, ASTM A182-F316, SPLIT FLANGE CLAMP (ONE PAIR)	1/2	MWABAWYHZZIARALS	1

OTHER THAN SHOP MATERIALS - ERECTION ONLY

PT NO	DESCRIPTION	NPD (IN)	CMDTY CODE	QTY
4	TUBE REDUCER (FRACTIONAL TO METRIC), SWGLKE END x PE, SWAGELOK PART NO.SS-810-R-12M, ASTM A182-F316	1/2	MWAOZZYDZZAKWALS	1
5	GASKET, ORING, 0.125" THK, AS 568, VITON HS90 (DUROMETER90)	1/2	MKBZZACIAQXANUS	1
6	SCREW, DRILLED HEX SOCKET HEAD CAP (2 HOLES), ASME B18.3, ASTM A193-B7, 18-UNC - 1.5 in. Length	5/16	MBCZZBRZZIALBEUZ	4
	INSTRUMENTS			
7	SPRINKLER ASSEMBLY BODY SET FOR HI-FOG® 1000 SERIES SPRINKLER AND SPRAY HEADS, A1000S1CT00D12S10 SPRINKLER ASSEMBLY BODY SET	1/2		1
	PIPE SUPPORTS			
8	PIPE SUPPORT	1/2	**64A0A-PS-70	1
9	NOZZLE MOUNTING ADAPTER SUPPORT	1/2	**64A0A-PS-69	1
	MISCELLANEOUS COMPONENTS			
10	LOCK WIRE, 1mm DIA x 1m LENGTH, ASTM A580-304	1/2	MWRZZZZBAIASXAPU	1
	SPOOLS			
	**64A0A-52			

CUT PIPE LENGTH

PIECE NO	LENGTH (FT-INS)	NPD (IN)	REMARKS	END ONE	END TWO	ITEM CODE
<1>	2'8 5/16"	1/2	PLD BEND(2)	SQ.CUT	SQ.CUT	MTAAZZBPZZZZZXAQ

ACTUAL QUANTITY OF WASHERS SHALL BE CONSIDERED AS HALF OF THE QUANTITY SHOWN ON ISOMETRICS

NOTES:
1. THE X AND Y COORDINATES ARE WITH RESPECT TO THE INTERSECTION OF THE CENTERLINE OF THE GAS TURBINE (GT) AND THE INLET MANIFOLD (IM). THE Z (EL) COORDINATE IS WITH RESPECT TO THE TOP OF CONCRETE, WHICH IS 15'-9" [4800mm] BELOW THE CENTERLINE OF THE GT AND IS 0'-0".
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4. ALL WELDING TO BE DONE AND INSPECTED PER LATEST APPLICABLE EDITION OF ASME B31.1 AND SECTION IX AVAILABLE ON THE PROJECT AWARD DATE.
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7. XXX## ### INDICATES PIPING TERMINAL POINT.

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WELD NO	WELD TYPE	WELD CAT	NPD	QTY
1	BW	S	1/2"	1

PIPING SPEC:	SDQ	PAINTING SPEC: PER PROJECT SPECIFICATION	REVISION RECORD				
FLUID CODE:	HW, FIRE FIGHTING WATER	INTERIOR SURFACE TREATMENT SPEC: LU-00129	REV. #	DESCRIPTION	BY	CHK	APP	DATE
MAX. PRESS.:	2030.00 psig / 14.00 MPag							
TEST PRESS.:	3045.00 psig / 20.99 MPag							
MAX. TEMP:	100 F / 38 C							
INSULATION SPEC:	NOT INSULATED							

WELD TYPE IDENTIFIER:
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SOB - SET-ON BRANCH
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S - SHOP
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SYMBOLOGY:
WELD NUMBER
PART NUMBER (PT NO)
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SYMBOL IDENTIFIER:
F - FLANGE
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			FOR FABRICATION & ERECTION		
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PIPELINE:	FFS_SPACE-IN_MD2.2_3	PAGE NUMBER:	1 OF 1		

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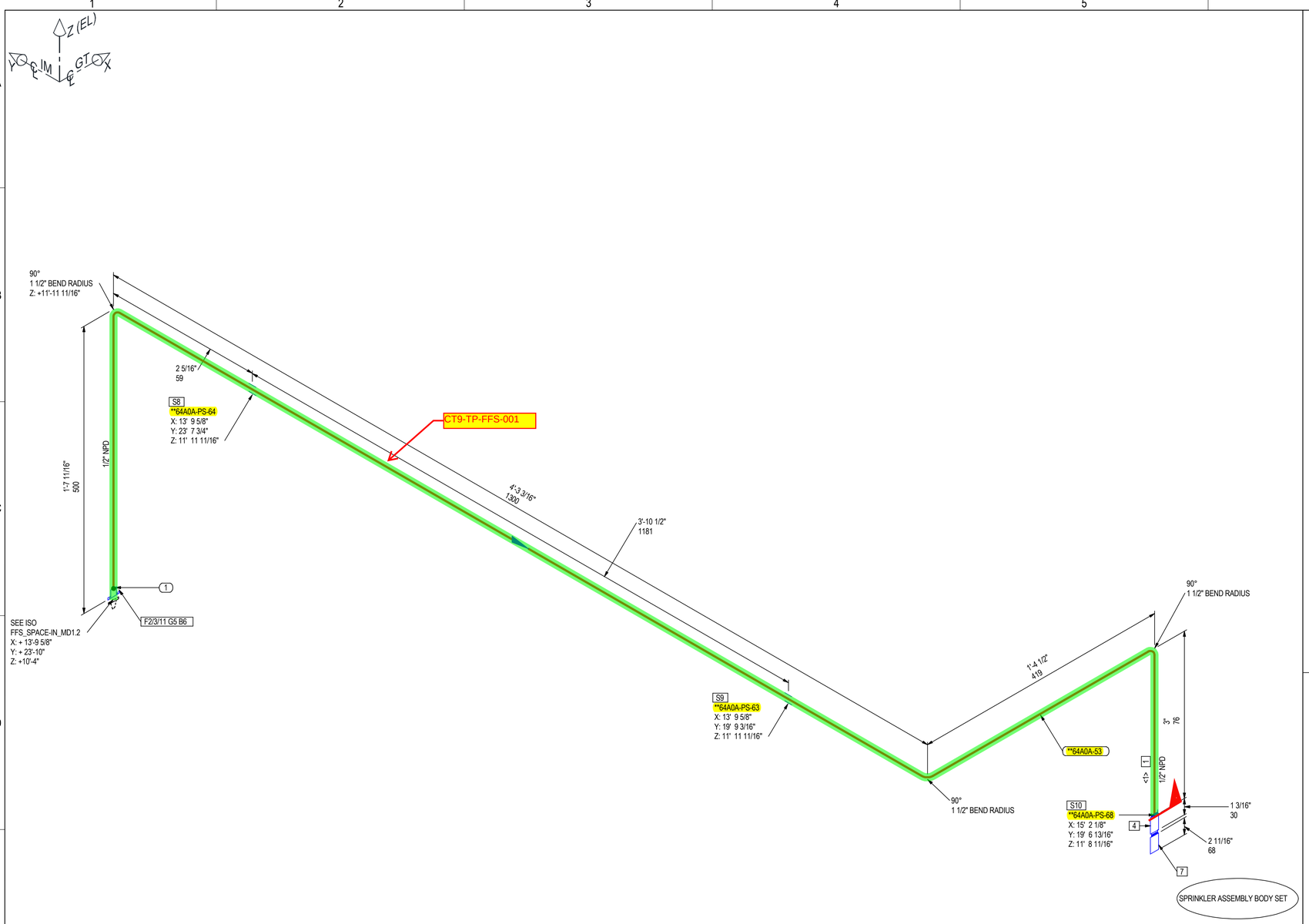

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	MISCELLANEOUS COMPONENTS			
3	FLANGE HYDRAULIC, 3000 PSI, SAE J518-1 CODE 61, ASTM A182-F316, SPLIT FLANGE CLAMP (ONE PAIR)	1/2	MWABAWYHZZIARALS	1

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PT NO	DESCRIPTION	NPD (IN)	CMDTY CODE	QTY
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5	GASKET, ORING, 0.125" THK, AS 568, VITON HS90 (DUROMETER90)	1/2	MKBZZACIAQXANUS	1
6	SCREW, DRILLED HEX SOCKET HEAD CAP (2 HOLES), ASME B18.3, ASTM A193-B7, 18-UNC - 1.5 in. Length	5/16	MBCZZBRZZIALBEUZ	4
	INSTRUMENTS			
7	SPRINKLER ASSEMBLY BODY SET FOR HI-FOG® 1000 SERIES SPRINKLER AND SPRAY HEADS, A1000S1CT00D12S10 SPRINKLER ASSEMBLY BODY SET	1/2		1
	PIPE SUPPORTS			
8	PIPE SUPPORT	1/2	**64A0A-PS-64	1
9	PIPE SUPPORT	1/2	**64A0A-PS-63	1
10	NOZZLE MOUNTING ADAPTER SUPPORT	1/2	**64A0A-PS-68	1
	MISCELLANEOUS COMPONENTS			
11	LOCK WIRE, 1mm DIA x 1m LENGTH, ASTM A580-304	1/2	MWRZZZZBAIASXAPU	1
	SPOOLS			
	**64A0A-53			

CUT PIPE LENGTH

PIECE NO	LENGTH (FT-INS)	NPD (IN)	REMARKS	END ONE	END TWO	ITEM CODE
<1>	7'4 1/4"	1/2	PLD BEND(3)	SQ.CUT	SQ.CUT	MTAAZZBPZZZZZXAQ

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MITSUBISHI POWER

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APPROVED BY:	N.KOZAN	07/25/2024	INTERNAL TO GT PACKAGE		
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			GRAND RIVER ENERGY CENTER UNIT 4		
			GRAND RIVER DAM AUTHORITY		
			M501JAC GAS TURBINE		
PROJECT CODE:	K-GREC1	P&ID:			
PIPELINE:	FFS_SPACE-IN_MD2.2_4	PAGE NUMBER:	1 OF 1		

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NO	SUPPORT TAG	SPOOL NO	REMARK
1	**23A0A-PS-05	**23A0A-9	
2	**23A0B-PS-03	**23A0B-16	
3	**23A0B-PS-04	**23A0B-22	
4	**23A0B-PS-05	**23A0B-22	
5	**23A0B-PS-12	**23A0B-15	SEE NOTE 5
6	**23A0C-PS-10	**23A0C-16	
7	**23A0C-PS-11	**23A0C-16	
8	**23A0C-PS-17	**23A0C-24	
9	**23A0D-PS-01	**23A0D-2	
10	**23A0E-PS-01	**23A0E-3	
11	**23A0G-PS-01	**23A0G-12	
12	**23A0G-PS-02	**23A0G-12	
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14	**23A0H-PS-07	**23A0H-6	
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16	**23A0H-PS-09	**23A0H-5	
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18	**23A0H-PS-25	**23A0H-21	
19	**23A0S-PS-07	**23A0S-24	SEE NOTE 5
20	**23A0U-PS-07	**23A0U-24	
21	**24A0A-PS-01	**24A0A-22	
22	**24A0A-PS-02	**24A0A-22	
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24	**24A0A-PS-04	**24A0A-25	
25	**24A0A-PS-05	**24A0A-27	
26	**24A0A-PS-06	**24A0A-28	
27	**24A0A-PS-100	**24A0A-13	
28	**24A0A-PS-108	**24A0A-24	
29	**24A0A-PS-109	**24A0A-24	
30	**24A0A-PS-122	AIR CHAMBER	
31	**24A0A-PS-123	AIR CHAMBER	
32	**24A0A-PS-124	AIR CHAMBER	
33	**24A0A-PS-125	AIR CHAMBER	
34	**24A0A-PS-13	**24A0A-40	
35	**24A0A-PS-14	**24A0A-8	
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47	**24A0C-PS-03	**24A0C-6	
48	**24A0C-PS-04	**24A0C-6	
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59	**27A0B-PS-09	**27A0B-24	
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61	**45A0A-PS-01	**45A0A-59	
62	**45A0A-PS-02	**45A0A-47	SEE NOTE 5
63	**45A0A-PS-03	**45A0A-59	
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72	**45A0A-PS-29	**45A0A-50	
73	**45A0A-PS-30	**45A0A-58	
74	**45A0A-PS-31	**45A0A-58	
75	**45A0A-PS-36	**45A0A-58	
76	**45A0A-PS-40	**45A0A-58	
77	**45A0A-PS-46	**45A0A-58	
78	**45A0A-PS-51	**45A0A-59	
79	**45A0A-PS-53	**45A0A-58	
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81	**45A0A-PS-57	**45A0A-58	
82	**45A0A-PS-58	**45A0A-58	
83	**45A0A-PS-63	**45A0A-58	
84	**45A0A-PS-66	**45A0A-59	
85	**45A0A-PS-67	**45A0A-58	
86	**45A0A-PS-68	**45A0A-58	
87	**45A0A-PS-69	**45A0A-58	
88	**45A0A-PS-70	**45A0A-58	
89	**45A0A-PS-71	**45A0A-57	
90	**45A0C-PS-01	**45A0A-59 **45A0C-1	
91	**45A0C-PS-02	**45A0C-1	
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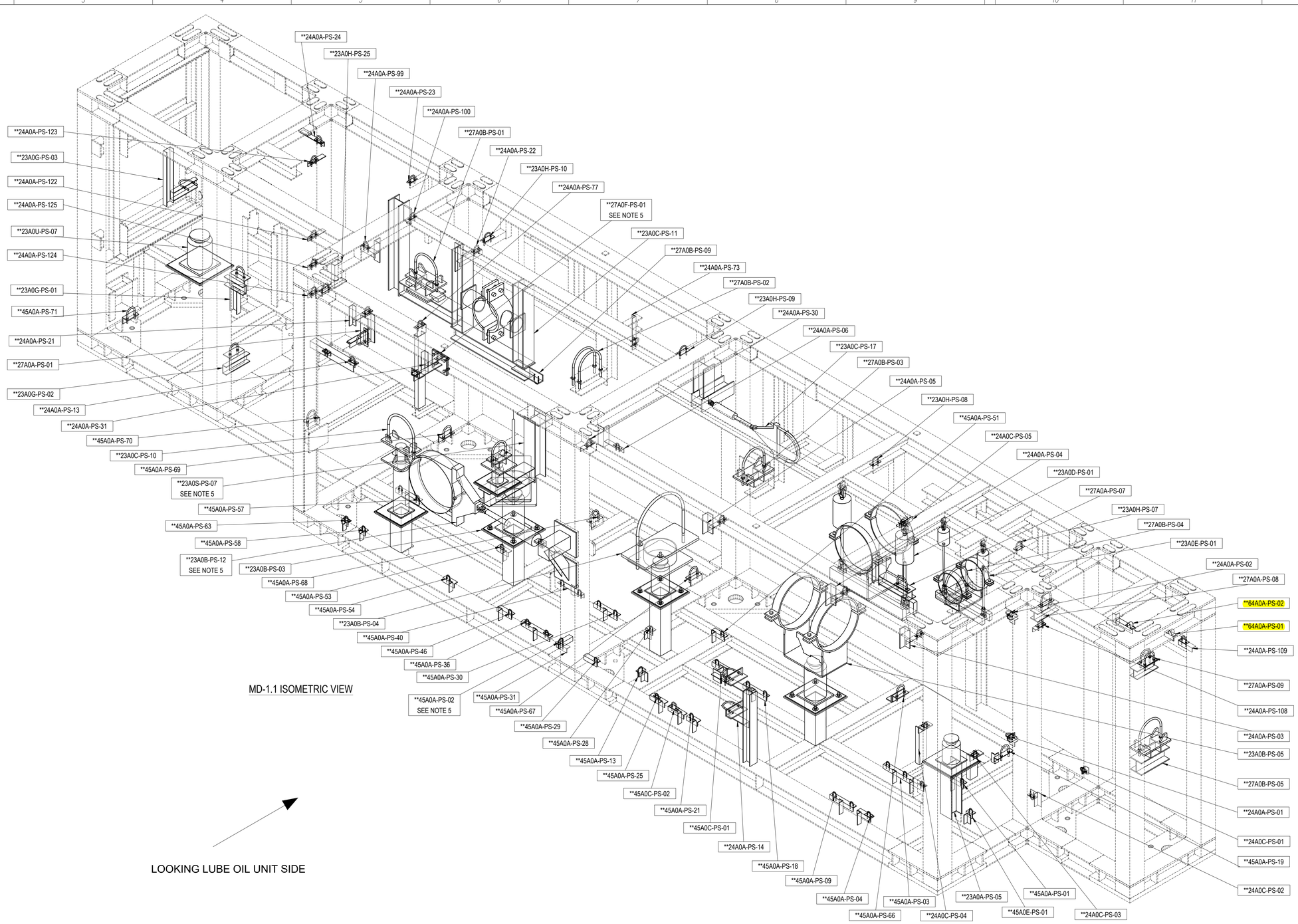
94	**64A0A-PS-02	FFS	



MD-1.1 ISOMETRIC VIEW

LOOKING LUBE OIL UNIT SIDE

REFERENCES

MDL ITEM #	MHPSA MDL DESCRIPTION	DETAILS FOR
N/A	LOW PRESSURE BLEED AIR PIPING (23A0A)	PIPE SUPPORT DRAWINGS
N/A	MIDDLE PRESSURE BLEED AIR PIPING (23A0B)	"
N/A	HIGH PRESSURE BLEED AIR PIPING (23A0C)	"
N/A	TURBINE COOLING AIR SUPPLY PIPING (23A0D)	"
N/A	TURBINE COOLING AIR RETURN PIPING (23A0E)	"
N/A	TCA FILTER BLEED AIR PIPING (23A0G)	"
N/A	SEAL AIR PIPING (23A0H)	"
N/A	ENHANCED COOLING AIR SUPPLY PIPING (23A0S)	"
N/A	TURBINE BYPASS PIPING (23A0U)	"
N/A	INSTRUMENT AIR PIPING (24A0A)	"
N/A	GT COOLING AIR LINE PURGE AIR PIPING (24A0C)	"
N/A	LUBE OIL SUPPLY PIPING (27A0A)	"
N/A	LUBE OIL RETURN PIPING (27A0B)	"
N/A	LUBE OIL TURBINE SUPPORT COOLING SUPPLY PIPING (27A0F)	"
N/A	BLADE WASHING DRAIN PIPING (45A0A)	"
N/A	COMBUSTOR CYLINDER DRAIN PIPING (45A0C)	"
N/A	INLET MANIFOLD DRAIN PIPING (45A0E)	"
34	GT ENCLOSURE MODULE 1-1 DETAIL DRAWINGS	STRUCTURE

NOTES

1. DIMENSIONS : MM [INCHES]
2. **XXXXX-XX-XX → SUPPORT TAG
3. GRATINGS ARE HIDDEN FOR A MORE CLEAR VIEW WHERE IT IS CONSIDERED NECESSARY
4. THE DIMENSIONS GIVEN IN THESE GENERAL ARRANGEMENT DRAWINGS ARE GIVEN FROM THE NEAREST PROFILES AND ARE THEORETICAL. DUE TO PRODUCTION TOLERANCES OF THE PROFILES AND MANUFACURING & ASSEMBLY TOLERANCES OF THE STEEL CONSTRUCTION SOME DIFFERENCES CAN BE OBSERVED IN REGARDS TO ORIGINAL SUPPORT THEORITICAL WORKING POINT COORDINATES AND ELEVATIONS. OUR SUGGESTION IS IN SUCH INCONVENIENT CASES, A CHECK SHALL BE MADE ACCORDING TO THE ELEVATION AND COORDINATES IN THE ORIGINAL SUPPORT DRAWINGS.
5. OFF MODULE SUPPORTS ARE GOING TO BE SITE MANUFACTURED AND ASSEMBLED.

REV.	DESCRIPTION	DATE	BY	CHECK	APPROVAL
0	ISSUED FOR CONSTRUCTION	04.07.2024	S.C.K.	S.S.	S.O.

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MITSUBISHI POWER

CUSTOMER JOB NO: -

ITEM NO: MDL-150

DWG NAME: MD-1.1 SUPPORTS GENERAL ARRANGEMENT DRAWING

REF.DWG.NO.: -

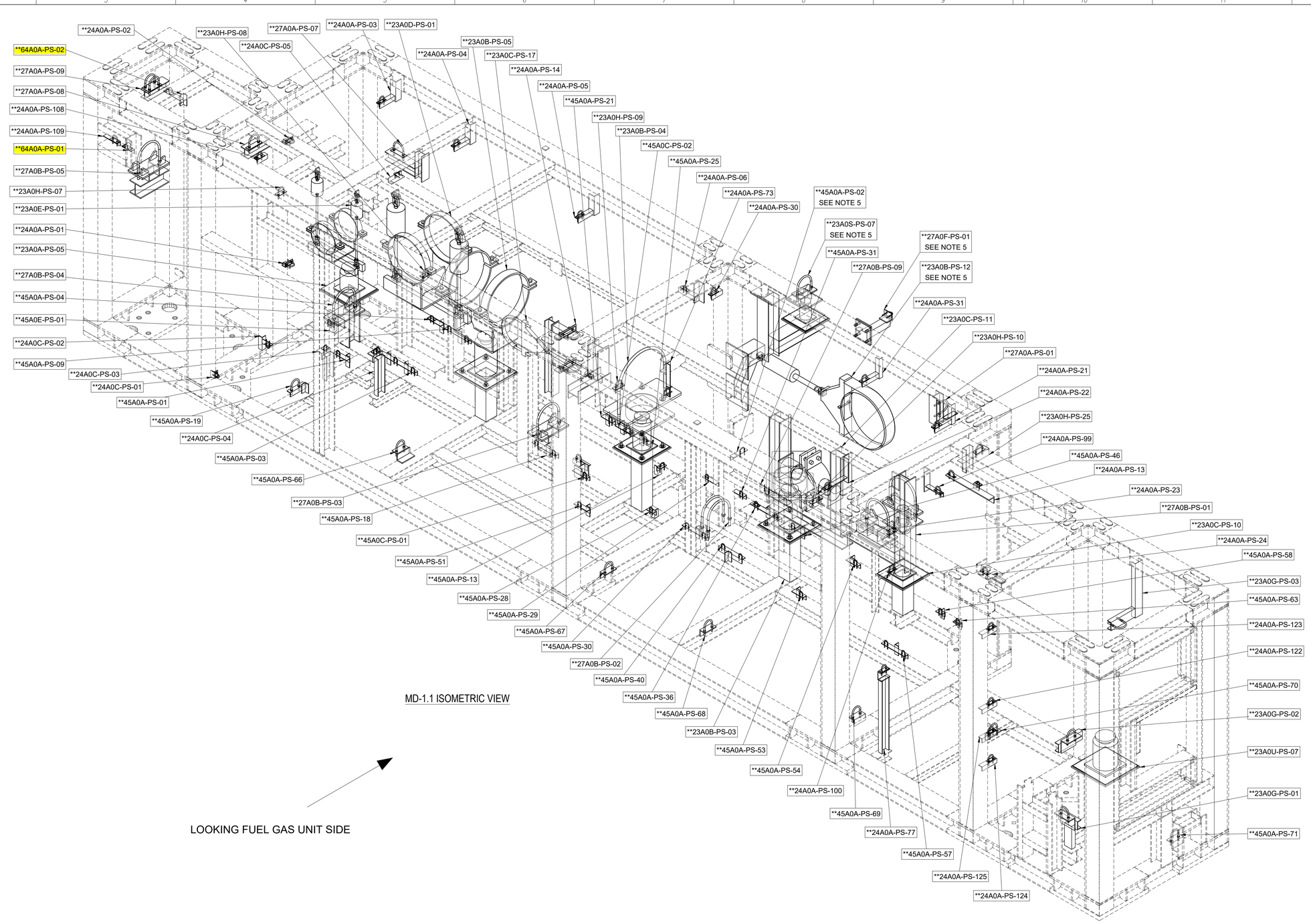
PREPARED BY: cimtas

JOB NO	SHEET	SCALE	SIZE
2E00140	1/6	1/20	A1

DRAWING NO: 2E00140-CI-026-150

REV.: 0

NO	SUPPORT TAG	SPOOL NO	REMARK
1	**23A0A-PS-05	**23A0A-9	
2	**23A0B-PS-03	**23A0B-16	
3	**23A0B-PS-04	**23A0B-22	
4	**23A0B-PS-05	**23A0B-22	
5	**23A0B-PS-12	**23A0B-15	SEE NOTE 5
6	**23A0C-PS-10	**23A0C-16	
7	**23A0C-PS-11	**23A0C-16	
8	**23A0C-PS-17	**23A0C-24	
9	**23A0D-PS-01	**23A0D-2	
10	**23A0E-PS-01	**23A0E-3	
11	**23A0G-PS-01	**23A0G-12	
12	**23A0G-PS-02	**23A0G-12	
13	**23A0G-PS-03	**23A0G-12	
14	**23A0H-PS-07	**23A0H-6	
15	**23A0H-PS-08	**23A0H-6	
16	**23A0H-PS-09	**23A0H-5	
17	**23A0H-PS-10	**23A0H-5	
18	**23A0H-PS-25	**23A0H-21	
19	**23A0S-PS-07	**23A0S-24	SEE NOTE 5
20	**23A0U-PS-07	**23A0U-24	
21	**24A0A-PS-01	**24A0A-22	
22	**24A0A-PS-02	**24A0A-22	
23	**24A0A-PS-03	**24A0A-25	
24	**24A0A-PS-04	**24A0A-25	
25	**24A0A-PS-05	**24A0A-27	
26	**24A0A-PS-06	**24A0A-28	
27	**24A0A-PS-100	**24A0A-13	
28	**24A0A-PS-108	**24A0A-24	
29	**24A0A-PS-109	**24A0A-24	
30	**24A0A-PS-122	AIR CHAMBER	
31	**24A0A-PS-123	AIR CHAMBER	
32	**24A0A-PS-124	AIR CHAMBER	
33	**24A0A-PS-125	AIR CHAMBER	
34	**24A0A-PS-13	**24A0A-40	
35	**24A0A-PS-14	**24A0A-8	
36	**24A0A-PS-21	**24A0A-49	
37	**24A0A-PS-22	**24A0A-50	
38	**24A0A-PS-23	**24A0A-52	
39	**24A0A-PS-24	**24A0A-52	
40	**24A0A-PS-30	**24A0A-28	
41	**24A0A-PS-31	**24A0A-28	
42	**24A0A-PS-73	**24A0A-50	
43	**24A0A-PS-77	**24A0A-51	
44	**24A0A-PS-99	**24A0A-14	
45	**24A0C-PS-01	**24A0C-4	
46	**24A0C-PS-02	**24A0C-3	
47	**24A0C-PS-03	**24A0C-6	
48	**24A0C-PS-04	**24A0C-6	
49	**24A0C-PS-05	**24A0C-6	
50	**27A0A-PS-01	**27A0A-6	
51	**27A0A-PS-07	**27A0A-10	
52	**27A0A-PS-08	**27A0A-10	
53	**27A0A-PS-09	**27A0A-10	
54	**27A0B-PS-01	**27A0B-1	
55	**27A0B-PS-02	**27A0B-3	
56	**27A0B-PS-03	**27A0B-3	
57	**27A0B-PS-04	**27A0B-4	
58	**27A0B-PS-05	**27A0B-5	
59	**27A0B-PS-09	**27A0B-24	
60	**27A0F-PS-01	**27A0F-6	SEE NOTE 5
61	**45A0A-PS-01	**45A0A-59	
62	**45A0A-PS-02	**45A0A-47	SEE NOTE 5
63	**45A0A-PS-03	**45A0A-59	
64	**45A0A-PS-04	**45A0A-59	
65	**45A0A-PS-09	**45A0A-59	
66	**45A0A-PS-13	**45A0A-48	
67	**45A0A-PS-18	**45A0A-59	
68	**45A0A-PS-19	**45A0A-59	
69	**45A0A-PS-21	**45A0A-59	
70	**45A0A-PS-25	**45A0A-59	
71	**45A0A-PS-28	**45A0A-59	
72	**45A0A-PS-29	**45A0A-50	
73	**45A0A-PS-30	**45A0A-58	
74	**45A0A-PS-31	**45A0A-58	
75	**45A0A-PS-36	**45A0A-58	
76	**45A0A-PS-40	**45A0A-58	
77	**45A0A-PS-46	**45A0A-58	
78	**45A0A-PS-51	**45A0A-59	
79	**45A0A-PS-53	**45A0A-58	
80	**45A0A-PS-54	**45A0A-58	
81	**45A0A-PS-57	**45A0A-58	
82	**45A0A-PS-58	**45A0A-58	
83	**45A0A-PS-63	**45A0A-58	
84	**45A0A-PS-66	**45A0A-59	
85	**45A0A-PS-67	**45A0A-58	
86	**45A0A-PS-68	**45A0A-58	
87	**45A0A-PS-69	**45A0A-58	
88	**45A0A-PS-70	**45A0A-58	
89	**45A0A-PS-71	**45A0A-57	
90	**45A0C-PS-01	**45A0A-59 **45A0C-1	
91	**45A0C-PS-02	**45A0C-1	
92	**45A0E-PS-01	**45A0E-1	
93	**64A0A-PS-01	FFS	
94	**64A0A-PS-02	FFS	



MD-1.1 ISOMETRIC VIEW

LOOKING FUEL GAS UNIT SIDE

REFERENCES

MDL ITEM #	MHPSA MDL DESCRIPTION	DETAILS FOR
N/A	LOW PRESSURE BLEED AIR PIPING (23A0A)	PIPE SUPPORT DRAWINGS
N/A	MIDDLE PRESSURE BLEED AIR PIPING (23A0B)	"
N/A	HIGH PRESSURE BLEED AIR PIPING (23A0C)	"
N/A	TURBINE COOLING AIR SUPPLY PIPING (23A0D)	"
N/A	TURBINE COOLING AIR RETURN PIPING (23A0E)	"
N/A	TCA FILTER BLEED AIR PIPING (23A0G)	"
N/A	SEAL AIR PIPING (23A0H)	"
N/A	ENHANCED COOLING AIR SUPPLY PIPING (23A0S)	"
N/A	TURBINE BYPASS PIPING (23A0U)	"
N/A	INSTRUMENT AIR PIPING (24A0A)	"
N/A	GT COOLING AIR LINE PURGE AIR PIPING (24A0C)	"
N/A	LUBE OIL SUPPLY PIPING (27A0A)	"
N/A	LUBE OIL RETURN PIPING (27A0B)	"
N/A	LUBE OIL TURBINE SUPPORT COOLING SUPPLY PIPING (27A0F)	"
N/A	BLADE WASHING DRAIN PIPING (45A0A)	"
N/A	COMBUSTOR CYLINDER DRAIN PIPING (45A0C)	"
N/A	INLET MANIFOLD DRAIN PIPING (45A0E)	"
34	GT ENCLOSURE MODULE 1-1 DETAIL DRAWINGS	STRUCTURE

NOTES

1. DIMENSIONS : MM [INCHES]
2. **XXXX-XX-XX → SUPPORT TAG
3. GRATINGS ARE HIDDEN FOR A MORE CLEAR VIEW WHERE IT IS CONSIDERED NECESSARY
4. THE DIMENSIONS GIVEN IN THESE GENERAL ARRANGEMENT DRAWINGS ARE GIVEN FROM THE NEAREST PROFILES AND ARE THEORETICAL. DUE TO PRODUCTION TOLERANCES OF THE PROFILES AND MANUFACURING & ASSEMBLY TOLERANCES OF THE STEEL CONSTRUCTION SOME DIFFERENCES CAN BE OBSERVED IN REGARDS TO ORIGINAL SUPPORT THEORITICAL WORKING POINT COORDINATES AND ELEVATIONS. OUR SUGGESTION IS IN SUCH INCONVENIENT CASES, A CHECK SHALL BE MADE ACCORDING TO THE ELEVATION AND COORDINATES IN THE ORIGINAL SUPPORT DRAWINGS.
5. OFF MODULE SUPPORTS ARE GOING TO BE SITE MANUFACTURED AND ASSEMBLED.

REV.	DESCRIPTION	DATE	BY	CHECK	APPROVAL
0	ISSUED FOR CONSTRUCTION	04.07.2024	S.C.K.	S.S.	S.O.

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MITSUBISHI POWER

CUSTOMER JOB NO: -

ITEM NO: MDL-150

DWG NAME: MD-1.1 SUPPORTS GENERAL ARRANGEMENT DRAWING

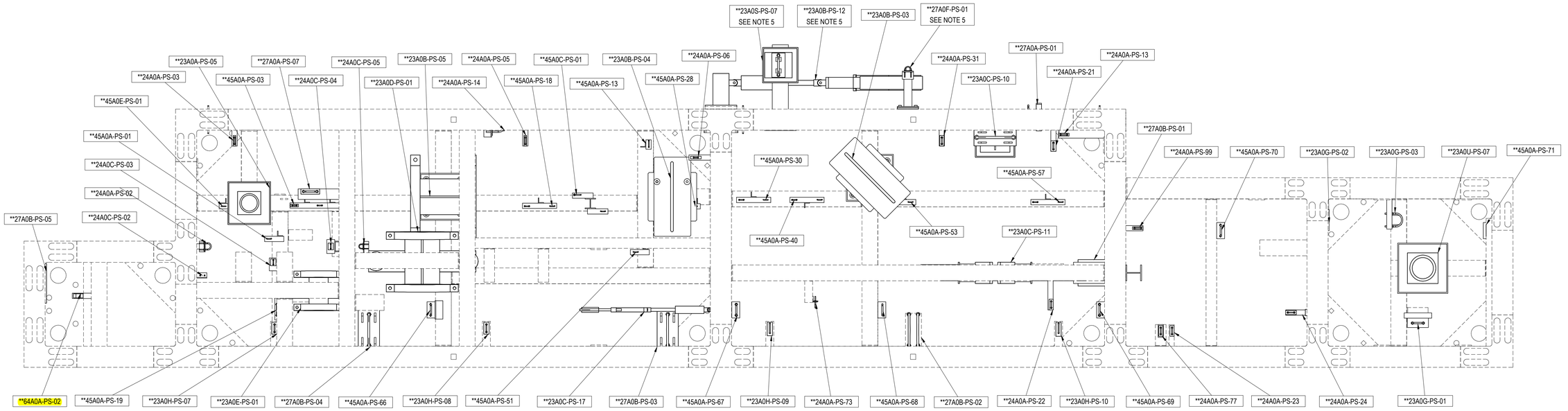
REF.DWG.NO.: -

PREPARED BY: cimtas

JOB NO	SHEET	SCALE	SIZE
2E00140	2/6	1/20	A1

DRAWING NO: 2E00140-CI-026-150

REV.: 0



MD-1.1 PLAN VIEW

REFERENCES

MDL ITEM #	MHPSA MDL DESCRIPTION	DETAILS FOR
N/A	LOW PRESSURE BLEED AIR PIPING (23A0A)	PIPE SUPPORT DRAWINGS
N/A	MIDDLE PRESSURE BLEED AIR PIPING (23A0B)	"
N/A	HIGH PRESSURE BLEED AIR PIPING (23A0C)	"
N/A	TURBINE COOLING AIR SUPPLY PIPING (23A0D)	"
N/A	TURBINE COOLING AIR RETURN PIPING (23A0E)	"
N/A	TCA FILTER BLEED AIR PIPING (23A0G)	"
N/A	SEAL AIR PIPING (23A0H)	"
N/A	ENHANCED COOLING AIR SUPPLY PIPING (23A0S)	"
N/A	TURBINE BYPASS PIPING (23A0U)	"
N/A	INSTRUMENT AIR PIPING (24A0A)	"
N/A	GT COOLING AIR LINE PURGE AIR PIPING (24A0C)	"
N/A	LUBE OIL SUPPLY PIPING (27A0A)	"
N/A	LUBE OIL RETURN PIPING (27A0B)	"
N/A	LUBE OIL TURBINE SUPPORT COOLING SUPPLY PIPING (27A0F)	"
N/A	BLADE WASHING DRAIN PIPING (45A0A)	"
N/A	COMBUSTOR CYLINDER DRAIN PIPING (45A0C)	"
N/A	INLET MANIFOLD DRAIN PIPING (45A0E)	"
34	GT ENCLOSURE MODULE 1-1 DETAIL DRAWINGS	STRUCTURE

NOTES

1. DIMENSIONS : MM [INCHES]
2. **XXXXX-XX-XX → SUPPORT TAG
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REV.	DESCRIPTION	DATE	BY	CHECK	APPROVAL
0	ISSUED FOR CONSTRUCTION	04.07.2024	S.C.K.	S.S.	S.O.

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MITSUBISHI POWER

CUSTOMER JOB NO: -

ITEM NO: MDL-150

DWG NAME: MD-1.1 SUPPORTS GENERAL ARRANGEMENT DRAWING

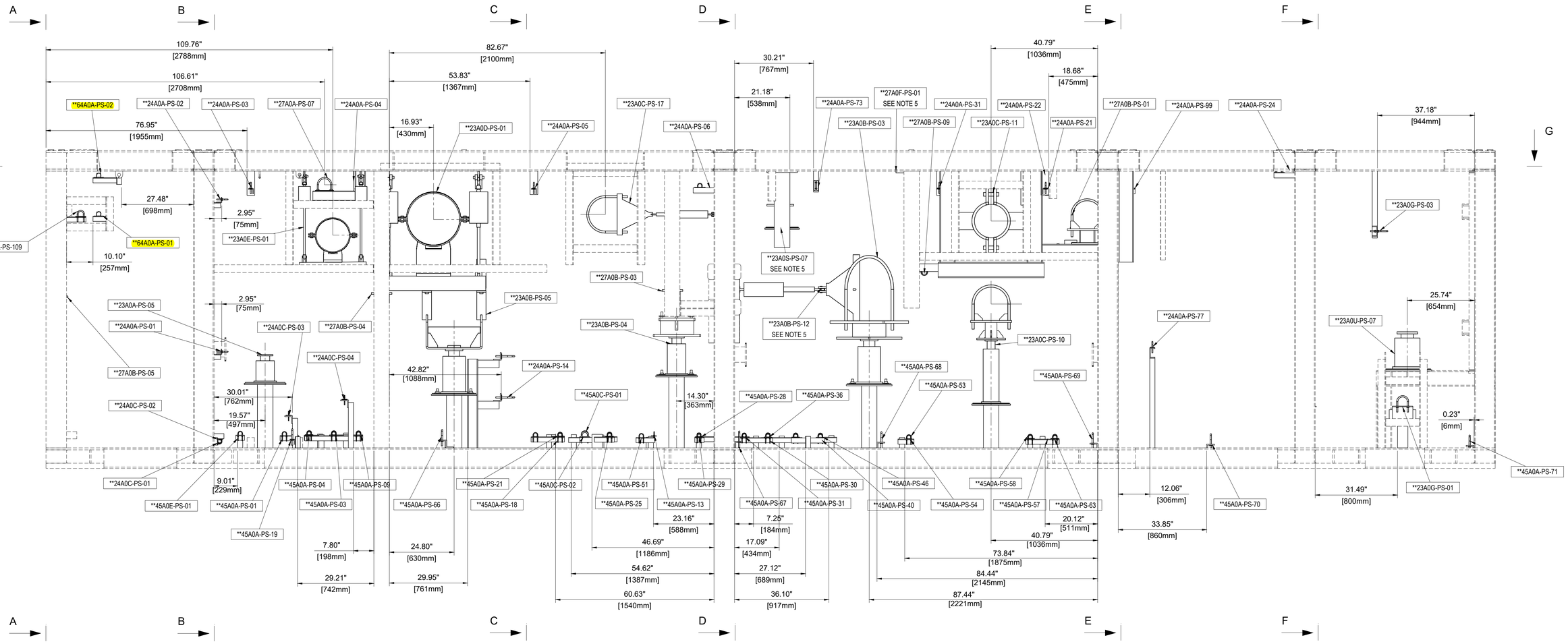
REF.DWG.NO.: -

PREPARED BY: cimtas

JOB NO	SHEET	SCALE	SIZE
2E00140	3/6	1/20	A1

DRAWING NO: 2E00140-CI-026-150

REV.: 0



MD-1.1 LOOKING FUEL GAS UNIT SIDE

REFERENCES

MDL ITEM #	MHPSA MDL DESCRIPTION	DETAILS FOR
N/A	LOW PRESSURE BLEED AIR PIPING (23A0A)	PIPE SUPPORT DRAWINGS
N/A	MIDDLE PRESSURE BLEED AIR PIPING (23A0B)	"
N/A	HIGH PRESSURE BLEED AIR PIPING (23A0C)	"
N/A	TURBINE COOLING AIR SUPPLY PIPING (23A0D)	"
N/A	TURBINE COOLING AIR RETURN PIPING (23A0E)	"
N/A	TCA FILTER BLEED AIR PIPING (23A0G)	"
N/A	SEAL AIR PIPING (23A0H)	"
N/A	ENHANCED COOLING AIR SUPPLY PIPING (23A0S)	"
N/A	TURBINE BYPASS PIPING (23A0U)	"
N/A	INSTRUMENT AIR PIPING (24A0A)	"
N/A	GT COOLING AIR LINE PURGE AIR PIPING (24A0C)	"
N/A	LUBE OIL SUPPLY PIPING (27A0A)	"
N/A	LUBE OIL RETURN PIPING (27A0B)	"
N/A	LUBE OIL TURBINE SUPPORT COOLING SUPPLY PIPING (27A0F)	"
N/A	BLADE WASHING DRAIN PIPING (45A0A)	"
N/A	COMBUSTOR CYLINDER DRAIN PIPING (45A0C)	"
N/A	INLET MANIFOLD DRAIN PIPING (45A0E)	"
34	GT ENCLOSURE MODULE 1-1 DETAIL DRAWINGS	STRUCTURE

NOTES

1. DIMENSIONS : MM [INCHES]
2. **XXXXX-XX-XX → SUPPORT TAG
3. GRATINGS ARE HIDDEN FOR A MORE CLEAR VIEW WHERE IT IS CONSIDERED NECESSARY
4. THE DIMENSIONS GIVEN IN THESE GENERAL ARRANGEMENT DRAWINGS ARE GIVEN FROM THE NEAREST PROFILES AND ARE THEORETICAL. DUE TO PRODUCTION TOLERANCES OF THE PROFILES AND MANUFACURING & ASSEMBLY TOLERANCES OF THE STEEL CONSTRUCTION SOME DIFFERENCES CAN BE OBSERVED IN REGARDS TO ORIGINAL SUPPORT THEORITICAL WORKING POINT COORDINATES AND ELEVATIONS. OUR SUGGESTION IS IN SUCH INCONVENIENT CASES, A CHECK SHALL BE MADE ACCORDING TO THE ELEVATION AND COORDINATES IN THE ORIGINAL SUPPORT DRAWINGS.
5. OFF MODULE SUPPORTS ARE GOING TO BE SITE MANUFACTURED AND ASSEMBLED.

REV.	DESCRIPTION	DATE	BY	CHECK	APPROVAL
0	ISSUED FOR CONSTRUCTION	04.07.2024	S.C.K.	S.S.	S.O.

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MITSUBISHI POWER

CUSTOMER JOB NO: -

ITEM NO: MDL-150

DWG NAME: MD-1.1 SUPPORTS GENERAL ARRANGEMENT DRAWING

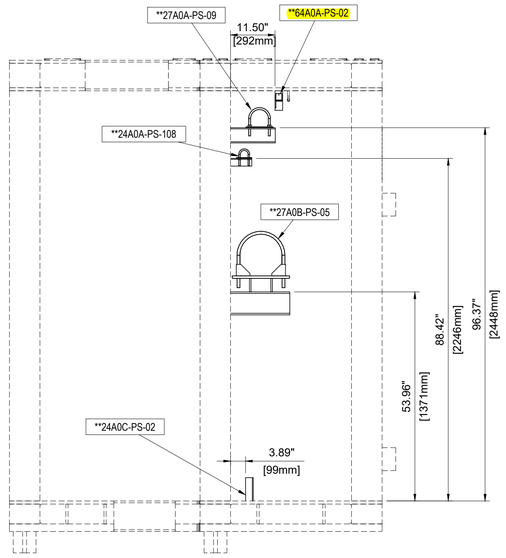
REF.DWG.NO.: -

PREPARED BY: cimtas

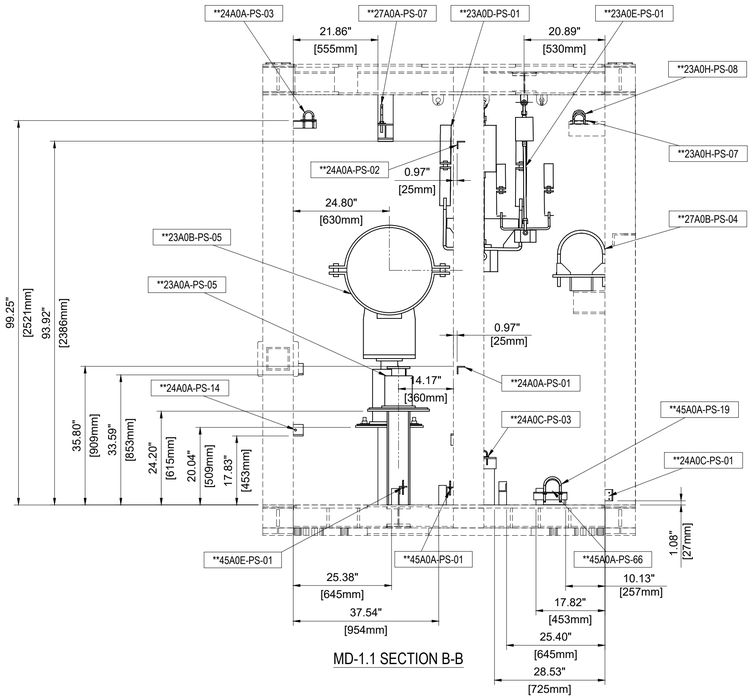
JOB NO	SHEET	SCALE	SIZE
2E00140	4/6	1/20	A1

DRAWING NO: 2E00140-CI-026-150

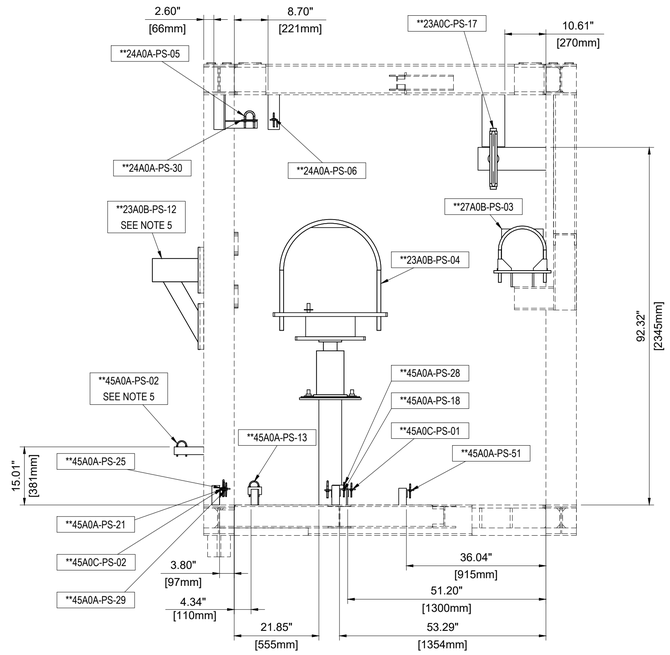
REV.: 0



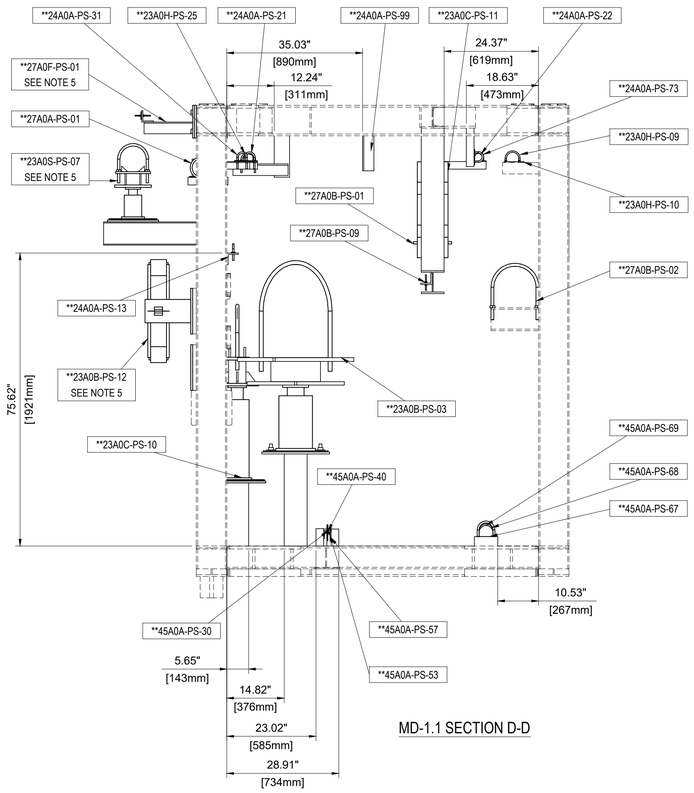
MD-1.1 SECTION A-A



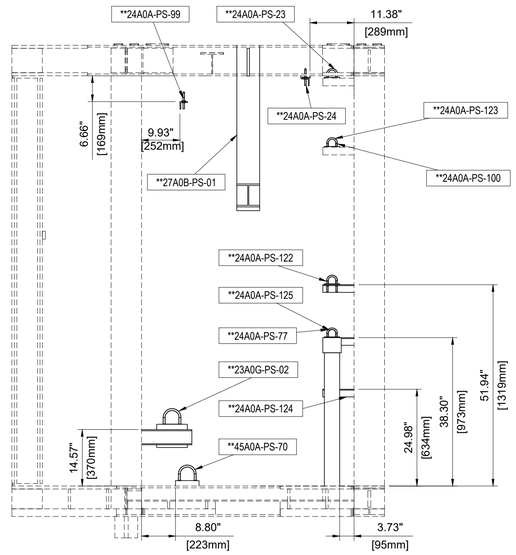
MD-1.1 SECTION B-B



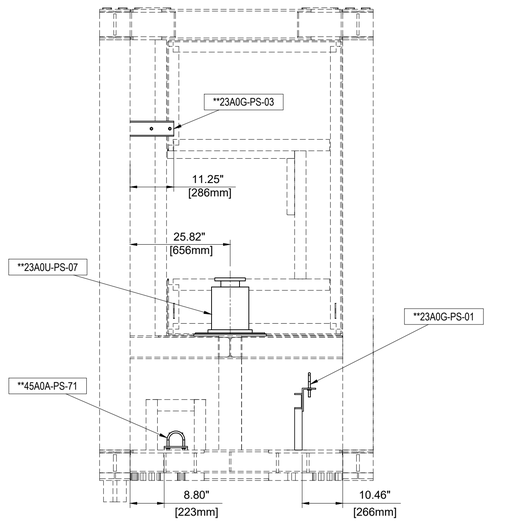
MD-1.1 SECTION C-C



MD-1.1 SECTION D-D



MD-1.1 SECTION E-E



MD-1.1 SECTION F-F

REFERENCES

MDL ITEM #	MHPSA MDL DESCRIPTION	DETAILS FOR
N/A	LOW PRESSURE BLEED AIR PIPING (23A0A)	PIPE SUPPORT DRAWINGS
N/A	MIDDLE PRESSURE BLEED AIR PIPING (23A0B)	"
N/A	HIGH PRESSURE BLEED AIR PIPING (23A0C)	"
N/A	TURBINE COOLING AIR SUPPLY PIPING (23A0D)	"
N/A	TURBINE COOLING AIR RETURN PIPING (23A0E)	"
N/A	TCA FILTER BLEED AIR PIPING (23A0G)	"
N/A	SEAL AIR PIPING (23A0H)	"
N/A	ENHANCED COOLING AIR SUPPLY PIPING (23A0S)	"
N/A	TURBINE BYPASS PIPING (23A0U)	"
N/A	INSTRUMENT AIR PIPING (24A0A)	"
N/A	GT COOLING AIR LINE PURGE AIR PIPING (24A0C)	"
N/A	LUBE OIL SUPPLY PIPING (27A0A)	"
N/A	LUBE OIL RETURN PIPING (27A0B)	"
N/A	LUBE OIL TURBINE SUPPORT COOLING SUPPLY PIPING (27A0F)	"
N/A	BLADE WASHING DRAIN PIPING (45A0A)	"
N/A	COMBUSTOR CYLINDER DRAIN PIPING (45A0C)	"
N/A	INLET MANIFOLD DRAIN PIPING (45A0E)	"
34	GT ENCLOSURE MODULE 1-1 DETAIL DRAWINGS	STRUCTURE

NOTES

1. DIMENSIONS : MM [INCHES]
2. **XXXXX-XX-XX** → SUPPORT TAG
3. GRATINGS ARE HIDDEN FOR A MORE CLEAR VIEW WHERE IT IS CONSIDERED NECESSARY
4. THE DIMENSIONS GIVEN IN THESE GENERAL ARRANGEMENT DRAWINGS ARE GIVEN FROM THE NEAREST PROFILES AND ARE THEORETICAL. DUE TO PRODUCTION TOLERANCES OF THE PROFILES AND MANUFACURING & ASSEMBLY TOLERANCES OF THE STEEL CONSTRUCTION SOME DIFFERENCES CAN BE OBSERVED IN REGARDS TO ORIGINAL SUPPORT THEORITICAL WORKING POINT COORDINATES AND ELEVATIONS. OUR SUGGESTION IS IN SUCH INCONVENIENT CASES, A CHECK SHALL BE MADE ACCORDING TO THE ELEVATION AND COORDINATES IN THE ORIGINAL SUPPORT DRAWINGS.
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REV.	DESCRIPTION	DATE	BY	CHECK	APPROVAL
0	ISSUED FOR CONSTRUCTION	04.07.2024	S.C.K.	S.S.	S.O.

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MITSUBISHI POWER

CUSTOMER JOB NO: -

ITEM NO: MDL-150

DWG NAME: MD-1.1 SUPPORTS GENERAL ARRANGEMENT DRAWING

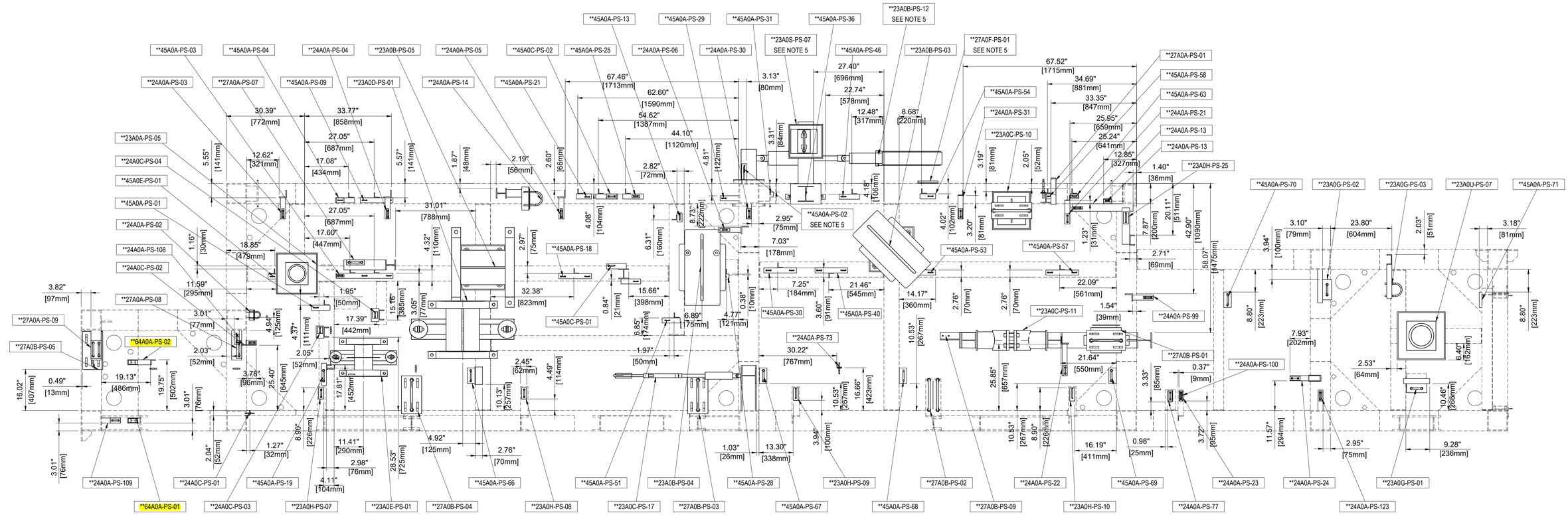
REF.DWG.NO.: -

PREPARED BY: cimtas

JOB NO	SHEET	SCALE	SIZE
2E00140	5/6	1/20	A1

DRAWING NO: 2E00140-CI-026-150

REV.: 0



MD-1.1 SECTION G-G

REFERENCES

MDL ITEM #	MHPSA MDL DESCRIPTION	DETAILS FOR
N/A	LOW PRESSURE BLEED AIR PIPING (23A0A)	PIPE SUPPORT DRAWINGS
N/A	MIDDLE PRESSURE BLEED AIR PIPING (23A0B)	"
N/A	HIGH PRESSURE BLEED AIR PIPING (23A0C)	"
N/A	TURBINE COOLING AIR SUPPLY PIPING (23A0D)	"
N/A	TURBINE COOLING AIR RETURN PIPING (23A0E)	"
N/A	TCA FILTER BLEED AIR PIPING (23A0G)	"
N/A	SEAL AIR PIPING (23A0H)	"
N/A	ENHANCED COOLING AIR SUPPLY PIPING (23A0S)	"
N/A	TURBINE BYPASS PIPING (23A0U)	"
N/A	INSTRUMENT AIR PIPING (24A0A)	"
N/A	GT COOLING AIR LINE PURGE AIR PIPING (24A0C)	"
N/A	LUBE OIL SUPPLY PIPING (27A0A)	"
N/A	LUBE OIL RETURN PIPING (27A0B)	"
N/A	LUBE OIL TURBINE SUPPORT COOLING SUPPLY PIPING (27A0F)	"
N/A	BLADE WASHING DRAIN PIPING (45A0A)	"
N/A	COMBUSTOR CYLINDER DRAIN PIPING (45A0C)	"
N/A	INLET MANIFOLD DRAIN PIPING (45A0E)	"
34	GT ENCLOSURE MODULE 1-1 DETAIL DRAWINGS	STRUCTURE

NOTES

1. DIMENSIONS : MM [INCHES]
2. **XXXX-XX-XX → SUPPORT TAG
3. GRATINGS ARE HIDDEN FOR A MORE CLEAR VIEW WHERE IT IS CONSIDERED NECESSARY
4. THE DIMENSIONS GIVEN IN THESE GENERAL ARRANGEMENT DRAWINGS ARE GIVEN FROM THE NEAREST PROFILES AND ARE THEORETICAL. DUE TO PRODUCTION TOLERANCES OF THE PROFILES AND MANUFACURING & ASSEMBLY TOLERANCES OF THE STEEL CONSTRUCTION SOME DIFFERENCES CAN BE OBSERVED IN REGARDS TO ORIGINAL SUPPORT THEORITICAL WORKING POINT COORDINATES AND ELEVATIONS. OUR SUGGESTION IS IN SUCH INCONVENIENT CASES, A CHECK SHALL BE MADE ACCORDING TO THE ELEVATION AND COORDINATES IN THE ORIGINAL SUPPORT DRAWINGS.
5. OFF MODULE SUPPORTS ARE GOING TO BE SITE MANUFACTURED AND ASSEMBLED.

REV.	DESCRIPTION	DATE	BY	CHECK	APPROVAL
0	ISSUED FOR CONSTRUCTION	04.07.2024	S.C.K.	S.S.	S.O.

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REVISIONS

CUSTOMER

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MITSUBISHI POWER

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